

Section 8 Mycobacterial Diseases

178 Tuberculosis

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Tuberculosis (TB), which is caused by bacteria of the *Mycobacterium tuberculosis* complex, is one of the oldest diseases known to affect humans and the top cause of infectious death worldwide excluding COVID-19. Population genomic studies suggest that *M. tuberculosis* may have emerged ~70,000 years ago in Africa and subsequently disseminated along with anatomically modern humans, expanding globally during the Neolithic Age as human density started to increase. Progenitors of *M. tuberculosis* are likely to have affected prehumanoids. This disease most often affects the lungs, although other organs are involved in up to one-third of cases. If properly treated, TB caused by drug-susceptible strains is curable in the vast majority of cases. If untreated, the disease may be eventually fatal in over 70% of people. Transmission usually takes place through the airborne spread of droplet nuclei produced by patients with infectious pulmonary TB. Through pharmacological prophylaxis the development of the disease can be prevented in those who have contracted TB infection.

ETIOLOGIC AGENT

Mycobacteria belong to the family Mycobacteriaceae and the order Actinomycetales. Of the pathogenic species belonging to the *M. tuberculosis* complex, which comprises eight distinct subgroups, the most common and important agent of human disease by far is *M. tuberculosis* (sensu stricto). A closely related organism isolated from cases in West, Central, and East Africa is *M. africanum*. The complex includes some zoonotic members, such as *M. bovis* (the bovine tubercle bacillus—characteristically resistant to pyrazinamide, once an important cause of TB transmitted by unpasteurized milk, and currently responsible for 140,000 human cases worldwide in 2019, half of them in Africa) and *M. caprae* (related to *M. bovis*). In addition, other organisms that have been reported rarely as causing TB include *M. pinnipedii* (a bacillus infecting seals and sea lions in the southern hemisphere and recently isolated from humans), *M. mungi* (isolated from banded mongooses in southern Africa), *M. orygis* (described in oryxes and other Bovidae in Africa and Asia and a potential cause of infection in humans), and *M. microti* (the “vole” bacillus, a less virulent organism). Finally, *M. canettii* is a rare isolate from East African cases that produces unusual smooth colonies on solid media and is considered closely related to a supposed progenitor type. There is no known environmental reservoir for any of these organisms.

M. tuberculosis is a rod-shaped, non-spore-forming, thin aerobic bacterium measuring 0.5 μm by 3 μm . Mycobacteria, including *M. tuberculosis*, are often neutral on Gram's staining. However, once stained, the bacilli cannot be decolorized by acid alcohol; this characteristic justifies their classification as acid-fast bacilli (AFB; Fig. 178-1). Acid fastness is due mainly to the organisms' high content of mycolic acids, long-chain cross-linked fatty acids, and other cell-wall lipids. Microorganisms other than mycobacteria that display some acid fastness include species of *Nocardia* and *Rhodococcus*, *Legionella micdadei*, and the protozoa *Isospora* and *Cryptosporidium*. In the mycobacterial cell wall, lipids (e.g., mycolic acids) are linked to underlying arabinogalactan and peptidoglycan. This structure results in very low permeability of the cell wall, thus reducing the effectiveness of most antibiotics. Another molecule in the mycobacterial cell wall, lipoarabinomannan, is involved in the pathogen–host interaction and facilitates the survival of *M. tuberculosis* within macrophages.



The complete genome sequence of *M. tuberculosis* comprises 4.4 million base pairs, 4043 genes encoding 3993 proteins, and 50 genes encoding stable RNAs; its high guanine-plus-cytosine

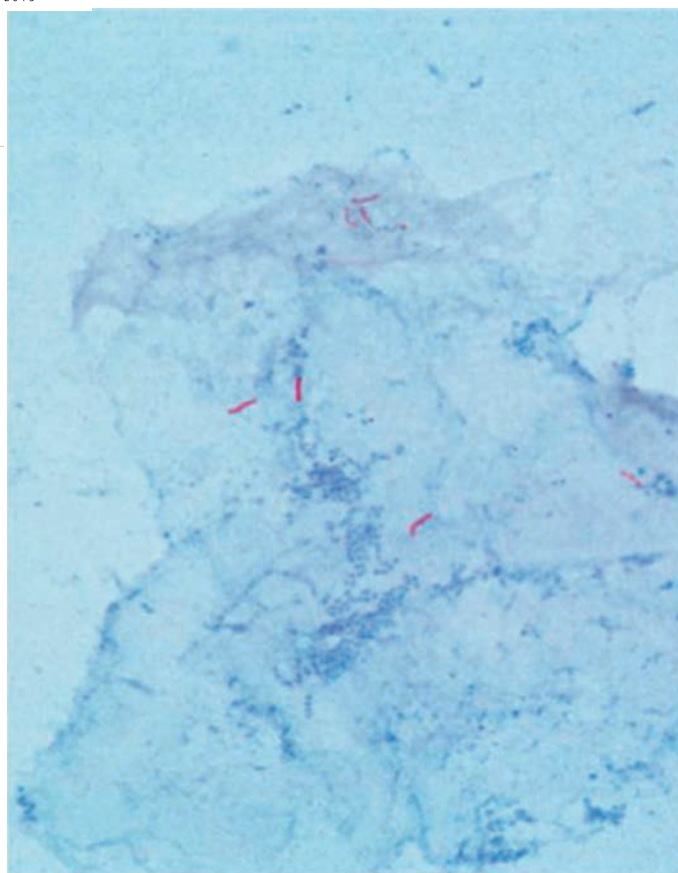


FIGURE 178-1 Acid-fast bacillus smear showing *M. tuberculosis* bacilli. (Courtesy of the Centers for Disease Control and Prevention, Atlanta.)

content (65.6%) is indicative of an aerobic “lifestyle.” A large proportion of genes are devoted to the production of enzymes involved in cell wall metabolism. Substantial genetic variability exists among the innumerable *M. tuberculosis* strains from different parts of the world. Based on such genetic variability it is possible to distinguish and compare different strains. Their distinction is important to study transmission dynamics and identify outbreaks. Starting in the 1990s, reproducible genotyping methods were developed to type the bacterium. Initially, they included insertion sequence 6110 (IS6110), restriction fragment length polymorphism (RFLP) typing, and spoligotyping. Lately, most studies utilize mycobacterial interspersed repetitive unit variable number tandem repeats (MIRU-VNTRs) and whole genome sequencing analysis.

EPIDEMIOLOGY

In 2019, 7.1 million new cases of TB (all forms, both pulmonary and extrapulmonary) were reported to the World Health Organization (WHO) by its member states; 97% of cases were reported from low- and middle-income countries. However, because of insufficient case detection and incomplete notification, reported cases represent only about two-thirds of the total estimated cases. The WHO estimated that 10 million (range 9–11 million; rate 130 per 100,000 people) new (incident) cases of TB occurred worldwide in 2019, 97% of them in low- and middle-income countries of Asia (6.1 million), Africa (2.4 million), the Middle East (0.8 million), and Latin America (0.28 million). Eight countries accounted for two-thirds of all new cases: India, Indonesia, China, the Philippines, Pakistan, Nigeria, Bangladesh, and South Africa. Of all cases, 57% occurred in male patients, 32% in female patients, and 11% in children. It is further estimated that 1.4 million (range, 1.3–1.6 million) deaths from TB, including 0.21 million among people with HIV co-infection, occurred in 2019; 98% of these deaths were in low- and middle-income countries. Estimates of TB incidence rates (per 100,000 population) and numbers of TB-related deaths in 2018 are depicted in Figs. 178-2 and 178-3, respectively. During the

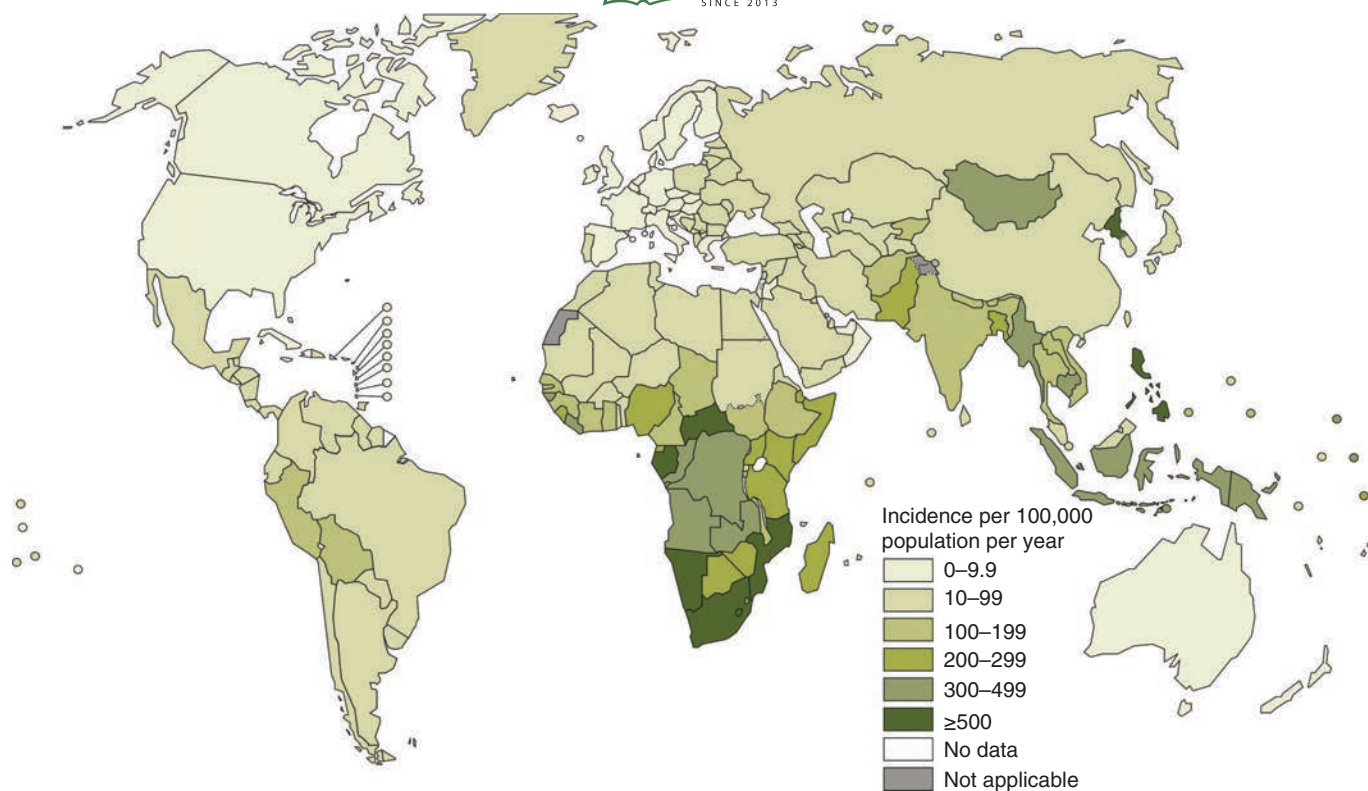


FIGURE 178-2 Estimated tuberculosis (TB) incidence rates (per 100,000 population) in 2018. The designations used and the presentation of material on this map do not imply the expression of any opinion whatsoever on the part of the World Health Organization (WHO) concerning the legal status of any country, territory, city, or area or of its authorities or concerning the delimitation of its frontiers or boundaries. Dotted, dashed, and white lines represent approximate border lines for which there may not yet be full agreement. (Reproduced with permission from *Global Tuberculosis Report 2019*. Geneva, World Health Organization; 2019.)

late 1980s and early 1990s, numbers of reported cases of TB increased in high-income countries after years of decline. These increases were related largely to immigration from countries with a high incidence of TB; the worldwide spread of the HIV epidemic; social problems, such as increase in urbanization and the related increased urban poverty, homelessness, and drug abuse; and dismantling of TB services.

During the past few years, numbers of reported cases have begun to decline again or have stabilized in most industrialized nations. In the United States, with the re-establishment of stronger control programs, the decline resumed in 1993, and during the period 2007–2012, the decline rate was 6.5% annually on average. Later, between 2012 and 2019 this annual rate slowed down to 2.1%. In 2019, 8920 cases of TB

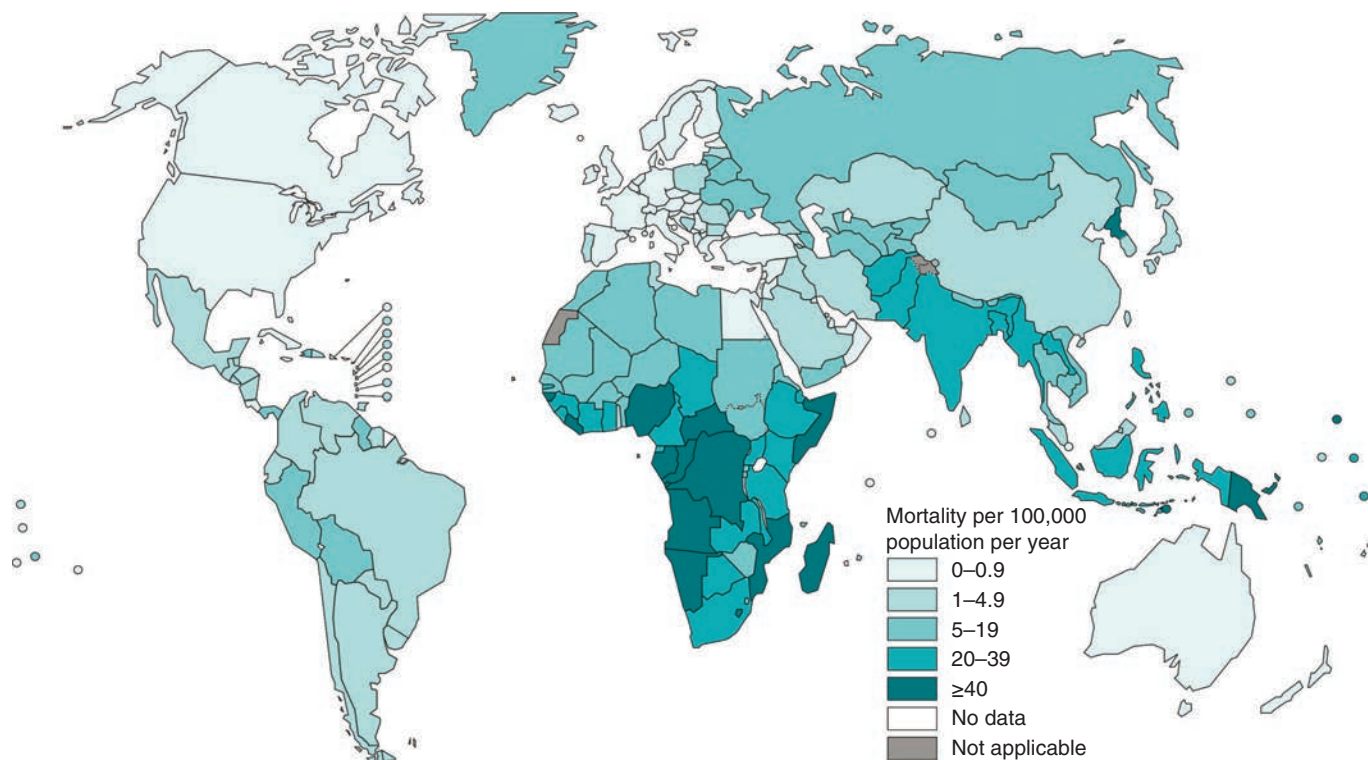


FIGURE 178-3 Estimated tuberculosis (TB) mortality rates in HIV-negative people in 2018. (See disclaimer in Fig. 178-2. Reproduced with permission from *Global Tuberculosis Report 2019*. Geneva, World Health Organization; 2019.)

(2.7 cases/100,000 population) were preliminarily reported to the U.S. Centers for Disease Control and Prevention (CDC).

In the United States, TB is uncommon among young white adults of European descent, who have only rarely been exposed to *M. tuberculosis* infection during recent decades. In contrast, because of a high risk of transmission in the past, the prevalence of *M. tuberculosis* infection is relatively high among elderly whites. In general, adults ≥ 65 years of age have the highest incidence rate per capita and children <14 years of age the lowest. Among U.S.-born persons, African Americans accounted for the highest proportion of cases (35%; 905 in 2019), followed by white persons (756 cases), and Hispanic/Latinos (628). TB in the United States is also a disease of adult members of the HIV-infected population (4.9% of all cases), the foreign-born population (71% of all cases in 2019), and disadvantaged/marginalized populations. Of the 6322 cases reported among non-U.S.-born persons in the United States in 2019, 33% occurred in Hispanic/Latinos and 47% in persons born in Asia. Overall, the highest rates per capita were among non-U.S.-born Asians (26 cases/100,000 population) and native Hawaiian/Pacific Islanders (25 cases/100,000 population). A total of 515 deaths was caused by TB in the United States in 2017. In Canada, TB cases and rates per 100,000 population have been increasing between 2014 and 2017 (from 1615/4.5 to 1796/4.9). In 2017, 1796 TB cases were reported (4.9 cases/100,000 population); 72% (1290) of these cases occurred in foreign-born persons, and 17.4% (313 cases) occurred in Canadian-born Indigenous Peoples, whose per capita rate is disproportionately high (21.5 cases/100,000 population). The highest rate was found in the territory of Nunavut, at 265 cases/100,000 population—a rate similar to that in many highly endemic countries. Similarly, in Europe, TB has reemerged as an important public health problem, mainly as a result of cases among immigrants from high-incidence countries and among marginalized populations, often in large urban settings such as London. In 2018, 36% of all cases reported from England occurred in London, 82% of them among people born abroad; although decreasing, the rate per capita (19 cases/100,000 population) is twice as high as that of England with a borough (Newham) reaching 47 cases per 100,000 population. Likewise, in most Western European countries, there are more cases annually among foreign-born than native populations.

Recent data on global trends indicate that in 2019 the TB incidence was stable or falling in most regions; this trend began in the early 2000s and appears to have continued, with an average annual decline of 1.7% globally and 2.3% between 2018 and 2019. This global decrease is explained largely by the reduction in TB incidence in sub-Saharan Africa, where rates had risen steeply since the 1980s as a result of the HIV epidemic and the lack of capacity of health systems and services to deal with the problem effectively, and, less so, in Eastern Europe, where incidence increased rapidly during the 1990s because of a deterioration in socioeconomic conditions and the health care infrastructure (although, after peaking in 2001, incidence in Eastern Europe has since declined slowly).

Of the estimated 10 million new cases of TB in 2019, 8.2% (0.82 million) were associated with HIV co-infection, and 73% of these HIV-associated cases occurred in Africa. An estimated 0.21 million persons with HIV-associated TB died in 2019. Furthermore, an estimated 465,000 (range 400,000–535,000) cases of rifampin- (also called rifampicin) resistant TB (RR-TB) and multidrug-resistant TB (MDR-TB)—a form of the disease caused by bacilli resistant at least to isoniazid and rifampin—occurred in 2019, representing 3.3% and 18%, respectively, of all new and previously treated cases. Only 44% of these cases were diagnosed because of a lack of culture and drug susceptibility testing (DST) capacity in many settings worldwide. As a consequence, an estimated 200,000 people with MDR/RR-TB died in 2019. The countries of the former Soviet Union reported the highest proportions of MDR/RR disease among new TB cases (37% in Belarus, 35% in Russia, 29% in Kyrgyzstan, Moldova, and Ukraine). Overall, half of all MDR/RR-TB cases occur in India (27%), China (14%), and the Russian Federation (9%). Since 2006, 131 countries, including the United States, have reported cases of extensively drug-resistant TB (XDR-TB), in which MDR-TB is compounded by additional resistance to any fluoroquinolones and at least one of the injectable drugs amikacin, kanamycin, and capreomycin. (N.B: In January 2021, the

WHO published the following new definitions: (i) Pre-XDR-TB, as TB caused by *Mycobacterium tuberculosis* strains that fulfill the definition of MDR/RR-TB and that are also resistant to any fluoroquinolone. (ii) XDR-TB, as TB caused by *Mycobacterium tuberculosis* strains that fulfill the definition of MDR/RR-TB and that are also resistant to any fluoroquinolone and at least one additional Group A drug including levofloxacin or moxifloxacin, bedaquiline and linezolid.) About 6.2% of the MDR-TB cases worldwide may be XDR-TB, but the vast majority of XDR-TB cases remain undiagnosed because reliable methods for DST are lacking and laboratory capacity is limited mainly in low-income countries. Lately, a few cases deemed resistant to all anti-TB drugs have been reported; however, this information must be interpreted with caution because susceptibility testing for several second-line drugs is neither accurate nor reproducible.

FROM EXPOSURE TO INFECTION

M. tuberculosis is most commonly transmitted from a person with infectious pulmonary TB by droplet nuclei containing *M. tuberculosis* bacteria, which are aerosolized by coughing, sneezing, or speaking. The tiny droplets dry rapidly; the smallest ($<5\text{--}10\text{ }\mu\text{m}$ in diameter) may remain suspended in the air for several hours and may reach the terminal air passages when inhaled. There may be as many as 3000 infectious nuclei per cough. Other routes of transmission of tubercle bacilli (e.g., through the skin or the placenta) are uncommon and of no epidemiologic significance. The risk of transmission and of subsequent acquisition of *M. tuberculosis* infection is determined mainly by exogenous factors, although endogenous factors may also play a role. The probability of contact with a person who has an infectious form of TB, the intimacy and duration of that contact, the degree of infectiousness of the case, and the shared environment in which the contact takes place are all important determinants of the likelihood of transmission. Several studies of close-contact situations have clearly demonstrated that TB patients whose sputum contains AFB visible by microscopy (sputum smear-positive cases) are the most likely to transmit the infection. The most infectious patients have cavitary pulmonary disease or, much less commonly, laryngeal TB and produce sputum containing as many as $10^5\text{--}10^7$ AFB/mL. Patients with sputum smear-negative/culture-positive TB are less infectious, although they have been responsible for up to 20% of transmission in some studies in the United States. Those with culture-negative pulmonary TB and extrapulmonary TB are essentially noninfectious. Because persons with both HIV infection and TB are less likely to have cavitations, they may be less infectious than persons without HIV co-infection. Crowding in poorly ventilated rooms is one of the most important factors in the transmission of tubercle bacilli because it increases the intensity of contact with a case. The virulence of the transmitted organism is also an important factor in establishing infection. Endogenous factors such as the degree of immune competence are also important. In particular, HIV-infected patients, persons undergoing cancer treatment, or those administered immunosuppressive drugs may be at higher risk of TB infection acquisition.

Because of delays in seeking care and in making a diagnosis, it has been estimated that, in high-prevalence settings, up to 20 contacts (or 3–10 people per year) may be infected by each AFB-positive case before the index case is diagnosed. Attempts to estimate the basic reproductive number R_0 for TB have resulted in a wide range of values depending on environmental conditions and social behaviors of populations: from 0.24 in the Netherlands during the period 1933–2007 to 4.3 in China in 2012 reflecting the status of disease control.

FROM INFECTION TO DISEASE

Unlike the risk of acquiring infection with *M. tuberculosis*, the risk of developing disease after being infected depends largely on endogenous factors, such as the individual's innate immunologic and nonimmunologic defenses and the level at which the individual's cell-mediated immunity is functioning. Clinical illness directly following infection is classified as primary TB and is common among children in the first few years of life and among immunocompromised persons. Although primary TB may be severe and disseminated, it generally is not associated

TABLE 178-1 Risk Factors for Active Tuberculosis in Persons Who Have Been Infected with Tubercle Bacilli

FACTOR	RELATIVE RISK/ODDS ^a
Recent infection (<1 year)	12.9
Fibrotic lesions (spontaneously healed)	2–20
Comorbidities and iatrogenic causes	
HIV infection	21–>30
Silicosis	30
Chronic renal failure/hemodialysis	10–25
Diabetes	2–4
IV drug use	10–30
Excessive alcohol use	3
Immunosuppressive treatment	10
Tumor necrosis factor α inhibitors	4–5
Gastrectomy	2–5
Jejunioileal bypass	30–60
Posttransplantation period (renal, cardiac)	20–70
Tobacco smoking	2–3
Malnutrition and severe underweight	2

^aOld infection = 1.

with high-level transmissibility. When infection is acquired later in life, the chance is greater that the mature immune system will contain it at least temporarily. Bacilli, however, may persist for years before reactivating to produce *secondary* (or *postprimary*) TB, which, because of frequent cavitation, is more often infectious than is primary disease. Overall, it is estimated that up to 10% of infected persons will eventually develop active TB in their lifetime—half of them during the first 18 months after infection. The risk is much higher among immunocompromised individuals and, particularly, HIV-infected persons. Reinfection of a previously infected individual, which is common in areas with high rates of TB transmission, may also favor the development of disease. At the height of the TB resurgence in the United States in the early 1990s, molecular typing and comparison of strains of *M. tuberculosis* suggested that up to one-third of cases of active TB in some inner-city communities were due to recent transmission rather than to reactivation of old infection. Age is an important determinant of the risk of disease after infection. Among infected persons, the incidence of TB is highest during late adolescence and early adulthood; the reasons are unclear. The incidence among women peaks at 25–34 years of age. In this age group, rates among women may be higher than those among men, whereas at older ages the opposite is true. The risk increases in the elderly, possibly because of waning immunity and comorbidity.

A variety of diseases and conditions favor the development of active TB (Table 178-1). In absolute terms, the most potent risk factor for TB among infected individuals is clearly HIV co-infection, which suppresses cellular immunity. The risk that infection will proceed to active disease is directly related to the patient's degree of immunosuppression. In a study of HIV-co-infected, tuberculin skin test (TST)-positive persons, this risk varied from 2.6 to 13.3 cases/100 person-years and increased as the CD4+ T-cell count decreased.

■ NATURAL HISTORY OF DISEASE

Studies conducted in various countries before the advent of antimicrobial TB therapy showed that untreated TB is often fatal. About one-third of patients died within 1 year after diagnosis. Historical data also show that 55% of sputum smear-positive cases were dead within 5 years and up to 86% (weighted mean 70%) within 10 years. A lower case fatality rate, around 20%, was estimated for untreated paucibacillary smear-negative cases at 5 years. Of the survivors at 5 years, ~60% had undergone spontaneous remission, while the remainder were still excreting tubercle bacilli. With effective, timely, and proper antimicrobial TB treatment, patients have a very high chance of being cured. However, improper use of anti-TB drugs, while reducing mortality rates, may also result in large numbers of chronic infectious cases, often with drug-resistant bacilli.

PATHOGENESIS AND IMMUNITY

■ INFECTION AND MACROPHAGE INVASION

The interaction of *M. tuberculosis* with the human host begins when droplet nuclei containing viable microorganisms, propelled into the air by infectious patients, are inhaled by a close bystander. Although the majority of inhaled bacilli are trapped in the upper airways and expelled by ciliated mucosal cells, a fraction (usually <10%) reach the alveoli, a unique immunoregulatory environment. There, in the very early phases of infection, the predominant cells infected by *M. tuberculosis* are myeloid dendritic cells. Subsequently, alveolar macrophages that have not yet been activated (prototypic alternatively activated macrophages) phagocytose the bacilli. Adhesion of mycobacteria to macrophages results largely from binding of the bacterial cell wall to a variety of macrophage cell-surface receptor molecules, including complement receptors, the mannose receptor, the immunoglobulin G Fc γ receptor, and type A scavenger receptors. Surfactants may also play a role in the early phase of interaction between the host and the pathogen, and surfactant protein D can prevent phagocytosis. Phagocytosis is enhanced by complement activation leading to opsonization of bacilli with C3 activation products such as C3b and C3bi. Concomitantly, binding of certain receptors, such as the mannose receptor, regulates postphagocytic events like phagosome-lysosome fusion and inflammatory cytokine production. After a phagosome forms, the survival of *M. tuberculosis* in the cell seems to depend in part on reduced acidification due to lack of assembly of a complete vesicular proton-adenosine triphosphatase. A complex series of events is generated by the bacterial cell-wall lipoglycan lipoarabinomannan, which inhibits the intracellular increase of Ca²⁺. Thus, the Ca²⁺/calmodulin pathway (leading to phagosome-lysosome fusion) is impaired, and the bacilli survive within the phagosomes by blocking fusion. The *M. tuberculosis* phagosome inhibits the production of phosphatidylinositol 3-phosphate, which normally earmarks phagosomes for membrane sorting and maturation (including phagolysosome formation), which would destroy the bacteria. Bacterial factors block the host defense of autophagy, in which the cell sequesters the phagosome in a double-membrane vesicle (*autophagosome*) that is destined to fuse with lysosomes. If the bacilli are successful in arresting phagosome maturation, then replication begins and the macrophage eventually ruptures and releases its bacillary contents. This process is mediated by the ESX-1 secretion system that is encoded by genes contained in the region of difference 1 (RD1). Other uninfected phagocytic cells are then recruited to continue the infection cycle by ingesting dying macrophages and their bacillary content, thus, in turn, becoming infected themselves and expanding the infection.

■ VIRULENCE OF TUBERCLE BACILLI



M. tuberculosis must be viewed as a complex formed by a multitude of strains that differ in virulence and are capable of producing a variety of manifestations of disease. Since the elucidation of the *M. tuberculosis* genome in 1998, large mutant collections have been generated, and many bacterial genes that contribute to *M. tuberculosis* virulence have been found. Moreover, different patterns of virulence defects have been defined in various animal models—predominantly mice but also guinea pigs, rabbits, and nonhuman primates. The *katG* gene encodes for a catalase/peroxidase enzyme that protects against oxidative stress and is required for isoniazid activation and subsequent bactericidal activity. RD1 is a 9.5-kb locus that encodes two key small protein antigens—6-kDa early secretory antigen (ESAT-6) and culture filtrate protein-10 (CFP-10)—as well as a putative secretion apparatus that may facilitate their egress; the absence of this locus in the vaccine strain *M. bovis* bacille Calmette-Guérin (BCG) is a key attenuating mutation. In *M. marinum*, a mutation in the RD1 virulence locus encoding the ESX-1 secretion system impairs the capacity of apoptotic macrophages to recruit uninfected cells for further rounds of infection. The results are less replication and fewer new granulomas. These observations in *M. marinum* are similar in part to events related to the virulence of *M. tuberculosis*; however, ESX-1, although necessary, is probably insufficient to explain virulence, and other mechanisms may be in play. Mutants lacking key enzymes of bacterial biosynthesis become auxotrophic for the

missing substrate and often are totally unable to proliferate in animals; these include the *leuCD* and *panCD* mutants, which require leucine and pantothenic acid, respectively. The *isocitrate lyase gene (icl1)* encodes a key step in the glyoxylate shunt that facilitates bacterial growth on fatty acid substrates; this gene is required for long-term persistence of *M. tuberculosis* infection in mice with chronic TB. *M. tuberculosis* mutants in regulatory genes such as sigma factor C and sigma factor H (*sigC* and *sigH*) are associated with normal bacterial growth in mice, but they fail to elicit full tissue pathology. Finally, the mycobacterial protein CarD (expressed by the *carD* gene) seems essential for the control of rRNA transcription that is required for mycobacterial replication and persistence in the host cell. Its loss exposes mycobacteria to oxidative stress, starvation, DNA damage, and ultimately sensitivity to killing by a variety of host mutagens and defense mechanisms.

■ INNATE RESISTANCE TO INFECTION

 Several observations suggest that genetic factors play a key role in innate resistance to infection with *M. tuberculosis* and the development of disease. The existence of this resistance, which is polygenic in nature, is suggested by the differing degrees of susceptibility to TB in different populations. This mechanism of elimination of the pathogen may be accompanied by negative results in the TST and interferon- γ (IFN- γ) release assays (IGRAs). In mice, a gene called *Nramp1* (natural resistance-associated macrophage protein 1) plays a regulatory role in resistance/susceptibility to mycobacteria. The human homologue NRAMP1, which maps to chromosome 2q, may play a role in determining susceptibility to TB, as is suggested by a study among West Africans. Studies of mice identified a novel host resistance gene, *ipr1*, that is encoded within the *sst1* locus; *ipr1* encodes an IFN-inducible nuclear protein that interacts with other nuclear proteins in macrophages primed with IFNs or infected by *M. tuberculosis*. In addition, polymorphisms in multiple genes, such as those encoding for various major histocompatibility complex alleles, IFN- γ , T-cell growth factor β , interleukin (IL) 10, mannose-binding protein, IFN- γ receptor, Toll-like receptor 2, vitamin D receptor, and IL-1, have been associated with susceptibility to TB.

■ THE HOST RESPONSE, GRANULOMA FORMATION, AND “LATENCY”

In the initial stage of host–bacterium interaction, prior to the onset of an acquired cell-mediated immune (CMI) response, *M. tuberculosis* disseminates widely through the lymph vessels, spreading to other sites in the lungs and other organs, and undergoes a period of extensive growth within naïve inactivated macrophages; additional naïve macrophages are recruited to the early granuloma. How the bacillus accesses the parenchymal tissue still needs to be elucidated: it may directly infect epithelial cells or transigrate through infected macrophages across the epithelium. Infected dendritic cells or monocytes then begin to transport bacilli to the lymphatic system. Studies suggest that *M. tuberculosis* uses specific virulence mechanisms to subvert host cellular signaling and to elicit an early regulated proinflammatory response that promotes granuloma expansion and bacterial growth during this key early phase. A study of *M. marinum* infection in zebrafish has delineated one molecular mechanism by which mycobacteria induce granuloma formation. The mycobacterial protein ESAT-6 induces secretion of matrix metalloproteinase 9 (MMP9) by nearby epithelial cells that are in contact with infected macrophages. MMP9 in turn stimulates recruitment of naïve macrophages, thus inducing granuloma maturation and bacterial growth. Disruption of MMP9 function results in reduced bacterial growth. Another study has shown that *M. tuberculosis*-derived cyclic AMP is secreted from the phagosome into host macrophages, subverting the cell's signal transduction pathways and stimulating an elevation in the secretion of tumor necrosis factor α (TNF- α) as well as further proinflammatory cell recruitment. Ultimately, the chemoattractants and bacterial products released during the repeated rounds of cell lysis and infection of newly arriving macrophages enable dendritic cells to access bacilli; these cells migrate to the draining lymph nodes and present mycobacterial antigens to T lymphocytes. At this point, the development of cell-mediated and humoral immunity begins. These initial stages of infection are usually asymptomatic.

About 2–4 weeks after infection, two host responses to *M. tuberculosis* develop: a macrophage-activating CMI response and a tissue-damaging response. The macrophage-activating response is a T-cell mediated phenomenon resulting in the activation of macrophages that are capable of killing and digesting tubercle bacilli. The tissue-damaging response is the result of a delayed-type hypersensitivity reaction to various bacillary antigens; it destroys inactivated macrophages that contain multiplying bacilli but also causes caseous necrosis of the involved tissues (see below). Although both of these responses can inhibit mycobacterial growth, it is the balance between the two that determines the forms of TB that will develop subsequently. With the development of specific immunity and the accumulation of large numbers of activated macrophages at the site of the primary lesion, granulomatous lesions (tubercles) are formed. These lesions consist of accumulations of lymphocytes and activated macrophages that evolve toward epithelioid and giant cell morphologies. Initially, the tissue-damaging response can limit mycobacterial growth within macrophages. As stated above, this response, mediated by various bacterial products, not only destroys macrophages but also produces early solid necrosis in the center of the tubercle. Although *M. tuberculosis* can survive, its growth is inhibited within this necrotic environment by low oxygen tension and low pH. At this point, some lesions may heal by fibrosis, with subsequent calcification, whereas inflammation and necrosis occur in other lesions. Some observations have challenged the traditional view that any encounter between mycobacteria and macrophages results in chronic infection. It is possible that an immune response capable of eradicating early infection may sometimes develop as a consequence, for instance, of disabling mutations in mycobacterial genomes rendering their replication ineffective. Individual granulomas that are formed during this phase of infection can vary in size and cell composition; some can contain the spread of mycobacteria, while others cannot. TB infection ensues as a result of this dynamic balance between the microorganism and the host. For many years, TB infection has been called “latent TB infection (LTBI).” This terminology was used to define a state of persistent immune response to stimulation by *M. tuberculosis* antigens with no evidence of clinically manifest, active TB. The qualification “latent” may offer some convenience of distinguishing infection from disease, albeit an inaccurate description of a process that encompasses bacterial generations that are not dormant. It has been speculated that latency may therefore be an inaccurate term because bacilli may remain active during this “latent” stage, forming biofilms in necrotic areas within which they temporarily hide. Thus some have proposed the term *persist* as more accurate to indicate the behavior of the bacilli in this phase. It is important to recognize that infection and disease represent not a binary state but rather a continuum along which infection will eventually move in the direction of full containment or disease. The ability to predict, through systemic biomarkers, which affected individuals will progress toward disease would be of immense value in devising prophylactic interventions at scale.

■ MACROPHAGE-ACTIVATING RESPONSE

Cell-mediated immunity is critical at this early stage. In the majority of infected individuals, local macrophages are activated when bacillary antigens processed by macrophages stimulate T lymphocytes to release a variety of lymphokines. These activated macrophages aggregate around the lesion's center and effectively neutralize tubercle bacilli without causing further tissue destruction. In the central part of the lesion, the necrotic material resembles soft cheese (caseous necrosis)—a phenomenon that may also be observed in other conditions, such as neoplasms. Even when healing takes place, viable bacilli may remain dormant within macrophages or in the necrotic material for many years. These “healed” lesions in the lung parenchyma and hilar lymph nodes may later undergo calcification.

■ DELAYED-TYPE HYPERSENSITIVITY

In a minority of cases, the macrophage-activating response is weak, and mycobacterial growth can be inhibited only by intensified delayed hypersensitivity reactions, which lead to lung tissue destruction. The lesion tends to enlarge further, and the surrounding tissue is

progressively damaged. At the center of the lesion, the caseous material liquefies. Bronchial walls and blood vessels are invaded and destroyed, and cavities are formed. The liquefied caseous material, containing large amount of bacilli, is drained through bronchi. Within the cavity, tubercle bacilli multiply, spill into the airways, and are discharged into the environment through expiratory maneuvers such as coughing and talking. In the early stages of infection, bacilli are usually transported by macrophages to regional lymph nodes, from which they gain access to the central venous return; from there they reseed the lungs and may also disseminate beyond the pulmonary vasculature throughout the body via the systemic circulation. The resulting extrapulmonary lesions may undergo the same evolution as those in the lungs, although most tend to heal. In young children with poor natural immunity, hematogenous dissemination may result in fatal miliary TB or tuberculous meningitis.

■ ROLE OF MACROPHAGES AND MONOCYTES

While cell-mediated immunity confers partial protection against *M. tuberculosis*, humoral immunity plays a less well-defined role in protection (although evidence is accumulating on the existence of antibodies to lipoarabinomannan, which may prevent dissemination of infection in children). In cell-mediated immunity, two types of cells are essential: macrophages, which directly phagocytose tubercle bacilli, and T cells (mainly CD4⁺ T lymphocytes, although the role of CD8⁺ T cells has recently been the subject of much research), which induce protection through the production of cytokines, especially IFN- γ . After infection with *M. tuberculosis*, alveolar macrophages secrete various cytokines responsible for a number of events (e.g., the formation of granulomas) as well as systemic effects (e.g., fever and weight loss). However, alternatively activated alveolar macrophages may be particularly susceptible to *M. tuberculosis* growth early on, given their more limited proinflammatory and bactericidal activity, which is related in part to being bathed in surfactant. New monocytes and macrophages attracted to the site are key components of the immune response. Their primary mechanism is probably related to production of oxidants (such as reactive oxygen intermediates or nitric oxide) that have antimycobacterial activity and increase the synthesis of cytokines such as TNF- α and IL-1, which in turn regulate the release of reactive oxygen intermediates and reactive nitrogen intermediates. In addition, macrophages can undergo apoptosis—a defensive mechanism to prevent the release of cytokines and bacilli via their sequestration in the apoptotic cell. Recent work also describes the involvement of neutrophils in the host response, although the timing of their appearance and their effectiveness remain uncertain.

■ ROLE OF T LYMPHOCYTES

Alveolar macrophages, monocytes, and dendritic cells are also critical in processing and presenting antigens to T lymphocytes, primarily CD4⁺ and CD8⁺ T cells; the result is the activation and proliferation of CD4⁺ T lymphocytes, which are crucial to the host's defense against *M. tuberculosis*. Qualitative and quantitative defects of CD4⁺ T cells explain the inability of HIV-infected individuals to contain mycobacterial proliferation. Activated CD4⁺ T lymphocytes can differentiate into cytokine-producing T_H1 or T_H2 cells. T_H1 cells produce IFN- γ —an activator of macrophages and monocytes—and IL-2. T_H2 cells produce IL-4, IL-5, IL-10, and IL-13 and may also promote humoral immunity. The interplay of these various cytokines and their cross-regulation determine the host's response. The role of cytokines in promoting intracellular killing of mycobacteria, however, has not been entirely elucidated. IFN- γ may induce the generation of reactive nitrogen intermediates and regulate genes involved in bactericidal effects. TNF- α is also important. Although its precise mechanisms are complex and not yet fully clarified, a model has been suggested that foresees an ideal setting for TNF- α between excessive activation—with consequent worsening of immunopathological reactions—and insufficient activation—with resulting lack of containment—in the control of TB infection. Observations made originally in transgenic knockout mice and more recently in humans suggest that other T-cell subsets, especially CD8⁺ T cells, may play an important role. CD8⁺ T cells have been associated with protective activities via cytotoxic responses and lysis of infected

cells as well as with production of IFN- γ and TNF- α . Finally, natural killer cells act as co-regulators of CD8⁺ T-cell lytic activities, and $\gamma\delta$ T cells are increasingly thought to be involved in protective responses in humans.

■ MYCOBACTERIAL LIPIDS AND PROTEINS

Lipids are involved in mycobacterial recognition by the innate immune system, and lipoproteins (such as 19-kDa lipoprotein) trigger potent signals through Toll-like receptors present in blood dendritic cells. *M. tuberculosis* possesses various protein antigens. Some are present in the cytoplasm and cell wall; others are secreted. That the latter are more important in eliciting a T lymphocyte response is suggested by experiments documenting the appearance of protective immunity in animals after immunization with live, protein-secreting mycobacteria. Among the antigens that may play a protective role are the 30-kDa (or 85B) and ESAT-6 antigens. Protective immunity is probably the result of reactivity to many different mycobacterial antigens. These antigens are being incorporated into newly designed vaccines on various platforms.

■ SKIN-TEST REACTIVITY

Coincident with the appearance of immunity, delayed-type hypersensitivity to *M. tuberculosis* develops. This reactivity is the basis of the TST, which is used primarily for the diagnosis of *M. tuberculosis* infection in persons without symptoms. The cellular mechanisms responsible for TST reactivity are related mainly to previously sensitized CD4⁺ T lymphocytes, which are attracted to the skin-test site. There, they proliferate and produce cytokines. Although delayed hypersensitivity is associated with protective immunity (TST-positive persons are less susceptible to a new *M. tuberculosis* infection than TST-negative persons), it by no means guarantees protection against reactivation. In fact, cases of active TB are often accompanied by strongly positive skin-test reactions. There is also evidence of reinfection with a new strain of *M. tuberculosis* in patients previously treated for active disease. This evidence underscores the fact that previous infection or active TB may not confer fully protective immunity.

CLINICAL MANIFESTATIONS

TB is classified as pulmonary, extrapulmonary, or both. Depending on several factors linked to host immunological status and bacterial strains, extrapulmonary TB may occur in 10–40% of patients. Furthermore, up to two-thirds of HIV-infected patients with TB may have both pulmonary and extrapulmonary TB or extrapulmonary TB alone.

■ PULMONARY TB

Pulmonary TB is conventionally categorized as primary or postprimary (adult-type, secondary). This distinction has been challenged by molecular evidence from TB-endemic areas indicating that a large percentage of cases of adult pulmonary TB result from recent infection (either primary infection or reinfection) and not from reactivation.

Primary Disease Primary pulmonary TB occurs soon after the initial infection. It may be asymptomatic or may present with fever and occasionally pleuritic chest pain. In areas of high TB transmission, this form of disease is often seen in children. Because most inspired air is distributed to the middle and lower lung zones, these areas are most commonly involved in primary TB. The lesion forming after initial infection (*Ghon focus*) is usually peripheral and accompanied by transient hilar or paratracheal lymphadenopathy, which may or may not be visible on standard chest radiography (CXR) (Fig. 178-4). Some patients develop erythema nodosum on the legs (see Fig. A1-39) or phlyctenular conjunctivitis. In the majority of cases, the lesion heals spontaneously and becomes evident only as a small calcified nodule. Pleural reaction overlying a subpleural focus is also common. The Ghon focus, with or without overlying pleural reaction, thickening, and regional lymphadenopathy, is referred to as the *Ghon complex*.

In young children with immature cell-mediated immunity and in persons with impaired immunity (e.g., those with malnutrition or HIV infection), primary pulmonary TB may progress rapidly to clinical illness. The initial lesion increases in size and can evolve in different ways. Pleural effusion, which is found in up to two-thirds of cases,



FIGURE 178-4 Chest radiograph showing right hilar lymph node enlargement with infiltration into the surrounding lung tissue in a child with primary tuberculosis. (Courtesy of Prof. Robert Gie, Department of Paediatrics and Child Health, Stellenbosch University, South Africa; with permission.)

results from the penetration of bacilli into the pleural space from an adjacent subpleural focus. In severe cases, the primary site rapidly enlarges, its central portion undergoes necrosis, and cavitation develops (*progressive primary TB*). TB in young children is almost invariably accompanied by hilar or paratracheal lymphadenopathy due to the spread of bacilli from the lung parenchyma through lymphatic vessels. Enlarged lymph nodes may compress bronchi, causing total obstruction with distal collapse, partial obstruction with large-airway wheezing, or a ball-valve effect with segmental/lobar hyperinflation. Lymph nodes may also rupture into the airway with development of pneumonia, often including areas of necrosis and cavitation, distal to the obstruction. Bronchiectasis (**Chap. 290**) may develop in any segment/lobe damaged by progressive caseating pneumonia. Occult hematogenous dissemination commonly follows primary infection. However, in the absence of a sufficient acquired immune response, which usually contains the infection, disseminated or miliary disease may result (**Fig. 178-5**). Small granulomatous lesions develop in multiple organs



FIGURE 178-5 Chest radiograph showing bilateral miliary (millet-sized) infiltrates in a child. (Courtesy of Prof. Robert Gie, Department of Paediatrics and Child Health, Stellenbosch University, South Africa; with permission.)



FIGURE 178-6 Chest radiograph showing a right-upper-lobe infiltrate and a cavity with an air-fluid level in a patient with active tuberculosis. (Courtesy of Dr. Andrea Gori, Infectious Diseases Unit, Fondazione IRCCS Ca' Granda Ospedale Maggiore Policlinico, University of Milan, Milan, Italy; with permission.)

and may cause locally progressive disease or result in tuberculous meningitis; this is a particular concern in very young children and immunocompromised persons (e.g., patients with HIV infection).

Postprimary (Adult-Type) Disease Also referred to as *reactivation* or *secondary TB*, postprimary TB is probably most accurately termed *adult-type TB* because it may result from endogenous reactivation of distant or recent infection (primary infection or reinfection). It is usually localized to the apical and posterior segments of the upper lobes, where the substantially higher mean oxygen tension (compared with that in the lower zones) favors mycobacterial growth. The superior segments of the lower lobes are also frequently involved. The extent of lung parenchymal involvement varies greatly, from small infiltrates to extensive cavitory disease. With cavity formation, liquefied necrotic contents are ultimately discharged into the airways and may undergo bronchogenic spread, resulting in satellite lesions within the lungs that may in turn undergo cavitation (**Figs. 178-6 and 178-7**). Massive involvement of pulmonary segments or lobes,



FIGURE 178-7 CT scan showing a large cavity in the right lung of a patient with active tuberculosis. (Courtesy of Dr. Elisa Busi Rizzi, National Institute for Infectious Diseases, Spallanzani Hospital, Rome, Italy; with permission.)

1364 with coalescence of lesions, produces caseating pneumonia. While up to one-third of untreated patients reportedly succumb to severe pulmonary TB within a few months after onset (the classic “galloping consumption” of the past), others may undergo a process of spontaneous remission or proceed along a chronic, progressively debilitating course (“consumption” or *phthisis*). Under these circumstances, some pulmonary lesions become fibrotic and may later calcify, but cavities persist in other parts of the lungs. Individuals with such chronic disease continue to discharge tubercle bacilli into the environment. Most patients respond to treatment, with defervescence, decreasing cough, weight gain, and a general improvement in well-being within several weeks.

Early in the course of disease, symptoms and signs are often non-specific and insidious, consisting mainly of fever, often diurnal and night sweats due to defervescence, weight loss, anorexia, general malaise, and weakness. However, in up to 90% of cases, cough eventually develops—often initially nonproductive and limited to the morning and subsequently accompanied by the production of purulent sputum, sometimes with blood streaking. Hemoptysis develops in 20–30% of cases, and massive hemoptysis may ensue as a consequence of the erosion of a blood vessel in the wall of a cavity. Hemoptysis, however, may also result from rupture of a dilated vessel in a cavity (*Rasmussen’s aneurysm*) or from aspergilloma formation in an old cavity. Pleuritic chest pain sometimes develops in patients with subpleural parenchymal lesions or pleural disease. Extensive disease may produce dyspnea and, in rare instances, adult respiratory distress syndrome. Physical findings are of limited use in pulmonary TB. Many patients have no abnormalities detectable by chest examination, whereas others have detectable rales in the involved areas during inspiration, especially after coughing. Occasionally, rhonchi due to partial bronchial obstruction and classic amphoric breath sounds in areas with large cavities may be heard. Systemic features include fever (often low-grade and intermittent) in up to 80% of cases and wasting. Absence of fever, however, does not exclude TB. In some recurrent cases and among people with low Karnofsky score, finger clubbing has been reported. The most common hematologic findings are mild anemia, leukocytosis, and thrombocytosis with a slightly elevated erythrocyte sedimentation rate and/or C-reactive protein level. None of these findings is consistent or sufficiently accurate for diagnostic purposes. Hyponatremia due to the syndrome of inappropriate secretion of antidiuretic hormone has also been reported.

■ EXTRAPULMONARY TB

In descending order of frequency, the extrapulmonary sites most commonly involved in TB are the lymph nodes, pleura, genitourinary tract, bones and joints, meninges, peritoneum, and pericardium. However, virtually any organ system may be affected. As a result of hematogenous dissemination in HIV-infected individuals, extrapulmonary TB is seen more commonly today than in the past in settings of high HIV prevalence.

Lymph Node TB (Tuberculous Lymphadenitis) The most common presentation of extrapulmonary TB in both HIV-seronegative individuals and HIV-infected patients (35% of cases worldwide and >40% of cases in the United States in recent series), lymph node disease is particularly frequent among HIV-infected patients and among children (Fig. 178-8). In the United States, besides children, women (particularly non-Caucasians) seem to be especially susceptible. Once caused mainly by *M. bovis*, tuberculous lymphadenitis today is due largely to *M. tuberculosis*. Lymph node TB presents as painless swelling of the lymph nodes, most commonly at posterior cervical and supraclavicular sites (a condition historically referred to as *scrofula*). Lymph nodes are usually discrete in early disease but develop into a matted nontender mass over time; a fistulous tract draining caseous material may result. Associated pulmonary disease is present in fewer than 50% of cases, and systemic symptoms are uncommon except in HIV-infected patients. The diagnosis is established by fine-needle aspiration biopsy (with a yield of up to 80%) or surgical excision biopsy. Bacteriologic confirmation is achieved in the vast majority of cases,



FIGURE 178-8 Tuberculous lymphadenitis affecting the cervical lymph nodes in a 2-year-old child from Malawi. (Courtesy of Prof. S. Graham, Centre for International Child Health, University of Melbourne, Australia; with permission.)

granulomatous lesions with or without visible AFBs are typically seen, and cultures are positive in 70–80% of cases. Among HIV-infected patients, granulomas are less well organized and are frequently absent entirely, but bacterial loads are heavier than in HIV-seronegative patients, with higher yields from microscopy and culture. Differential diagnosis includes a variety of infectious conditions, neoplastic diseases such as lymphomas or metastatic carcinomas, and rare disorders like Kikuchi’s disease (necrotizing histiocytic lymphadenitis), Kimura’s disease, and Castleman’s disease.

Pleural TB Involvement of the pleura accounts for ~20% of extrapulmonary cases in the United States and elsewhere. Isolated pleural effusion usually reflects recent primary infection, and the collection of fluid in the pleural space represents a hypersensitivity response to mycobacterial antigens. Pleural disease may also result from contiguous parenchymal spread, as in many cases of pleurisy accompanying postprimary disease. Depending on the extent of reactivity, the effusion may be small, remain unnoticed, and resolve spontaneously or may be sufficiently large to cause symptoms such as fever, pleuritic chest pain, and dyspnea. Physical findings are those of pleural effusion: dullness to percussion and absence of breath sounds. CXR reveals the effusion and, in up to one-third of cases, also shows a parenchymal lesion. Thoracentesis is required to ascertain the nature of the effusion and to differentiate it from manifestations of other etiologies. The fluid is straw-colored and at times hemorrhagic; it is an exudate with a protein concentration >50% of that in serum (usually ~4–6 g/dL), a normal to low glucose concentration, a pH of ~7.3 (occasionally <7.2), and detectable white blood cells (usually 500–6000/μL). Neutrophils may predominate in the early stage, but lymphocyte predominance is the typical finding later. Mesothelial cells are generally rare or absent. AFBs are rarely seen on direct smear, and cultures often may be falsely negative for *M. tuberculosis*; positive cultures are more common among postprimary cases. Determination of the pleural concentration of adenosine deaminase may be a useful screening test, and TB may be excluded if the value is very low. Lysozyme is also present in the pleural effusion. Measurement of IFN-γ, either directly or through stimulation of sensitized T cells with mycobacterial antigens, can be diagnostically helpful. Needle biopsy of the pleura is often required for diagnosis and is recommended over pleural fluid analysis; it reveals granulomas and/or yields a positive culture in up to 80% of cases. Pleural biopsy can yield a positive result in ~75% of cases when real-time automated nucleic acid amplification is used (the Xpert MTB/RIF assay [Cepheid; Sunnyvale, CA]; see “Nucleic Acid Amplification Technology,” below); testing of pleural fluid with this assay is not recommended because of low sensitivity. This form of pleural TB responds rapidly to chemotherapy and may resolve spontaneously. Concurrent glucocorticoid

administration may reduce the duration of fever and/or chest pain but is not of proven benefit.

Tuberculous empyema is a less common complication of pulmonary TB. It is usually the result of the rupture of a cavity, with spillage of a large number of organisms into the pleural space. This process may create a bronchopleural fistula with evident air in the pleural space. CXR shows hydropneumothorax with an air-fluid level. The pleural fluid is purulent and thick and contains large numbers of lymphocytes. Acid-fast smears and mycobacterial cultures are often positive. Surgical drainage is usually required as an adjunct to chemotherapy. Tuberculous empyema may result in severe pleural fibrosis and restrictive lung disease. Removal of the thickened visceral pleura (*decortication*) is occasionally necessary to improve lung function.

TB of the Upper Airways Nearly always a complication of advanced cavitary pulmonary TB, TB of the upper airways may involve the larynx, pharynx, and epiglottitis. Symptoms include hoarseness, dysphonia, and dysphagia in addition to chronic productive cough. Findings depend on the site of involvement, and ulcerations may be seen on laryngoscopy. Acid-fast smear of the sputum is often positive, but biopsy may be necessary in some cases to establish the diagnosis. Carcinoma of the larynx may have similar features but is usually painless.

Genitourinary TB Genitourinary TB, which accounts for ~10–15% of all extrapulmonary cases in the United States and elsewhere, may involve any portion of the genitourinary tract. Clinical manifestations are cryptic and protean. Patients may be asymptomatic and their disease discovered only after destructive lesions of the kidneys have developed. Symptoms are often nonspecific, and include those of urinary tract infection with frequency, dysuria, nocturia and hematuria, and abdominal or flank pain. Without a high index of suspicion, this form of TB may result in delayed diagnosis with irreversible organ damage. Up to 75% of patients have abnormalities on CXR suggesting previous or concomitant pulmonary disease. Urinalysis gives abnormal results in 90% of cases, revealing pyuria and hematuria. The documentation of culture-negative pyuria in acidic urine should raise the suspicion of TB. IV pyelography, abdominal CT, or MRI (Fig. 178-9) may show deformities and obstructions; calcifications and ureteral strictures are suggestive findings. Culture of three morning urine specimens yields a definitive diagnosis in nearly 90% of cases. Severe ureteral strictures may lead to hydronephrosis, serious renal damage, and, ultimately, renal failure. Genital TB is diagnosed more commonly

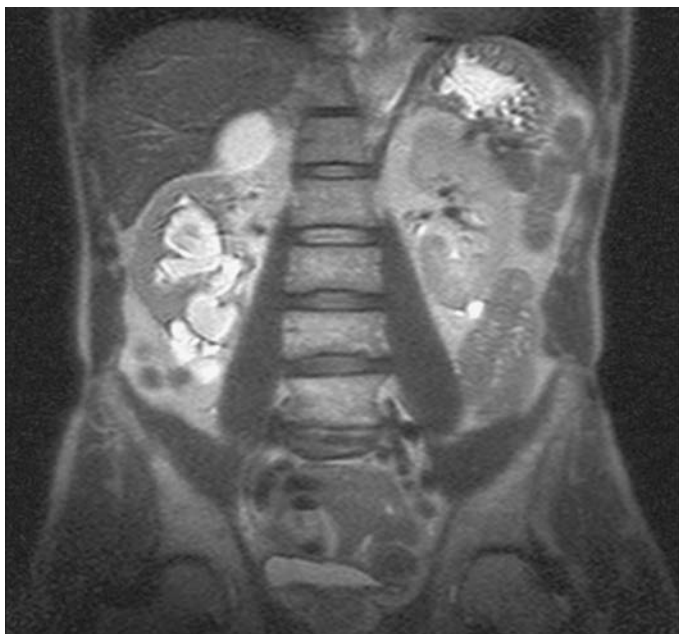


FIGURE 178-9 MRI of culture-confirmed renal tuberculosis. T2-weighted coronal plane: coronal sections showing several renal lesions in both the cortical and the medullary tissues of the right kidney. (Courtesy of Dr. Alberto Matteelli, Department of Infectious Diseases, University of Brescia, Italy; with permission.)

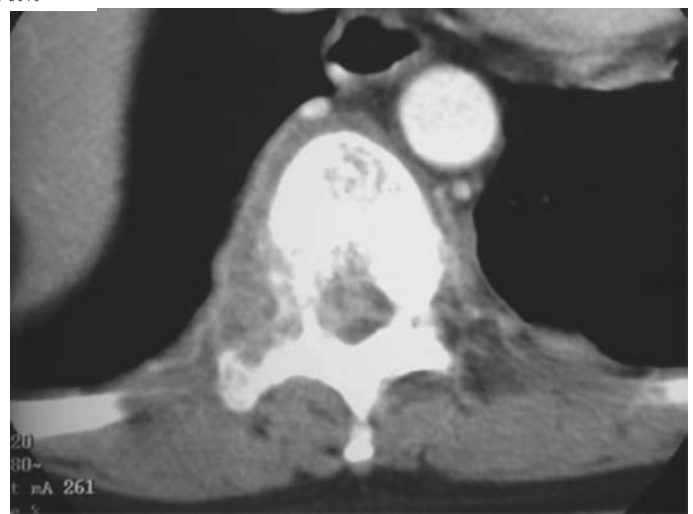


FIGURE 178-10 CT scan demonstrating destruction of the right pedicle of T10 due to Pott's disease. The patient, a 70-year-old Asian woman, presented with back pain and weight loss and had biopsy-proven tuberculosis. (Courtesy of Charles L. Daley, MD, University of California, San Francisco; with permission.)

in female than in male patients. In female patients, it affects the fallopian tubes and the endometrium and may cause infertility, pelvic pain, and menstrual abnormalities. Diagnosis requires biopsy or culture of specimens obtained by dilation and curettage. In male patients, genital TB preferentially affects the epididymis, producing a slightly tender mass that may drain externally through a fistulous tract; orchitis and prostatitis may also develop. In almost half of cases of genitourinary TB, urinary tract disease is also present. Genitourinary TB responds well to chemotherapy.

Skeletal TB In the United States, TB of the bones and joints is responsible for ~10% of extrapulmonary cases. In bone and joint disease, pathogenesis is related to reactivation of hematogenous foci or to spread from adjacent paravertebral lymph nodes. Weight-bearing joints (the spine in 40% of cases, the hips in 13%, and the knees in 10%) are most commonly affected. Spinal TB (Pott's disease or tuberculous spondylitis; Fig. 178-10) often involves two or more adjacent vertebral bodies. Whereas the upper thoracic spine is the most common site of spinal TB in children, the lower thoracic and upper lumbar vertebrae are usually affected in adults. From the anterior superior or inferior angle of the vertebral body, the lesion slowly reaches the adjacent body, later affecting the intervertebral disk. With advanced disease, collapse of vertebral bodies results in kyphosis (*gibbus*). A paravertebral "cold" abscess may also form. In the upper spine, this abscess may track to and penetrate the chest wall, presenting as a soft tissue mass; in the lower spine, it may reach the inguinal ligaments or present as a psoas abscess. CT or MRI reveals the characteristic lesion and suggests its etiology. The differential diagnosis includes tumors and other infections. Pyogenic bacterial osteomyelitis, in particular, involves the disk very early and produces rapid sclerosis. Aspiration of the abscess or bone biopsy confirms the tuberculous etiology, as cultures are usually positive and histologic findings highly typical. A catastrophic complication of Pott's disease is paraplegia, which is usually due to an abscess or a lesion compressing the spinal cord. Paraparesis due to a large abscess is a medical emergency and requires rapid drainage. TB of the hip joints, usually involving the head of the femur, causes pain; TB of the knee produces pain and swelling. If the disease goes unrecognized, the joints may be destroyed. Diagnosis requires examination of the synovial fluid, which is thick in appearance, with a high protein concentration and a variable cell count. Although synovial fluid culture is positive in a high percentage of cases, synovial biopsy and tissue culture may be necessary to establish the diagnosis. Skeletal TB responds to chemotherapy, but severe cases may require surgery.

Tuberculous Meningitis and Tuberculoma TB of the central nervous system (CNS) accounts for ~5% of extrapulmonary cases in



the United States. It is seen most often in young children but also develops in adults, especially those infected with HIV. Tuberculous meningitis results from the hematogenous spread of primary or postprimary pulmonary TB or from the rupture of a subependymal tubercle into the subarachnoid space. In more than half of cases, evidence of old pulmonary lesions or a miliary pattern is found on CXR. The disease often presents subtly as headache and slight mental changes after a prodrome of weeks of low-grade fever, malaise, anorexia, and irritability. If not recognized, tuberculous meningitis may evolve acutely with severe headache, confusion, lethargy, altered sensorium, and neck rigidity. Typically, the disease evolves over 1–2 weeks, a course longer than that of bacterial meningitis. Because meningeal involvement is pronounced at the base of the brain, paresis of cranial nerves (ocular nerves in particular) is a frequent finding, and the involvement of cerebral arteries may produce focal ischemia. The ultimate evolution is toward coma, with hydrocephalus and intracranial hypertension.

Lumbar puncture is the cornerstone of diagnosis. In general, examination of cerebrospinal fluid (CSF) reveals a high leukocyte count (up to 1000/μL), usually with a predominance of lymphocytes but sometimes with a predominance of neutrophils in the early stage; a protein content of 1–8 g/L (100–800 mg/dL); and a low glucose concentration. However, any of these three parameters can be within the normal range. AFBs are infrequently seen on direct smear of CSF sediment, and repeated lumbar punctures increase the yield. Culture of CSF is diagnostic in up to 80% of cases and remains the gold standard. Real-time automated nucleic acid amplification (the Xpert MTB/RIF assay) has a sensitivity of up to 80% and is the preferred initial diagnostic option. Treatment should be initiated immediately upon a positive Xpert MTB/RIF result. A negative result does not exclude a diagnosis of TB and requires further diagnostic workup. Imaging studies (CT and MRI) may show hydrocephalus and abnormal enhancement of basal cisterns or ependyma. If unrecognized, tuberculous meningitis is uniformly fatal. This disease responds to chemotherapy; however, neurologic sequelae are documented in 25% of treated cases, in most of which the diagnosis has been delayed. Clinical trials have demonstrated that patients given adjunctive glucocorticoids may experience faster resolution of CSF abnormalities and elevated CSF pressure, resulting in lower rates of death or severe disability and relapse. In one study, adjunctive dexamethasone significantly enhanced the chances of survival among persons >14 years of age but did not reduce the frequency of neurologic sequelae. The dexamethasone schedule was (1) 0.4 mg/kg per day given IV with tapering by 0.1 mg/kg per week until the fourth week, when 0.1 mg/kg per day was administered; followed by (2) 4 mg/d given by mouth with tapering by 1 mg per week until the fourth week, when 1 mg/d was administered. The WHO now recommends that adjuvant glucocorticoid therapy with either dexamethasone or prednisolone, tapered over 6–8 weeks, should be used in CNS TB.

Tuberculoma, an uncommon manifestation of TB of the CNS, presents as one or more space-occupying lesions and usually causes seizures and focal signs. CT or MRI reveals contrast-enhanced ring lesions, but biopsy is necessary to establish the diagnosis.

Gastrointestinal TB Gastrointestinal TB is uncommon, making up only 3.5% of extrapulmonary cases in the United States. Various pathogenetic mechanisms are involved: swallowing of sputum with direct seeding, hematogenous spread, or (largely in developing areas) ingestion of milk from cows affected by bovine TB. Although any portion of the gastrointestinal tract may be affected, the terminal ileum and the cecum are the sites most commonly involved. Abdominal pain (at times similar to that associated with appendicitis) and swelling, obstruction, hematochezia, and a palpable mass in the abdomen are common findings at presentation. Fever, weight loss, anorexia, and night sweats are also common. With intestinal wall involvement, ulcerations and fistulae may simulate Crohn's disease; the differential diagnosis of this entity is always difficult. Anal fistulae should prompt an evaluation for rectal TB. Because surgery is required in most cases, the diagnosis can be established by histologic examination and culture of specimens obtained intraoperatively.

Tuberculous peritonitis follows either the direct spread of tubercle bacilli from ruptured lymph nodes and intraabdominal organs (e.g., genital TB in women) or hematogenous seeding. Nonspecific abdominal pain, fever, and ascites should raise the suspicion of tuberculous peritonitis. The coexistence of cirrhosis (Chap. 342) in patients with tuberculous peritonitis complicates the diagnosis. In tuberculous peritonitis, paracentesis reveals an exudative fluid with a high protein content and leukocytosis that is usually lymphocytic (although neutrophils occasionally predominate). The yield of direct smear and culture is relatively low; culture of a large volume of ascitic fluid can increase the yield, but peritoneal biopsy (with a specimen best obtained by laparoscopy) is often needed to establish the diagnosis.

Pericardial TB (Tuberculous Pericarditis) Due either to direct extension from adjacent mediastinal or hilar lymph nodes or to hematogenous spread, pericardial TB has often been a disease of the elderly in countries with low TB prevalence. However, it also develops frequently in HIV-infected patients. Case-fatality rates are as high as 40% in some series. The onset may be subacute, although an acute presentation, with dyspnea, fever, dull retrosternal pain, and a pericardial friction rub, is possible. An effusion eventually develops in many cases; cardiovascular symptoms and signs of cardiac tamponade may ultimately appear (Chap. 270). In the presence of effusion, TB must be suspected if the patient belongs to a high-risk population (HIV-infected, originating in a high-prevalence country); if there is evidence of previous TB in other organs; or if echocardiography, CT, or MRI shows effusion and thickness across the pericardial space. A definitive diagnosis can be obtained by pericardiocentesis under echocardiographic guidance. The pericardial fluid must be submitted for biochemical, cytologic, and microbiologic evaluation. The effusion is exudative in nature, with a high count of lymphocytes and monocytes. Hemorrhagic effusion is common. Direct smear examination is very rarely positive. Culture of pericardial fluid reveals *M. tuberculosis* in up to two-thirds of cases, whereas pericardial biopsy has a higher yield. High levels of adenosine deaminase, lysozyme, and IFN-γ may suggest a tuberculous etiology.

Without treatment, pericardial TB is usually fatal. Even with treatment, complications may develop, including chronic constrictive pericarditis with thickening of the pericardium, fibrosis, and sometimes calcification, which may be visible on a chest radiograph. Systematic reviews and meta-analyses show a trend toward benefit from glucocorticoid treatment with regard to death and constrictive pericarditis. However, the largest and most recent study—the IMPI study—failed to show such a benefit. Of the patients enrolled in this trial, 67% were infected with HIV, and only a fraction were receiving antiretroviral treatment (ART). A supplemental analysis among HIV-negative people showed a small mortality benefit, as did another small study among HIV-infected people. The WHO currently recommends that, in patients with tuberculous pericarditis, initial adjuvant glucocorticoid therapy may be used. The 2016 guidelines of the American Thoracic Society (ATS), the CDC, and the Infectious Diseases Society of America (IDSA), on the other hand, suggest that glucocorticoid therapy should not be routinely administered.

Caused by direct extension from the pericardium or by retrograde lymphatic extension from affected mediastinal lymph nodes, tuberculous myocarditis is an extremely rare disease. Usually, it is fatal and is diagnosed postmortem.

Miliary or Disseminated TB Miliary TB is due to hematogenous spread of tubercle bacilli. Although in children it is often the consequence of primary infection, in adults it may be due to either recent infection or reactivation of old disseminated foci. The lesions are usually yellowish granulomas 1–2 mm in diameter that resemble millet seeds (thus the term *miliary*, coined by nineteenth-century pathologists). Clinical manifestations are nonspecific and protean, depending on the predominant site of involvement. Fever, night sweats, anorexia, weakness, and weight loss are presenting symptoms in the majority of cases. At times, patients have a cough and other respiratory symptoms due to pulmonary involvement as well as abdominal symptoms.

Physical findings include hepatomegaly, splenomegaly, and lymphadenopathy. Eye examination may reveal choroidal tubercles, which are pathognomonic of miliary TB, in up to 30% of cases. Meningismus occurs in fewer than 10% of cases.

A high index of suspicion is required for the diagnosis of miliary TB. Frequently, CXR (Fig. 178-5) reveals a miliary reticulonodular pattern (more easily seen on underpenetrated film), although no radiographic abnormality may be evident early in the course and among HIV-infected patients. Other radiologic findings include large infiltrates, interstitial infiltrates (especially in HIV-infected patients), and pleural effusion. Sputum-smear microscopy is negative in most cases. Various hematologic abnormalities may be seen, including anemia with leukopenia, lymphopenia, neutrophilic leukocytosis and leukemoid reactions, and polycythemia. Disseminated intravascular coagulation has been reported. Elevation of alkaline phosphatase levels and other abnormal values in liver function tests are detected in patients with severe hepatic involvement. TST results may be negative in up to half of cases, but reactivity may be restored during chemotherapy. Bronchoalveolar lavage and transbronchial biopsy are more likely to provide bacteriologic confirmation, and granulomas are evident in liver or bone-marrow biopsy specimens from many patients. If it goes unrecognized, miliary TB is lethal; with proper early treatment, however, it is amenable to cure. Glucocorticoid therapy has not proved beneficial.

A rare presentation seen in the elderly, *cryptic miliary TB* has a chronic course characterized by mild intermittent fever, anemia, and—ultimately—meningeal involvement preceding death. An acute septicemic form, *nonreactive miliary TB*, occurs very rarely and is due to massive hematogenous dissemination of tubercle bacilli. Pancytopenia is common in this form of disease, which is rapidly fatal. At postmortem examination, multiple necrotic but nongranulomatous (“nonreactive”) lesions are detected.

Less Common Extrapulmonary Forms TB may cause chorioretinitis, uveitis, panophthalmitis, and painful hypersensitivity-related phlyctenular conjunctivitis. Tuberculous otitis is rare and presents as hearing loss, otorrhea, and tympanic membrane perforation. In the nasopharynx, TB may simulate granulomatosis with polyangiitis. Cutaneous manifestations of TB include primary infection due to direct inoculation, abscesses and chronic ulcers, scrofuloderma, lupus

vulgaris (a smoldering disease with nodules, plaques, and fissures), miliary lesions, and erythema nodosum. Tuberculous mastitis results from retrograde lymphatic spread, often from the axillary lymph nodes. Adrenal TB is a manifestation of disseminated disease presenting rarely as adrenal insufficiency. Finally, congenital TB results from transplacental spread of tubercle bacilli to the fetus or from ingestion of contaminated amniotic fluid. This rare disease affects the liver, spleen, lymph nodes, and various other organs.

Post-TB Complications TB may cause persisting pulmonary damage in patients whose infection has been considered cured on clinical grounds. Chronic impairment of lung functions, bronchiectasis, aspergillomas, and chronic pulmonary aspergillosis (Chap. 217) have been associated with TB. Chronic pulmonary aspergillosis may manifest as simple aspergilloma (fungal ball) or chronic cavitary aspergillosis. Early studies revealed that, especially in the presence of large residual cavities, *Aspergillus fumigatus* may colonize the lesion and produce symptoms such as respiratory impairment, hemoptysis, persistent fatigue, and weight loss, often resulting in the erroneous diagnosis of TB recurrence. The detection of *Aspergillus* precipitins (IgG) in the blood suggests chronic pulmonary aspergillosis, as do radiographic abnormalities such as thickening of the pleura and cavitary walls or the presence of a fungal ball inside the cavity. Treatment is difficult. Recent preliminary studies on the use of itraconazole for ≥ 6 months indicate improvement or stabilization of 60–75% of the radiologic and clinical manifestations. Surgical removal of lesions is risky except in simple aspergilloma.

HIV-Associated TB (See also Chap. 202) TB is one of the most common diseases among HIV-infected persons worldwide. Responsible for up to 30% of all HIV-related mortality (208,000 deaths per year), TB is likely the main cause of death in this population. In certain urban settings in some African countries, the prevalence of HIV infection among TB patients reaches 70–80% (Fig. 178-11).

A person with a positive TST who acquires HIV infection has a 3–13% annual risk of developing active TB, with the exact risk depending on the degree of immunosuppression when observation begins. Furthermore, a new TB infection acquired by an HIV-infected individual may evolve into active disease in a matter of weeks rather

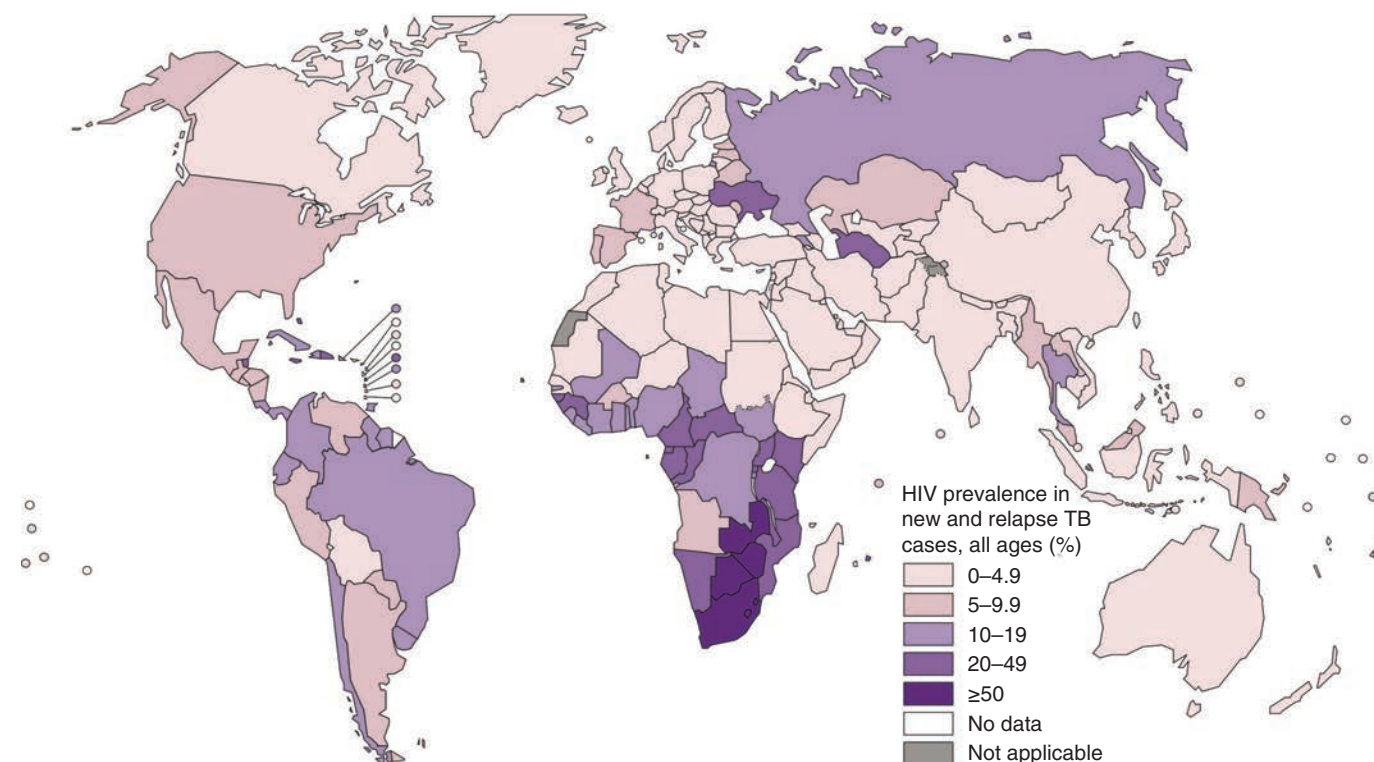


FIGURE 178-11 Estimated HIV prevalence in new and relapse tuberculosis (TB) cases in 2018. (See disclaimer in Fig. 178-2. Reproduced with permission from Global Tuberculosis Report 2019. Geneva, World Health Organization; 2019.)

than months or years. TB can appear at any stage of HIV infection, and its presentation varies with the stage. When cell-mediated immunity is only partially compromised, pulmonary TB presents in a typical manner (Figs. 178-6 and 178-7), with upper-lobe infiltrates and cavitation and without significant lymphadenopathy or pleural effusion. In late stages of HIV infection, when the CD4+ T-cell count is <200/ μ L, a primary TB-like pattern, with diffuse interstitial and subtle infiltrates, little or no cavitation, pleural effusion, and intrathoracic lymphadenopathy, is more common. However, these forms are becoming less common because of the expanded use of ART. Overall, sputum smears are less frequently positive among TB patients with HIV infection than among those without; thus the diagnosis of TB with traditional technology may be difficult, especially in view of the variety of HIV-related pulmonary conditions mimicking TB. Extrapulmonary TB is common among HIV-infected patients. In various series, extrapulmonary TB—alone or in association with pulmonary disease—has been documented in 40–60% of all cases in individuals co-infected with HIV. The most common forms are lymphatic, disseminated, pleural, and pericardial. Mycobacteremia and meningitis are also common, particularly in advanced HIV disease. The diagnosis of TB in HIV-infected patients may be complicated not only by the increased frequency of sputum-smear negativity (up to 40% in culture-proven pulmonary cases) but also by atypical radiographic findings, a lack of classic granuloma formation in the late stages, and a negative TST. The Xpert MTB/RIF assay is the preferred initial diagnostic option for pulmonary TB ensuring a sensitivity of 81% and a specificity of 98%, and therapy should be started on the basis of a positive result because treatment delays may be fatal. A negative Xpert MTB/RIF result, however, does not exclude a diagnosis of TB. Culture remains the gold standard. Recent assessment of a test based on the detection of mycobacterial lipoarabinomannan antigen in urine has shown favorable results in assisting with the detection of TB in HIV-positive people (see “Additional Diagnostic Procedures,” below).

The immune reconstitution inflammatory syndrome (IRIS) or TB immune reconstitution disease consists of exacerbations in systemic manifestations (lymphadenopathy, fever) or respiratory signs (worsening of pulmonary infiltrations, pleural effusion) as well as laboratory or radiographic manifestations of TB. This syndrome has been associated with the administration of ART and occurs in ~10% of HIV-infected TB patients. Usually developing 1–3 months after initiation of ART, IRIS is more common among patients with advanced immunosuppression and extrapulmonary TB. “Unmasking IRIS” may develop after the initiation of ART in patients with undiagnosed subclinical TB. The earlier ART is started and the lower the baseline CD4+ T-cell count, the greater the risk of IRIS. Death due to IRIS is relatively infrequent and occurs mainly among patients who have a high preexisting mortality risk. The presumed pathogenesis of IRIS consists of an immune response that is elicited by antigens released as bacilli are killed during effective chemotherapy and that is temporally associated with improving immune function. There is no diagnostic test for IRIS, and its confirmation relies heavily upon case definitions incorporating clinical and laboratory data; a variety of case definitions have been suggested. The first priority in the management of a possible case of IRIS is to ensure that the clinical syndrome does not represent a failure of TB treatment or the development of another infection. Mild paradoxical reactions can be managed with symptom-based treatment and do not worsen outcomes of treatment for TB. However, IRIS can result in serious neurologic complications or death in patients with CNS TB. Therefore, ART should not be initiated during the first 8 weeks of TB treatment in patients with TB meningitis. Glucocorticoids have been used for severe paradoxical reactions; prednisolone given for 4 weeks at a low dosage (1.5 mg/kg per day for 2 weeks and half that dose for the remaining 2 weeks) has reduced the need for hospitalization and therapeutic procedures and has hastened alleviation of symptoms, as reflected by Karnofsky performance scores, quality-of-life assessments, radiographic response, and C-reactive protein levels. The effectiveness of glucocorticoids in alleviating the symptoms of IRIS is probably linked to suppression of proinflammatory cytokine concentrations as these medications reduce serum concentrations of

IL-6, IL-10, IL-12p40, TNF- α , IFN- γ , and IFN- γ -inducible protein 10. Recommendations for the prevention and treatment of TB in HIV-infected individuals are provided below.

DIAGNOSIS

The key to the early diagnosis of TB is a high index of suspicion. Diagnosis is not difficult in persons belonging to high-risk populations who present with typical symptoms and a classic chest radiograph showing upper-lobe infiltrates with cavities (Fig. 178-6). On the other hand, the diagnosis can easily be missed in an elderly nursing-home resident or a teenager with a focal infiltrate. Often, the diagnosis is first entertained when the chest radiograph of a patient being evaluated for respiratory symptoms is abnormal. If the patient has no complicating medical conditions that cause immunosuppression, the chest radiograph may show typical upper-lobe infiltrates with cavitation (Fig. 178-6). The longer the delay between the onset of symptoms and the diagnosis, the more likely is the finding of cavitory disease. In contrast, immunosuppressed patients, including those with HIV co-infection, may have “atypical” findings on CXR—e.g., lower-zone infiltrates without cavity formation—or interstitial disease only.

The several approaches to the diagnosis of TB require, above all, a well-organized microbiology laboratory network with an appropriate distribution of tasks at different levels of the health care system. Besides clinical assessment and radiography, screening and referral are the principal tasks at the peripheral and community levels. Diagnosis at a secondary level (e.g., a traditional district hospital in a high-incidence setting) can be accomplished nowadays through real-time automated nucleic acid amplification technology (e.g., the Xpert MTB/RIF assay, which also allows detection of drug resistance) or through traditional AFB microscopy, where new tools have not yet been introduced. At a tertiary level, additional technology is necessary, including molecular tests, rapid culture, and DST.

■ NUCLEIC ACID AMPLIFICATION TECHNOLOGY

Several test systems based on amplification of mycobacterial nucleic acid have become available in the past few years and are now the preferred first-line diagnostic tests. These tests are progressively replacing smear microscopy, as they ensure rapid confirmation of all types of TB. One system that permits rapid diagnosis of TB with high specificity and sensitivity (approaching that of liquid culture) is the fully automated, real-time nucleic acid amplification technology known as the Xpert MTB/RIF assay. Xpert MTB/RIF can simultaneously detect TB and rifampin resistance in <2 h and has minimal biosafety and training requirements. Therefore, it can be housed in nonconventional laboratory settings as long as a stable and uninterrupted power supply can be assured. The WHO recommends its use worldwide as the first-line diagnostic test in all adults and children with signs or symptoms of active TB. Given the test’s high sensitivity, the WHO also recommends its use as the initial diagnostic test for people living with HIV in whom TB is suspected. In the diagnosis of pulmonary TB, this test has an overall sensitivity of 85% reaching 98% among AFB-positive cases and ~70% among AFB-negative specimens; its specificity is 98%. When compared to phenotypic drug susceptibility testing for simultaneous detection of rifampin resistance, Xpert MTB/RIF has an overall sensitivity of 96% and a specificity of 98%. The newer Xpert MTB/RIF Ultra assay (Ultra), which uses the same GeneXpert diagnostic platform, has an overall sensitivity of 90% including “trace calls” (i.e., the “noise” produced by detection of DNA from nonviable bacilli) as positive with the greatest increases among smear-negative, culture-positive cases (+17%) and among HIV-infected persons (+12%). If “trace calls” are excluded, sensitivity decreases to 86%. Because of this greater sensitivity and the capacity to also detect nonviable bacilli, the new Ultra cartridge has 2% lower specificity than the original test. However, excluding “trace calls,” specificity increases to 98%. Among people with HIV co-infection, Ultra sensitivity is 88% and specificity 95%. Sensitivity and specificity for detection of rifampin resistance by Ultra are 94% and 99%, respectively, similar to those by the Xpert MTB/RIF assay.

In the diagnosis of extrapulmonary TB, Xpert MTB/RIF and Ultra should be the initial test applied to CSF from patients in whom TB

meningitis is suspected as well as a replacement test (preferable to conventional microscopy, culture, and histopathology) for selected nonrespiratory specimens—those obtained by gastric lavage, fine-needle aspiration, or pleural or other biopsies. Sensitivity varies according to specimen type being the lowest in pleural fluid (50% with Xpert MTB/RIF and 71% with Ultra) and the highest in synovial fluid (97%) and lymph node biopsy (100% with Ultra). “Trace calls” in specimens from persons with extrapulmonary TB, as well as for HIV-infected patients and children, should be considered true positives, given the high risk of severe morbidity and premature death, while among other cases they warrant additional tests to confirm the diagnosis of TB and prevent overtreatment. Among patients with a recent history of TB, “trace calls” may represent false positivity due to DNA from dead bacilli under degradation.

Truenat MTB and MTC Plus are two newly introduced rapid molecular tests with a sensitivity of 83% and 89%, respectively, if compared to bacteriological culture, and with specificity of 98% and 99%, respectively. Truenat MTB-Rif Dx detects rifampin resistance with a sensitivity of 93% and a specificity of 95%. These rapid tests, developed in India by MolBio Diagnostics Pvt Ltd Goa, have accuracy comparable to that of Xpert MTB/RIF and Ultra. Being portable and battery-operated, they can be used in the most peripheral care settings. New high-throughput automated platforms for TB diagnosis and drug-resistant variants are becoming available (Abbott RealTime MTB and RIF/INH, FluoroType MTBDR, BD Max MDR-TB). These platforms are suitable for centralized laboratories and have the advantage of processing a large number of samples in a reasonable time. Sensitivity is higher than 91% and specificity ranges from 97 to 100%. Head-to-head studies with Xpert MTB/RIF have shown comparable performance. Another available molecular test for detection of *M. tuberculosis* is based on the loop-mediated isothermal amplification (LAMP) temperature-independent technology that amplifies DNA, is relatively simple to use, and is interpreted through a visual display. The TB-LAMP assay (LoopampTM *M. tuberculosis* complex detection kit; Eiken Chemical Company, Japan) requires minimal laboratory infrastructure and has few biosafety requirements. It may be used as a replacement for sputum-smear microscopy for the diagnosis of adult pulmonary TB and as a follow-up test to smear microscopy for the further investigation of smear-negative specimens from adults with suspected pulmonary TB. The TB-LAMP assay should not replace rapid molecular tests that detect both TB and rifampin resistance, and its usefulness in HIV-infected people in whom TB is suspected remains unclear.

■ AFB MICROSCOPY

In many low- and middle-income settings, a presumptive diagnosis is still commonly based on the finding of AFB on microscopic examination of a diagnostic specimen, such as a smear of expectorated sputum or of tissue (e.g., a lymph node biopsy). Although inexpensive, AFB microscopy has relatively low sensitivity (40–60%) in culture-confirmed cases of pulmonary TB. The traditional method—light microscopy of specimens stained with Ziehl-Neelsen basic fuchsin dyes—is satisfactory, although time consuming and operator dependent. Most modern laboratories processing large numbers of diagnostic specimens use auramine–rhodamine staining and fluorescence microscopy; this approach is more sensitive than the Ziehl-Neelsen method. However, it is expensive because it requires high-cost mercury vapor light sources and a darkroom. Less expensive light-emitting diode (LED) fluorescence microscopes are now recommended by the WHO as the microscopy tool of choice. They are as sensitive as—or more sensitive than—traditional fluorescence microscopes. As a result, conventional light and fluorescence microscopes are being replaced with this more recent technology, especially in developing countries. For patients with signs or symptoms of pulmonary TB, it has been recommended that one or two sputum specimens, preferably collected early in the morning, should be submitted to the laboratory for AFB smear and mycobacterial culture. If tissue is obtained, it is critical that the portion of the specimen intended for culture not be put in preservation fluid such as formaldehyde. The use of AFB microscopy in examining urine or gastric lavage fluid is limited by the low numbers

of organisms, which can cause false-negative results, or the presence of commensal mycobacteria, which can cause false-positive results.

■ MYCOBACTERIAL CULTURE

Definitive diagnosis depends on the isolation and identification of *M. tuberculosis* from a clinical specimen. Commercial liquid-culture systems such as the Mycobacterial Growth Indicator Tube (MGIT) system (Becton Dickinson; Franklin Lakes, NJ) are recommended by the WHO as the reference standard for culture. The MGIT system uses a fluorescent compound sensitive to the presence of oxygen dissolved in the liquid medium. The appearance of fluorescence, detected by fluorometric technology, indicates active growth of mycobacteria. MGIT cultures usually become positive after a period ranging from 10 days to 2–3 weeks; the tubes are read weekly until the eighth week of incubation before the result is declared to be negative. Specimens may also be inoculated onto egg- or agar-based medium (e.g., Löwenstein-Jensen or Middlebrook 7H10 or 7H11) and incubated at 37°C (under 5% CO₂ for Middlebrook medium). Because most species of mycobacteria, including *M. tuberculosis*, grow slowly, 4–8 weeks may be required before growth is detected on these conventional culture media. Although *M. tuberculosis* may be identified presumptively on the basis of growth time and colony pigmentation and morphology, a variety of biochemical tests have traditionally been used to speciate mycobacterial isolates. In modern, well-equipped laboratories, commercial liquid culture for isolation and species identification by molecular methods or high-pressure liquid chromatography of mycolic acids has replaced isolation on solid media and identification by biochemical tests. A low-cost, rapid immunochromatographic lateral-flow assay based on detection of MTP64 antigen may also be used for species identification of the *M. tuberculosis* complex in culture isolates. These new methods, which are increasingly used in limited-resource settings, have decreased the time required for bacteriologic confirmation of TB to 2–3 weeks.

■ DRUG SUSCEPTIBILITY TESTING

Universal DST is considered by the WHO as the current standard of care for all TB patients and should consist of DST to at least rifampin for all initial isolates of *M. tuberculosis*, as rifampin resistance is an excellent proxy for MDR-TB diagnosis. Susceptibility testing is particularly important if one or more risk factors for drug resistance are identified or if the patient either fails to respond to initial therapy or has a relapse after the completion of treatment (see “Treatment Failure and Relapse,” below). In addition, expanded and rapid susceptibility testing for isoniazid and key second-line anti-TB drugs (especially the fluoroquinolones and the injectable drugs) is mandatory when RR-TB is found in order to guide selection of the appropriate treatment regimens. Susceptibility testing may be conducted directly by molecular techniques (with the clinical specimen) or indirectly (with mycobacterial cultures) on solid or liquid medium. Results are obtained rapidly by direct susceptibility testing on liquid medium, with an average reporting time of 3 weeks. With indirect testing on solid medium, results may not be available for ≥8 weeks. Highly reliable genotypic methods for the rapid identification of genetic mutations in gene regions known to be associated with resistance to rifampin (such as those in *rpoB*) and isoniazid (such as those in *katG* and *inhA*) have been developed and are being widely implemented for screening of patients at increased risk of drug-resistant TB. Apart from the Xpert MTB/RIF, Xpert MTB/RIF Ultra, and Truenat MTB-Rif Dx assays, which, as mentioned above, effectively detect rifampin resistance, the most widely used tests are molecular line probe assays (LPAs). LPAs are a family of DNA strip-based tests capable of detecting bacterial DNA and identifying drug resistance-associated mutations. After extraction of DNA from *M. tuberculosis* isolates or from clinical specimens, the resistance gene regions are amplified by polymerase chain reaction (PCR), and labeled and probe-hybridized PCR products are detected by colorimetric development. This assay reveals the presence of *M. tuberculosis* as well as mutations in target resistance-gene regions. Given the rapidity and accuracy of commercially available LPAs, the WHO recommends that they (rather than phenotypic culture-based tests) may be used to detect resistance to isoniazid and rifampin when



patients have sputum smear–positive specimens or a cultured isolate of *M. tuberculosis*. These recommendations do not eliminate the need for conventional culture-based testing to identify resistance to other drugs and to monitor emergence of additional drug resistance. A similar approach has been developed for second-line anti-TB drugs, such as the fluoroquinolones and the injectable drugs kanamycin, amikacin, and capreomycin. Therefore, second-line LPAs (instead of phenotypic culture-based DST) are now recommended by the WHO as the initial test for rapid detection of resistance to the fluoroquinolones or the second-line injectable drugs in isolates from patients with confirmed RR-TB or MDR-TB. As with first-line LPAs, these recommendations do not eliminate the need for conventional phenotypic, culture-based testing to identify resistance to other drugs and to monitor for the emergence of additional resistance. Detection of pyrazinamide resistance is important among persons with MDR/RR-TB. The WHO has recently recommended the use of a LPA with reverse hybridization-based technology in culture isolates rather than phenotypic culture-based DST. Finally, a few noncommercial, inexpensive culture and susceptibility testing methods (e.g., microscopically observed drug susceptibility, nitrate reductase, and colorimetric redox indicator assays) have been used in resource-limited settings. Their use is restricted to national reference laboratories with proven proficiency and adequate external quality control as an interim solution while genotypic or automated liquid-culture technology is introduced.

Whole genome sequencing (WGS) of *M. tuberculosis* provides comprehensive information on mutations conferring resistance and is a promising alternative to existing phenotypic and molecular DST methods. Recent studies have confirmed the potential for WGS to identify genetic polymorphisms that reliably predict drug susceptibility phenotype within a clinically relevant timeframe and a comparable cost range. The clinical use of WGS, however, has been hampered by the requirement for a culture sample before DNA processing. Recently, amplification and sequencing of relevant genomic targets directly from sputum samples have been successfully tested and targeted next-generation sequencing (tNGS) is now a possible option. Evidence is accumulating supporting the clinical application of NGS-based diagnostic systems for TB to replace traditional diagnostic tests in the future.

■ RADIOGRAPHIC PROCEDURES

CXR is a rapid imaging technique that has historically been used as a primary tool to detect pulmonary TB. CXR has high sensitivity but poor specificity. Although TB may often present with typical patterns strongly suggesting the disease, some abnormalities seen in TB are also present in several other lung conditions. The initial suspicion of pulmonary TB is often based on abnormal CXR findings in a patient undergoing triage for respiratory symptoms. The presence of lesions suggestive of TB should prompt bacteriologic investigations in all cases, without exception. Although the “classic” picture is that of upper-lobe disease with infiltrates and cavities (Fig. 178-6), virtually any radiographic pattern—from a normal film or a solitary pulmonary nodule to diffuse alveolar infiltrates in a patient with adult respiratory distress syndrome—may be seen. In the era of HIV/AIDS, no radiographic pattern can be considered pathognomonic, but CXR can assist in diagnosing TB or ruling it out before initiation of any preventive treatment. CXR is also helpful as a screening test used preceding rapid molecular assays to improve their predictive value. Digital CXR technology, which allows display of images in a digital format on a computer screen instead of on x-ray film, offers several advantages: the procedure time is reduced, the running costs are lower, the imaging is of superior quality, and telemedicine assistance is available, including computer-aided detection (CAD) and interpretation of findings using software programs that analyze digital imaging for abnormalities compatible with TB. However, the limited evidence available suggests that while sensitivity may be high, specificity is variable. A recent systematic review of CAD studies concluded that the diagnostic accuracy of this technology is still limited and that generalizability to low-prevalence settings is still uncertain.

CT (Fig. 178-7) may be useful in interpreting questionable findings on plain CXR and in diagnosing some forms of extrapulmonary TB (e.g., intrabdominal disease, Pott’s disease; Fig. 178-10). A recent study

has shown the potential of positron emission tomography combined with CT for detection of subclinical disease that may be progressing toward full-blown TB in HIV-infected people. MRI is useful in the diagnosis of bone lesions and intracranial TB.

■ ADDITIONAL DIAGNOSTIC PROCEDURES

Other diagnostic tests may be used when pulmonary TB is suspected. Sputum induction by ultrasonic nebulization of hypertonic saline may be useful for patients who cannot produce a sputum specimen spontaneously. Frequently, patients with radiographic abnormalities that are consistent with other diagnoses (e.g., bronchogenic carcinoma) undergo fiberoptic bronchoscopy with bronchial brushings and endobronchial or transbronchial biopsy of the lesion. Bronchoalveolar lavage of a lung segment containing an abnormality may also be performed. In all cases, it is essential that specimens be submitted for molecular testing with the Xpert MTB/RIF assay, mycobacterial culture, and AFB smear. For the diagnosis of primary pulmonary TB in children, who often do not expectorate sputum, induced sputum specimens and specimens from early-morning gastric lavage may yield positive results in the Xpert MTB/RIF assay or on culture.

Invasive diagnostic procedures are indicated for patients with suspected extrapulmonary TB. In addition to testing of specimens from involved sites (e.g., CSF for tuberculous meningitis, pleural fluid and biopsy samples for pleural disease), biopsy and culture of bone marrow and liver tissue have a good diagnostic yield in disseminated (miliary) TB, particularly in HIV-infected patients, who also have a high frequency of positive blood cultures. Xpert MTB/RIF should always be the initial diagnostic test in patients where TB meningitis is suspected; any positive results should prompt immediate treatment initiation, while negative results should be followed up by additional testing. In some cases, the results of culture or Xpert MTB/RIF are negative but a clinical diagnosis of TB is supported by consistent epidemiologic evidence (e.g., a history of close contact with an infectious patient) and a compatible clinical and radiographic response to treatment. In the United States and other industrialized countries with low rates of TB, some patients with limited abnormalities on CXR and sputum positive for AFB are infected with nontuberculous mycobacteria, most commonly organisms of the *M. avium* complex or *M. kansasii* (Chap. 180). Factors favoring the diagnosis of nontuberculous mycobacterial disease over TB include an absence of risk factors for TB and the presence of underlying chronic pulmonary disease.

Patients with HIV-associated TB pose several diagnostic problems (see “HIV-Associated TB,” above). HIV-infected patients with sputum culture–positive, AFB-positive TB may present with a normal chest radiograph. The Xpert MTB/RIF assay is the preferred rapid diagnostic test in this population of patients because of its simplicity and increased sensitivity (~60–70% among AFB-negative, culture-positive cases and 97–98% among AFB-positive cases). With the advent of ART, the occurrence of disseminated *M. avium* complex disease that can be confused with TB has become much less common. A test based on the detection of mycobacterial lipoarabinomannan antigen in urine has emerged as a potentially useful point-of-care test for TB in HIV-infected persons with low CD4+ T-cell counts. The lateral-flow urine lipoarabinomannan assay can be performed manually and read by eye. After a systematic review of the evidence, the WHO recommends that this assay be used to assist in the diagnosis of TB in HIV-positive adults who have signs and symptoms of TB and a CD4+ T-cell count of ≤ 100 cells/ μ L or in HIV-positive patients who are seriously ill regardless of CD4+ T-cell count or with an unknown CD4+ count. The WHO recommends that this test not be used, pending information on recent promising technological test advances, for TB diagnosis or as a screening test for TB in any other patient categories. One limitation of the available lipoarabinomannan point-of-care test, AlereLAM (Alere Determine TB LAM Ag), is the low sensitivity (45%). A novel assay, FujiLAM (SILVAMP TB LAM) has recently shown a sensitivity of 70%.



■ SEROLOGIC AND OTHER DIAGNOSTIC TESTS FOR ACTIVE TB

Several serologic tests based on detection of antibodies to a variety of mycobacterial antigens have been carefully assessed by the WHO

and found not to be useful as diagnostic aids because of their low sensitivity and specificity and their poor reproducibility. In 2011, after a rigorous evaluation of these tests, the WHO issued a “negative” recommendation in order to prevent their abuse in the private sector of many resource-limited countries. Various methods aimed at detection of mycobacterial antigens in diagnostic specimens are being investigated but are limited at present by low sensitivity. Determinations of adenosine deaminase and IFN- γ levels in pleural fluid may be useful adjunctive tests in the diagnosis of pleural TB; their utility in the diagnosis of other forms of extrapulmonary TB (e.g., pericardial, peritoneal, and meningeal) is less clear.

■ BIOMARKERS

In view of the limitations of current diagnostics, research on TB biomarkers and multiple marker biosignatures that could be used as a point-of-care test for disease or triage is a high priority and has been crystallized in well-defined target product profiles by the WHO. Recent systematic reviews revealed that promising host biomarkers under study, such as antibodies, cytokines, chemokines, and RNA signatures, by far exceed pathogen biomarkers that can be obtained from urine or blood. However, currently, candidate biomarkers require additional studies to fully assess their performance.

■ DIAGNOSIS OF *M. TUBERCULOSIS* INFECTION

Two tests currently exist for identification of individuals with TB infection: the TST and IGRA, both of which measure host immunological response to TB antigens. These tests have limitations, especially in settings or populations with high TB and/or HIV prevalence.

Tuberculin Skin Testing In 1891, Robert Koch discovered that components of *M. tuberculosis* in a concentrated liquid-culture medium, subsequently named “old tuberculin,” were capable of eliciting a skin reaction when injected subcutaneously into patients with TB. In 1932, Seibert and Munday purified this product by ammonium sulfate precipitation to produce an active protein fraction known as *tuberculin purified protein derivative* (PPD). In 1941, PPD-S, developed by Seibert and Glenn, was chosen as the international standard. Later, the WHO and UNICEF sponsored large-scale production of a master batch of PPD (RT23) and made it available for general use. The greatest limitation of PPD is its lack of mycobacterial species specificity, a property due to the large number of proteins in this product that are highly conserved in the various species. In addition, subjectivity of the skin-reaction interpretation that is dependent on the operator, deterioration of the product, and batch-to-batch variations limit the usefulness of PPD.

The skin test with tuberculin PPD (TST) is most widely used in screening for TB infection. It probably measures the response to antigenic stimulation by T cells that reside in the skin rather than the response of recirculating memory T cells. The test is of limited value in the diagnosis of active TB because of its relatively low sensitivity and specificity and its inability to discriminate between TB infection and active disease. False-negative reactions are common in immunosuppressed patients and in those with overwhelming TB. False-positive reactions may be caused by infections with nontuberculous mycobacteria (Chap. 180) and by BCG vaccination. A repeated TST can produce larger reaction sizes due to either boosting or true conversion. The “boosting phenomenon” is a spurious TST conversion resulting from boosting of reactivity on a subsequent TST 1–5 weeks after the initial test. Distinguishing boosting from true conversion is difficult yet important and can be based on clinical and epidemiologic considerations. For instance, true conversions are likely after BCG vaccination in a previously TST-negative person or in a close contact of an infectious patient.

IFN- γ Release Assays Two in vitro assays that measure T-cell release of IFN- γ in response to stimulation with the highly TB-specific RD1-encoded antigens ESAT-6 and CFP-10 were introduced in the early 2000s and are commercially available. The T-SPOT.TB test (Oxford Immunotec; Oxford, United Kingdom) is an enzyme-linked immunospot assay, and the QuantiFERON-TB Gold test (Qiagen GmbH; Hilden, Germany) is a whole-blood enzyme-linked immunosorbent

assay for measurement of IFN- γ . The QuantiFERON-TB Gold In-Tube (QFT-GIT) assay, which facilitates blood collection and initial incubation, also contains another specific antigen, TB7.7. These tests mainly measure the response of recirculating memory CD4+ T cells—normally part of a reservoir in the spleen, bone marrow, and lymph nodes—to persisting bacilli-producing antigenic signals. However, CD8+ cells can also release IFN- γ in vitro in response to stimulation with TB antigens, and they seem to do so especially in the early phase of infection and in the phase of reactivation. Therefore, a new version of the QFT-GIT assay, called QuantiFERON-TB Gold Plus (QFT-Plus), has been developed that operates through two antigen tubes: TB1, containing long peptides from ESAT-6 and CFP-10 and inducing a CD4+ T-cell response, and TB2 that also contains shorter peptides stimulating CD8+ cells. The QFT-Plus may have an increased sensitivity, compared to QFT-GIT, but this conclusion needs confirmation.

In settings or population groups with low TB and HIV burdens, IGRAs have previously been reported to be more specific than the TST as a result of less cross-reactivity with BCG vaccination and sensitization by nontuberculous mycobacteria; i.e., RD1 antigens are not encoded in the genome of either BCG strains or most nontuberculous mycobacteria. Recent studies suggest that IGRAs may not perform well in serial testing (e.g., among health care workers) and that interpretation of results depends on cutoff values used to define positivity. Potential advantages of IGRAs include logistical convenience, the need for fewer patient visits to complete testing, and the avoidance of somewhat subjective measurements (e.g., skin induration). However, IGRAs require that blood be drawn and then delivered to the laboratory in a timely fashion. IGRAs also require that testing be performed by specially trained technicians in a laboratory setting. These requirements pose challenges similar to those faced with the TST, including cold-chain requirements and batch-to-batch variations. Because of higher specificity and greater availability of resources, IGRAs have usually replaced the TST for TB infection diagnosis in low-incidence, high-income settings. However, in high-incidence TB and HIV settings and population groups, evidence about the performance and usefulness of IGRAs is still limited, and cost considerations may currently limit wider use.

A number of national guidelines on the use of IGRAs for TB infection testing have been issued. In the United States, an IGRA is preferred to the TST for most persons over the age of 5 years who are being screened for TB infection. However, for individuals at high risk of progression to active TB (e.g., HIV-infected persons), either test—or, to optimize sensitivity, both tests—may be used. Because of the paucity of data on the use of IGRAs in children, the TST is preferred for TB infection testing of children aged <5 years. In Canada and some European countries, a two-step approach for those with positive TSTs—i.e., an initial TST followed by an IGRA—is often recommended. However, a TST may boost an IGRA response if the interval between the two tests exceeds 3 days.

In conclusion, both the TST and IGRA, although useful as diagnostic aids, are imperfect tests for TB infection: while they can identify infected persons, they have low predictive value in identifying individuals with the highest risk of progression toward disease, cannot differentiate between active TB and TB infection, cannot distinguish new infections from reinfections, and display reduced sensitivity in immunocompromised patients.

TREATMENT

Tuberculosis

The two main aims of TB treatment are (1) to prevent morbidity and death by curing TB while preventing recurrences and emergence of drug resistance, and (2) to interrupt transmission by rendering patients noninfectious to others. Chemotherapy for TB became possible with the discovery of streptomycin in 1943. Randomized clinical trials clearly indicated that the administration of streptomycin to patients with chronic TB reduced mortality rates and led to cure in the majority of cases. However, monotherapy with streptomycin

was soon associated with the development of resistance to this drug and the resulting failure of treatment. With the introduction into clinical practice of para-aminosalicylic acid (PAS) and isoniazid, it became axiomatic in the early 1950s that cure of TB required the concomitant administration of at least two agents to which the organism was susceptible. Furthermore, early clinical trials demonstrated that a long period of treatment—i.e., 12–24 months—was required to prevent recurrence. The introduction of rifampin in the early 1970s heralded the era of effective short-course chemotherapy, with a treatment duration of <12 months. The discovery that pyrazinamide, which was first used in the 1950s, augmented the potency of isoniazid/rifampin regimens led to the use of a 6-month course of this triple-drug regimen as standard therapy. Streptomycin was added as the fourth drug mainly to prevent the emergence of drug resistance. These four drugs (with streptomycin replaced by ethambutol) still form the basis of the optimal treatment regimen for rifampin-susceptible TB. The emergence of drug-resistant TB in the 1990s prompted attempts to standardize the approach to treatment of this condition mainly on the basis of expert opinion. This event has also stimulated research on and development of new anti-TB agents in the past 15 years. In 2013 and 2014, respectively, bedaquiline and delamanid—the first two drugs specifically developed for TB during nearly half a century—received conditional approval by the US Food and Drug Administration (FDA) and other drug-regulatory authorities; approval was based on the results of phase 2b clinical trials in which the drugs were added to the 18- to 24-month WHO-recommended regimen for MDR-TB. Bedaquiline and delamanid are now being used increasingly for treatment of MDR-TB under specific conditions. In 2019, another new drug, pretomanid, was approved by the FDA as part of a new combination regimen with bedaquiline and linezolid for patients with MDR-TB caused by a strain with additional resistance to a fluoroquinolone or a second-line injectable drug, or were intolerant of therapy, or in whom treatment had failed.

DRUGS

Four major drugs are considered first-line agents for the treatment of TB: isoniazid, rifampin, pyrazinamide, and ethambutol. **Table 178-2** presents currently recommended dosages in adults and children. Some studies have suggested increased effectiveness when isoniazid, rifampin, and pyrazinamide are given at higher dosage; thus if these findings are confirmed, dosages may be revised in the future. These drugs are well absorbed after oral administration, with peak serum levels at 2–4 h and nearly complete elimination within 24 h. Isoniazid and rifampin, two key anti-TB drugs, are recommended on the basis of their bactericidal activity (i.e., their ability to rapidly reduce the number of viable organisms and render patients noninfectious). All four agents are recommended in light of their sterilizing activity (i.e., their ability to sterilize the affected tissues, measured in terms of the ability to prevent relapses) and the lowered risk that drug-resistant mutant bacilli will be selected when the drugs are used in combination. Two additional rifamycins,

ritapentine and rifabutin, are also available; however, their level of cross-resistance with rifampin is high. For a detailed discussion of the drugs used for the treatment of TB, see **Chap. 181**.

Because of a lower degree of effectiveness and tolerability, several classes of second-line drugs are generally used only for the treatment of patients with drug-resistant TB. These agents have previously been classified in various manners to facilitate a standardized approach to their use. In the latest WHO guidance on the treatment of MDR-TB, they are now grouped in three ranked categories for the purpose of designing more individualized regimens of 18–20 months' duration. **Group A drugs include three classes of oral agents: the fluoroquinolones levofloxacin and moxifloxacin; the oxazolidinone linezolid; and the recently introduced diarylquinoline bedaquiline, which was granted accelerated approval by the FDA in late 2012.** **Group B drugs include two other oral agents: clofazimine and cycloserine (or its analogue terizidone).** **Group C drugs include the nitroimidazole delamanid; imipenem-cilastatin or meropenem; the injectable aminoglycosides amikacin and streptomycin (the latter formerly a first-line agent, now rarely used for drug-resistant TB because resistance levels worldwide are high and it is more toxic than the other drugs in the same class); ethionamide or prothionamide; and PAS.** In addition, the first-line anti-TB drugs ethambutol and pyrazinamide (both included in Group C) as well as high-dose isoniazid (only for the shorter regimen; see below) are used for MDR-TB treatment. Information about drugs used in the treatment of drug-resistant TB (including dosages) can be found in the following WHO Handbook: http://apps.who.int/iris/bitstream/10665/130918/1/9789241548809_eng.pdf. Recent information from the phase 3 clinical trial of delamanid (a drug granted accelerated approval by the European Medicines Agency [EMA] in late 2013) added to an optimized longer WHO background regimen shows that treatment success is not different from that obtained with the addition of placebo. The future role of delamanid as a replacement drug in MDR-TB treatment remains to be assessed. The new classification scheme excludes the second-line injectable aminoglycoside kanamycin and the polypeptide capreomycin. Amithiozone, which has been associated with severe and at times fatal skin reactions—including Stevens-Johnson syndrome—among HIV-infected patients, is no longer recommended. Finally, amoxicillin-clavulanic acid is recommended only as an adjunct to carbapenems.

REGIMENS

Standard regimens are divided into an intensive (bactericidal) phase and a continuation (sterilizing) phase. During the intensive phase, the majority of tubercle bacilli are killed, symptoms resolve, and usually the patient becomes noninfectious. The continuation phase is required to eliminate persisting mycobacteria and prevent relapse.

The treatment regimen of choice for virtually all forms of drug-susceptible TB in adults consists of a 2-month initial (intensive) phase of isoniazid, rifampin, pyrazinamide, and ethambutol followed by a 4-month continuation phase of isoniazid and rifampin (**Table 178-3**). This regimen can cure TB in >90% of patients. In children, most forms of TB in the absence of HIV infection or suspected isoniazid resistance can be safely treated without ethambutol in the intensive phase. Treatment should be given daily throughout the course. Systematic reviews have demonstrated that the use of an intermittent thrice-weekly regimen in the intensive phase is associated with increased risk of treatment failure, relapse, and acquisition of drug resistance. Furthermore, a thrice-weekly regimen in the continuation phase only has also been associated with increased rates of failure and relapse, while a twice-weekly regimen in the continuation phase increased the risk of acquisition of drug resistance as well as rates of failure and relapse. Therefore, the WHO now recommends that TB treatment in all cases be administered daily. The 2016 guidelines by the ATS, the CDC, and the IDSA, while recommending daily administration of drugs, include a provision for use of intermittent thrice-weekly supervised regimens among patients who are not infected with HIV and are at low risk of relapse (i.e., have pulmonary TB caused by drug-susceptible

TABLE 178-2 Recommended Dosage^a for Initial Treatment of Tuberculosis in Adults and Children

DRUG	DAILY DOSE	
	ADULT	PEDIATRIC
Isoniazid	5 mg/kg, max 300 mg	10 (7–15) mg/kg, max 300 mg
Rifampin	10 mg/kg, max 600 mg	15 (10–20) mg/kg, max 600 mg
Pyrazinamide	25 mg/kg, max 2 g	35 (30–40) mg/kg
Ethambutol ^b	15 mg/kg	20 (15–25) mg/kg

^aThe duration of treatment with individual drugs varies by regimen, as detailed in Table 178-3. ^bIn certain settings, streptomycin (15 mg/kg daily, with a maximal dose of 1 g; or 25–30 mg/kg thrice weekly, with a maximal dose of 1.5 g) can replace ethambutol in the initial phase of treatment. However, streptomycin generally is no longer considered a first-line drug.

Source: Based on recommendations of the American Thoracic Society/Infectious Diseases Society of America/Centers for Disease Control and Prevention and the World Health Organization.

TABLE 178-3 Recommended Antituberculosis Treatment Regimens

INDICATION	INITIAL PHASE		CONTINUATION PHASE	
	DURATION, MONTHS	DRUGS	DURATION, MONTHS	DRUGS
New smear- or culture-positive cases	2	HRZE ^{a,b}	4	HR ^{a,c}
New culture-negative cases	2	HRZE ^a	4	HR ^{a,d}
Pregnancy	2	HRE ^a	7	HR
Relapses and treatment default ^f	Tailored according to rapid drug susceptibility testing			
Failures ^f	Tailored according to rapid drug susceptibility testing			
Resistance (or intolerance) to H	Throughout (6)	RZEQ		
Resistance (or intolerance) to R	Same as for MDR-TB; see below			
MDR-TB (resistance to at least H + R)	See Tables 178-4 and 178-5			
XDR-TB	See Table 178-4			
Intolerance to Z	2	HRE	7	HR

^aAll drugs should be given daily. ^bA 4-month regimen of 8 weeks of once-daily rifapentine, isoniazid, pyrazinamide and moxifloxacin followed by 9 weeks of once-daily rifapentine, isoniazid, and moxifloxacin is a possible alternative. ^cA clinical trial showed that HIV-negative patients with noncavitary pulmonary tuberculosis who have negative sputum AFB smears after the initial phase of treatment can be given once-weekly rifapentine/isoniazid in the continuation phase. However, this regimen is rarely used. ^dThe American Thoracic Society, the Centers for Disease Control and Prevention, and the Infectious Diseases Society of America suggest that a 2-month continuation phase could be used in HIV-seronegative patients with sputum smear-negative and culture-negative TB. ^eThe 6-month regimen with pyrazinamide can probably be used safely during pregnancy and is recommended by the WHO and the International Union Against Tuberculosis and Lung Disease. If pyrazinamide is not included in the initial treatment regimen, the minimal duration of therapy is 9 months. ^fThe availability of rapid molecular methods to identify drug resistance allows initiation of a proper regimen at the start of treatment.

Abbreviations: E, ethambutol; H, isoniazid; MDR-TB, multidrug-resistant tuberculosis; Q, a quinolone antibiotic; R, rifampin; WHO, World Health Organization; XDR-TB, extensively drug-resistant tuberculosis; Z, pyrazinamide.

organisms that, at the start of treatment, is noncavitary and/or sputum smear-negative). The same guidelines suggest that a 4-month regimen consisting of isoniazid, rifampin, pyrazinamide, and ethambutol may be adequate for treatment of HIV-negative adults with sputum smear-negative and culture-negative pulmonary TB (i.e., paucibacillary TB).

A continuation phase of once-weekly rifapentine and isoniazid is effective in HIV-seronegative patients without cavitation on CXR. In general, however, this regimen should be used with great caution. Patients with cavitary pulmonary TB and delayed sputum-culture conversion (i.e., those who remain culture-positive at 2 months) should be retested immediately for drug-resistant TB, and a change of regimen should be considered. A full course of 6 months with four-drug therapy should be performed not including interruptions of >4 weeks. In some developing countries where the ability to ensure adherence to treatment is limited, a continuation-phase regimen of daily isoniazid and ethambutol for 6 months has been used in the past. This regimen is clearly associated with a higher rate of relapse, treatment failure, and death, especially among HIV-infected patients, and is no longer recommended by the WHO. ~~Several studies attempting to reduce treatment duration to 4 months by using fluoroquinolones (with moxifloxacin replacing ethambutol or isoniazid, or gatifloxacin replacing ethambutol) were conducted over the last decade. The main finding was that shorter (4 month) fluoroquinolone-containing regimens are associated with significantly higher rates of relapse at 18 months than the standard 6-month rifampin-containing regimen. In addition, the studies showed no reduction in adverse events with the fluoroquinolone-containing regimen and no difference in all-cause and TB-related mortality rates. Therefore, shortening of the treatment duration to 4 months through the use of fluoroquinolones has not been recommended.~~ However, the recent Tuberculosis Trials Consortium Study 31/AIDS Clinical Trials Group A5349 (Study 31/A5349) showed that a 4-month daily regimen that included isoniazid, pyrazinamide, rifapentine at a daily dose of 1200 mg and moxifloxacin at a daily dose of 400 mg was noninferior to the standard 6-month regimen and had a similar adverse event profile. In early 2021, this option was therefore considered by the WHO as a possible alternative to the old standard provided that rigorous antibacterial stewardship is ensured especially to prevent fluoroquinolone resistance and that rifapentine becomes more widely available. Alternative regimens for patients who exhibit drug intolerance or adverse reactions are listed in Table 178-3. However, severe side effects prompting discontinuation of any of the first-line

drugs and use of these alternative regimens are uncommon. To prevent isoniazid-related neuropathy, pyridoxine (10–25 mg/d) should be added to the regimen given to persons at high risk of vitamin B₆ deficiency (e.g., alcoholics; malnourished persons; pregnant and lactating women; and patients with conditions such as chronic renal failure, diabetes, and HIV infection, which are also associated with neuropathy). Finally, to facilitate absorption of rifampin, the drug should be taken on an empty stomach and without meals.

PATIENT CARE AND SUPPORT

Poor adherence to treatment is one of the most important impediments to cure. Moreover, the tubercle bacilli harbored by patients who do not fully adhere to the prescribed regimen are likely to become resistant to the drugs to which they are irregularly exposed. Both patient- and provider-related factors may affect adherence. Patient-related factors include a lack of belief that the illness is worth the cost of adherence; the existence of concomitant medical conditions (notably alcohol or substance abuse); lack of social support; fear of the stigma and discrimination associated with TB; and poverty, with attendant joblessness and homelessness. Provider-related factors that may prevent adherence include lack of support, education, and encouragement of patients and inconvenient clinical services.

A variety of interventions to increase the chances of completion of the months-long treatment course are available. First, a package of social support interventions that are complementary and not mutually exclusive, consisting of educational, psychological, and material goods and services, may enable people with TB to address hurdles to treatment adherence. Health education and counseling on the disease's seriousness and solutions and on the importance of treatment adherence until cure should be provided to all patients at the start of and throughout the course of TB therapy. Psychological support (i.e., counseling sessions or peer-group support) can be particularly relevant in the context of the stigma and discrimination often affecting people with TB and their families. Material support (e.g., food or financial support in forms such as meals, food baskets, food supplements, food vouchers, transport subsidies, living allowances, housing incentives, or financial bonuses) reduces indirect costs incurred by patients or their attendants in accessing health services and mitigates the consequences of income loss related to the disease.

Second, it is paramount that health services be arranged to meet the needs and reasonable expectations of patients. Components of optimal health services include a suitable geographic location, a schedule responsive to patients' needs, functional channels of communication between patients and their health care providers (e.g.,

a telephone short-messaging system, audio/video call capability, home or workplace visits), and a staff willing and competent to care for people with TB, to address their concerns, and to base the care they provide on sound ethical standards.

Third, it is crucial to offer the patient a suitable option for treatment administration that minimizes the chance of nonadherence. Such options traditionally include unsupervised, self-administered therapy; in-person directly observed therapy (DOT); and nondaily DOT (e.g., supervision not for every dose but weekly or a few times per week) at a location mutually agreed on by patient and health care provider, with supervisory responsibility delegated to a qualified person. Direct supervision of adherence is crucial in view of the lack of tools to accurately predict adherence to self-administered treatment and of the public health importance of TB. The WHO, along with the ATS, the CDC, and the IDSA, states that ideally all patients should have their therapy directly supervised, especially during the initial phase, with proper social support based on a patient-centered approach as described above. In several countries, personnel to supervise therapy are usually available through TB control programs of local public health departments, often involving members of the community who are accepted by the patient and who have been properly trained and educated by health workers to undertake the supervisory role. Direct supervision with social support has been shown to significantly increase the proportion of patients completing treatment in all settings and to lessen the chances of treatment failure, relapse, and default. In general, community- or home-based DOT is recommended over health facility-based DOT or unsupervised treatment; DOT administered by trained lay providers or health care workers is recommended over DOT administered by family members. Recently, comparison of video-observed therapy (VOT) with in-person DOT has shown similar outcomes. In a multicenter, analyst-blinded, randomized controlled superiority trial of VOT through daily remote observation using a smartphone app versus DOT done 3–5 times weekly at home, community, or clinic settings, VOT was superior to DOT in ensuring scheduled observation of drug intake. Therefore, VOT can replace DOT when Internet access is good and video communication technology (e.g., smartphones, tablets, computers) is available. The system can be appropriately organized and operated by health care providers and patients. Other digital health tools can facilitate the monitoring of adherence, including digital medication monitors; these monitors can register when the pillbox is opened, with options to emit audio signals or a short message to remind patients to take medicines. These tools are customized to the needs and preferences of the individual patient and the provider.

In addition to the above measures promoting adherence, provision of fixed-dose combination products that reduce the number of tablets the patient needs to swallow is recommended over separate drug formulations. Various fixed-dose combination products are available (e.g., isoniazid/rifampin, isoniazid/rifampin/pyrazinamide, and isoniazid/rifampin/pyrazinamide/ethambutol). Fixed-dose combinations increase patient satisfaction and minimize the likelihood of prescription error or of development of drug resistance resulting from monotherapy if a drug is out of stock or the patient prefers one drug over others. In addition, these combinations facilitate programmatic management of procurement and supply. In the past, the bioavailability of rifampin was found to be substandard in some formulations of fixed-dose combinations. Medical regulatory authorities should ensure that combination products are of good quality; however, top standards for drug quality assurance are not always operative, especially in limited-resource countries. Prescribers should be aware of this potential problem.

MONITORING TREATMENT RESPONSE AND DRUG TOXICITY

Bacteriologic evaluation through commercial liquid-culture systems (or—when liquid-culture capacity is not yet available—through smear microscopy) is essential in monitoring the response to TB treatment. In addition, the patient's weight should be monitored regularly and the drug dosage adjusted with any significant weight change. Patients with pulmonary disease should have their sputum

examined monthly until cultures become negative to allow early detection of treatment failure. With the recommended 6-month standard first-line regimen, >80% of drug-susceptible TB patients will have negative sputum cultures at the end of the second month of treatment. By the end of the third month, the sputum of virtually all patients should be culture negative. In some patients, especially those with extensive cavitory disease and large numbers of organisms, AFB smear conversion may lag behind culture conversion as a result of the expectoration and microscopic visualization of dead bacilli. Therefore, as capacity is built, smear microscopy should be progressively abandoned as a monitoring tool in favor of liquid culture. As noted above, patients with cavitory disease in whom sputum culture conversion does not occur by 2 months require immediate testing or retesting for drug resistance. When a patient's sputum cultures or smears remain positive at ≥ 3 months despite good adherence, treatment failure caused by drug resistance is likely. The pattern of drug resistance should guide the choice of the best treatment option (see below). A sputum specimen should be collected at the end of treatment to document cure. In settings where mycobacterial cultures are not yet available, monitoring by AFB smear examination should be undertaken at 2, 5, and 6 months. Bacteriologic monitoring of patients with extrapulmonary TB is more difficult and often is not feasible. In these cases, the response to treatment must be assessed clinically with the help of medical imaging.

Monitoring of the response to chemotherapy by nucleic acid amplification technology, such as the Xpert MTB/RIF assay, is not suitable because these tests can produce positive results due to nonviable bacilli. Likewise, serial chest radiographs are not recommended because radiographic changes may lag behind bacteriologic response and are not highly sensitive. After the completion of treatment, neither sputum examination nor CXR is recommended for routine follow-up purposes. However, a chest radiograph obtained at the end of treatment may be useful for comparative purposes should the patient develop symptoms of recurrent TB months or years later. Patients should be instructed to report promptly for medical assessment if they develop any such symptoms.

During treatment, patients should also be monitored for drug toxicity. The most common adverse reaction of significance among people treated for drug-susceptible TB is hepatitis. Patients should be carefully educated about the signs and symptoms of drug-induced hepatitis (e.g., dark urine, loss of appetite, nausea) and should be instructed to discontinue treatment promptly and see their health care provider if these manifestations occur. Although biochemical monitoring is not routinely recommended, all adult patients should undergo baseline assessment of liver function (e.g., measurement of serum levels of hepatic aminotransferases and bilirubin). Older patients, those with concomitant diseases, those with a history of hepatic disease (especially hepatitis C), and those using alcohol daily should be monitored especially closely (i.e., monthly), with repeated measurements of aminotransferases, during the initial phase of treatment. Up to 20% of patients have small increases (up to three times the upper limit of normal) in serum levels of aspartate aminotransferase that are not accompanied by symptoms and are of no consequence. Suspension of treatment should be considered for patients with symptomatic hepatitis, especially when accompanied by at least a three-fold increase in serum levels of AST and/or ALT, and for patients without symptoms of hepatic injury who have marked (at least five-fold) elevations in serum levels of AST and/or ALT. Drugs should be reintroduced one at a time after liver function has returned to normal. Hypersensitivity reactions usually require the discontinuation of all drugs and rechallenge to determine which agent is the culprit. Because of the variety of regimens available, it usually is not necessary—although it is possible—to desensitize patients. Hyperuricemia and arthralgia caused by pyrazinamide can usually be managed by the administration of acetylsalicylic acid; however, pyrazinamide treatment should be stopped if the patient develops gouty arthritis. Individuals who develop autoimmune thrombocytopenia secondary to rifampin therapy should not receive the drug thereafter. Similarly, the occurrence of optic neuritis with ethambutol is an indication for permanent discontinuation of this drug. Other common

manifestations of drug intolerance, such as pruritus and gastrointestinal upset, can generally be managed without the interruption of therapy. Treatment with second-line agents for drug-resistant TB is associated with a variety of adverse drug reactions that are more frequent and severe than in patients receiving first-line TB regimens (see below). The likelihood of drug–drug interactions is also higher when second-line regimens are used.

TREATMENT FAILURE AND RELAPSE

As stated above, treatment failure should be suspected when a patient's cultures (or sputum smears, when cultures are not available) remain positive after 3 months of treatment. In the management of such patients, it is imperative that the current isolate be urgently retested (or tested for the first time if, for some reason, rapid molecular susceptibility testing was not performed at the start of treatment) for susceptibility to first-line agents and, if resistance to rifampin is detected, to second-line agents as well. The treatment approach should start with molecular testing for—at the least—resistance to rifampin and isoniazid. Because results are expected to become available within a few days, changes in the regimen can be postponed until that time. However, if the patient's clinical condition is deteriorating rapidly, an earlier change in regimen may be indicated. A cardinal rule in the latter situation is always to add more than one drug, preferably two or three, at a time to a failing regimen; in practice, starting an empirical regimen for MDR-TB (see “Drug-Resistant TB,” below) is warranted. The patient may continue to take isoniazid and rifampin along with these new agents pending the results of susceptibility tests.

Patients who experience a recurrence after apparently successful treatment (i.e., a relapse) are less likely to harbor drug-resistant strains than are patients in whom treatment has failed. Acquired resistance is uncommon among strains from patients in whom relapse follows the proper completion of a standard 6-month regimen. The treatment decision depends on a general assessment of the risk of drug resistance, the severity of the case, and the results of rapid susceptibility testing. Patients whose treatment has been interrupted and who have a high likelihood of MDR-TB should receive an empirical MDR-TB regimen that includes second-line

agents (Table 178-3). Once drug susceptibility results are available, the regimen can be adjusted accordingly.

DRUG-RESISTANT TB

Strains of *M. tuberculosis* resistant to individual drugs arise by spontaneous point mutations in the mycobacterial genome that occur at low but predictable rates (10^{-7} – 10^{-10} for the key drugs). Resistance to rifampin is associated with mutations in the *rpoB* gene in 95% of cases, that to isoniazid with mutations mainly in the *katG* gene (50–95% of cases) and the *inhA* gene promoter region (up to 45%), that to pyrazinamide in the *pncA* gene (up to 98%), that to ethambutol in the *embB* gene (50–65%), that to the fluoroquinolones in the *gyrA*–*gyrB* genes (75–95%), and that to the aminoglycosides mainly in the *rrs* gene (up to 80%); the C-12T mutation is the most common mutation in the *eis* promoter region associated with aminoglycoside resistance, especially in Eastern European countries. Because there is no cross-resistance among the commonly used classes of drugs, the probability that a strain will be resistant to two drug classes is the product of the probabilities of resistance to each drug class and thus is low. The development of drug-resistant TB almost invariably follows monotherapy—i.e., the failure of the health care provider to prescribe at least two drugs to which tubercle bacilli are susceptible; of the patient to absorb or take properly prescribed therapy; or of the bioavailability of poor-quality drugs or preparations (e.g., due to crushing of tablets). Drug-resistant TB may be either primary or acquired. In primary drug resistance, the patient is infected from the start by a drug-resistant strain. Acquired resistance develops in the infecting strain during treatment. In North America, Western Europe, most of Latin America, and the Persian Gulf states, rates of primary resistance are generally low and isoniazid resistance is most common. In the United States, although rates of primary isoniazid resistance have been stable at ~7–8%, the rate of primary MDR-TB has declined from 2.5% in 1993 to <1% since 2000. As described above, MDR-TB is an increasingly serious problem in some regions, especially in the countries of the former Soviet Union and some countries of Asia (Fig. 178-12). Even more serious is the occurrence of MDR strains that are also resistant to additional second-line agents used in treatment, such as the fluoroquinolones. Creation of

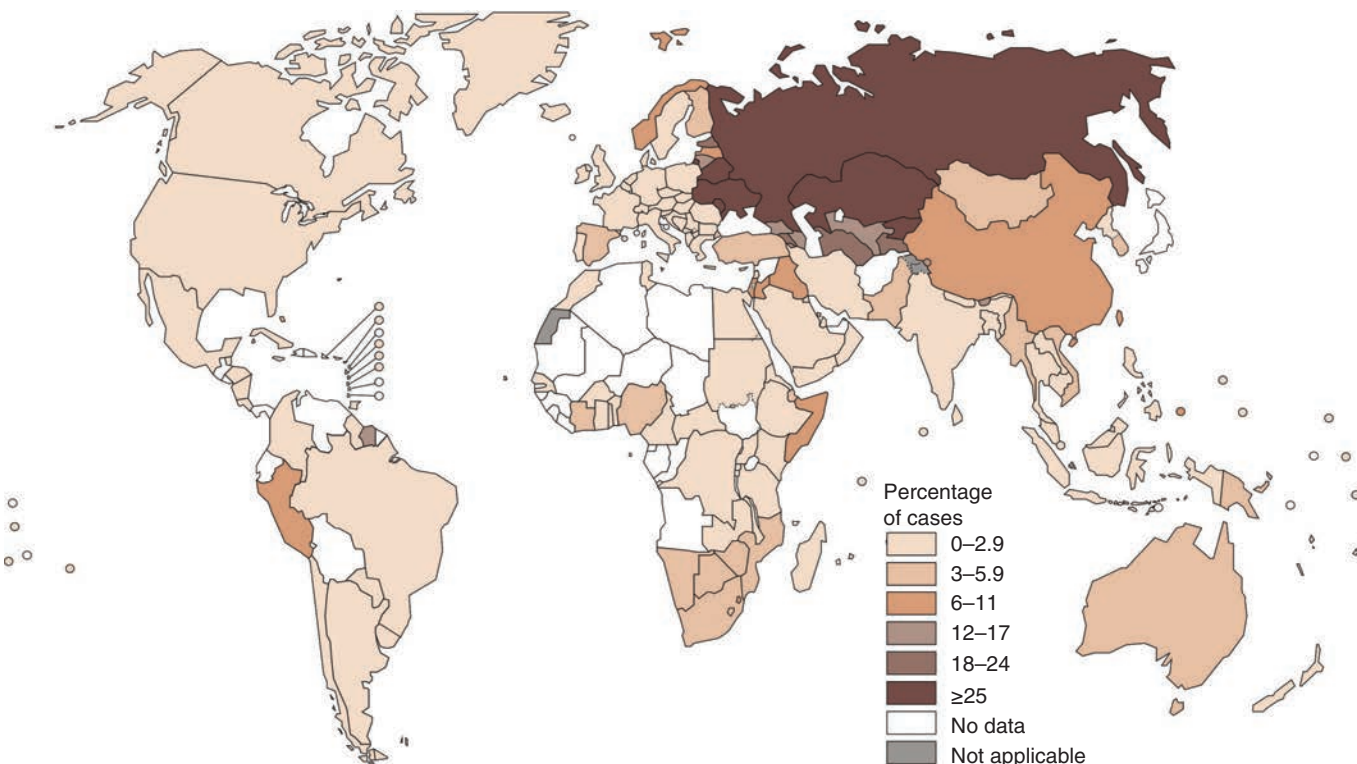


FIGURE 178-12 Percentage of new cases of multidrug-resistant/rifampin-resistant tuberculosis (TB) in all countries surveyed by the World Health Organization (WHO) Global Drug Resistance Surveillance Project during 1994–2018. Figures are based on the most recent year for which data have been reported, which varies among countries. Data reported before the year 2002 are not shown. (See disclaimer in Fig. 178-2. Reproduced with permission from Global Tuberculosis Report 2019. Geneva, World Health Organization; 2019.)

drug-resistant TB can be prevented by adherence to the principles of sound treatment: inclusion of at least two quality-assured, bactericidal drugs to which the organism is susceptible; use of effective combination regimens; supervision of treatment with patient support; and verification that patients complete the prescribed course. The use of fixed-dose combination products may prevent selective drug intake and therefore possibly protect against the creation of drug resistance. Transmission of drug-resistant strains can be prevented by the implementation of respiratory infection-control measures (see below) and by early detection of people with active TB followed by immediate initiation of treatment with an effective regimen.

Isoniazid-Resistant TB For the treatment of patients with isoniazid-resistant disease, a combination of rifampin, ethambutol, pyrazinamide, and levofloxacin for 6 months is recommended. This fluoroquinolone-containing regimen should not be used until rifampin resistance has been excluded by a reliable diagnostic test to avoid inadvertent treatment of MDR-TB with an inadequate regimen. Ideally, a laboratory test for susceptibility should also be done for the fluoroquinolones and pyrazinamide. If the fluoroquinolone is contraindicated because of intolerance or resistance, the patient can be given a 6-month regimen of rifampin, ethambutol, and pyrazinamide. Isoniazid probably does not contribute to a successful outcome in these regimens but may be retained (also to facilitate treatment with the four-drug fixed-dose formulation). Other drugs, such as the injectable aminoglycosides, are unlikely to play a role in the treatment of most isoniazid-resistant TB cases. However, they may be considered in the presence of additional resistance (e.g., to pyrazinamide or ethambutol) or of drug intolerance.

RR-, MDR-TB MDR-TB, in which bacilli are resistant to (at least) isoniazid and rifampin, is more difficult to manage than is disease caused by drug-susceptible organisms because these two bactericidal drugs are the most potent first-line agents available and because associated resistance to other first-line drugs as well (e.g., ethambutol) is not uncommon. Treatment for RR-TB and MDR-TB has traditionally been a topic of much debate, given its complexity, long duration, toxicity, and limited efficacy; the cost of most second-line drugs; and the lack of randomized controlled clinical trials to support combinations. Until recently, recommendations were therefore based largely on low-quality evidence from observational studies and on best-practice consensus among experts. Recent developments include the accrual of individual datasets for patients treated worldwide, the release of findings from two randomized controlled phase 3 clinical trials (the STREAM Stage 1 trial comparing a 9-month, shorter MDR-TB regimen with the previous optimized WHO background regimen; and Otsuka's phase 3 trial 213 comparing the addition of the new drug delamanid to the previous optimized WHO background regimen with the addition of placebo), the publication of results of an open-label, single group study enrolling highly drug-resistant cases on a regimen composed of three oral drugs (bedaquiline, pretomanid, and linezolid), and the assessment of programmatic data from South Africa on the large-scale use of a shorter all-oral bedaquiline-containing regimen. The assessment of this information resulted in an update of WHO guidance for the treatment of MDR-TB and all other RR-TB cases in which isoniazid resistance is absent or unknown.

As a result, two main approaches are now recommended by the WHO to treat MDR/RR-TB: (1) an individualized longer regimen of 18–20 months' duration (or 15–17 months after culture conversion) consisting of an optimal combination of oral drugs chosen according to a rational approach and using the WHO priority grouping of medicines (**Table 178-4**); and (2) a shorter, all-oral, bedaquiline-containing regimen of 9–12 months' duration. While these recommendations may change when new data indicate the need, all-oral regimens are now the preferred options and the use of either a shorter or a longer regimen depends on the assessment of the severity of disease, knowledge of drug resistance pattern, and history of previous treatment.

TABLE 178-4 Groups of Drugs Recommended for Use in Longer MDR-TB Regimens and Approach to the Design of a Longer Regimen for Adults and Children

GROUP	DRUG
Group A: Drugs to be prioritized and included in all regimens, unless they cannot be used	Levofloxacin <i>or</i> moxifloxacin Bedaquiline Linezolid
Group B: Drugs to be added in all regimens, unless they cannot be used	Clofazimine Cycloserine <i>or</i> terizidone
Group C: Drugs to be used to complete the regimen and when drugs from groups A and B cannot be used	Ethambutol Delamanid Pyrazinamide Imipenem-cilastatin <i>or</i> meropenem Amikacin <i>or</i> streptomycin Ethionamide <i>or</i> prothionamide <i>p</i> -Aminosalicylic acid

Source: Adapted from the World Health Organization, 2018.

Longer MDR-TB Regimen In MDR/RR-TB patients where infecting strains have or are presumed to have additional resistance (e.g., resistance to the fluoroquinolones), in those who have severe pulmonary or extrapulmonary disease, or those who have been treated previously with second-line drugs, a longer regimen is recommended. Table 178-4 shows the priority grouping of drugs recommended by the WHO and the approach to the design of a longer regimen for both adults and children. As much as possible, the regimen is composed of all group A agents and at least one group B agent. The use of bedaquiline and linezolid is promoted, together with a fluoroquinolone (levofloxacin or moxifloxacin), whenever possible, in all patients. Clofazimine and cycloserine (group B) are the two preferred options to be added to group A drugs. Group C drugs can replace group A and B agents that cannot be used, and the choice should be based on drug susceptibility testing, drug resistance levels in the population, the patient's history of previous use of these drugs, and potential intolerance or toxicity. The injectable agents (e.g., amikacin, streptomycin, and the carbapenems) are assigned a lower priority because of inconvenience of use and the toxicity of aminoglycosides. A fully oral regimen is thus also the first choice and most desirable option even for the most severe cases who are ineligible for a shorter regimen. Nevertheless, when injectables are necessary and DST supports their use, amikacin has been found to be the most effective drug in this class, followed by streptomycin, while kanamycin and capreomycin seem less effective, both having been associated with higher risks of failure and relapse when compared with longer regimens in which other agents were used instead. A treatment course of at least 18–20 months is recommended, but duration may depend on the patient's response. Important considerations when treating MDR-TB patients include the safety and effectiveness of bedaquiline use beyond 6 months. Likewise, the ideal duration of use of linezolid, which is known to be highly effective but also very frequently produces toxicity (e.g., peripheral and optic neuropathy, and bone marrow suppression), is unclear. Additional considerations concern the use of pyrazinamide and of the aminoglycosides amikacin and streptomycin, which is now restricted to cases with proven susceptibility to those agents. The role of delamanid in the treatment of MDR-TB remains to be assessed, although, as stated earlier, data from the phase 3 clinical trial of this agent as an addition to the longer regimen previously recommended by the WHO did not demonstrate a higher rate of treatment success than was obtained with the background regimen plus placebo. Furthermore, evidence on the safety and effectiveness of delamanid given for >6 months is presently incomplete.

Shorter MDR-TB Regimen In MDR/RR-TB patients with no extensive pulmonary disease or severe extrapulmonary disease, without history of previous treatment with second-line drugs, and whose

infecting strains are not resistant to the fluoroquinolones, a shorter, all-oral, bedaquiline-containing regimen is recommended. Recent observational programmatic data from South Africa, assessed by WHO in 2019, showed that a fully oral regimen starting with 6 months of bedaquiline accompanied by 4–6 months of levofloxacin or moxifloxacin, ethionamide, ethambutol, pyrazinamide, high-dose isoniazid (10–15 mg/kg per day), and clofazimine, and followed by 5 months of levofloxacin (or moxifloxacin), clofazimine, pyrazinamide and ethambutol, was associated with low toxicity and better outcomes than the older, standardized, injectable-containing regimen when used in patients, including the HIV-infected, with no extensive pulmonary disease or severe extrapulmonary disease, fluoroquinolone resistance, or previous history of second-line drug use. The WHO therefore now recommends the adoption of this regimen and the progressive phasing-out of the previously recommended injectable-containing shorter regimen. The WHO also recommends that any modifications of a shorter regimen, where injectables are replaced by drugs other than bedaquiline, should be tested under operational research conditions and not adopted on a large programmatic scale until evidence shows their safety, tolerability, and efficacy.

The criteria used to define eligible patients are listed in **Table 178-5**. Adults and children eligible for the shorter regimen may still be offered the option of a new longer regimen if their completion of the full duration is adequately supported; with the longer regimen, the likelihood of relapse-free cure could be increased, and its administration is fully oral. As with any anti-TB regimen, the risk of creating additional resistance is high if the regimen is used incorrectly (e.g., in someone with preexisting fluoroquinolone resistance).

As in past recommendations, informed consent should be sought from patients treated with all MDR-TB regimens, and active TB drug safety monitoring is recommended. Patients taking QT interval-prolonging drugs (bedaquiline, delamanid, clofazimine, and fluoroquinolones) should be closely monitored, with electrocardiography performed at the start of treatment and repeated during treatment; patients with a QTc interval >500 ms or a history of ventricular arrhythmias should not be given these drugs. Patients taking amikacin should undergo serial audiometry to detect any hearing loss early on. Incentives and other forms of support can encourage patients not to interrupt treatment.

MDR-TB patients with additional resistance to fluoroquinolones and other second-line medicines have fewer treatment options and a poorer prognosis. However, the new longer regimen offers more options for a reasonably effective and tolerable regimen. The design of regimens for complex patterns of MDR-TB follows the same principles outlined in Table 178-4 through the selection of agents likely to be effective and tolerated. Observational studies have shown that aggressive management in such patients, with early drug susceptibility testing, use of a rational combination of at least five effective drugs, strict adherence to directly observed therapy, monthly bacteriologic monitoring, and intensive patient support, may—besides interrupting transmission—increase the chances of

cure and avert death. For patients with localized disease and sufficient pulmonary reserve, lobectomy or wedge resection may be considered as part of treatment. A novel regimen composed of bedaquiline, the new nitroimidazole compound pretomanid, and linezolid (BPaL) administered orally for 6 months has been tested by the Global Alliance for TB Drug Development in South Africa in an open-label, single-group study enrolling 109 patients with MDR-TB caused by a strain with additional resistance to a fluoroquinolone or a second-line injectable drug, or were intolerant of therapy, or in whom treatment had failed. The cure rate was 90% after a 6-month course of treatment; the main toxicity, due to linezolid, consisted of peripheral neuropathy (81%) and myelosuppression (48%), which were manageable with dose reduction or interruption of the drug. Based on these findings, in August 2019 the FDA approved the new drug pretomanid under the Limited Population Pathway for Antibacterial and Antifungal Drugs (LPAD pathway) as part of a three-drug, 6-month, all-oral regimen for the treatment of a population of patients with MDR-TB caused by a strain with additional resistance to a fluoroquinolone or a second-line injectable drug, or who were intolerant of therapy, or in whom treatment had failed. Although approved by the FDA and proven as highly efficacious in the only trial available, in late 2019 the WHO suggested that this new 6–9 month regimen be used either under operational research conditions until more information is available on safety and efficacy or as a last resort in difficult-to-treat individual patients. A new, dose-blinded study reported in 2021 suggested that BPaL regimens where linezolid was used at reduced daily dose (from 1200 mg to 600 mg) and/or for a shorter period of time (from 6 to 2 months) were safer. Another regimen being tested by the Global Alliance is composed of bedaquiline, pretomanid, moxifloxacin, and pyrazinamide (BPaMZ). In a phase 2B trial, MDR-TB patients became culture-negative within 8 weeks of treatment three times faster than drug-sensitive TB patients treated with the standard regimen. The BPaMZ regimen is now being tested for both MDR-TB and drug-susceptible TB, with the aim of reducing treatment duration to 6 months and 4 months, respectively. Results are expected in late 2021.

Because the management of MDR-TB is complicated by both social and medical factors, care of seriously ill patients is ideally provided in specialized centers or, in their absence, in the context of programs with adequate resources and capacity, including community support. When patients are in stable condition, treatment and care on an ambulatory basis at a decentralized health care facility should be prioritized as this approach may increase treatment success and reduce loss to follow-up. This approach should not, however, preclude hospitalization when it is necessary. Respiratory infection-control measures should be observed throughout. As part of a patient-centered approach, palliative and end-of-life care should be provided as a priority when all recommended treatment options have been exhausted.

HIV-ASSOCIATED TB

Several observational studies and randomized controlled trials have shown that treatment of HIV-associated TB with anti-TB drugs and simultaneous use of ART is associated with significant reductions in mortality risk and AIDS-related events. Evidence from randomized controlled trials shows that early initiation of ART during anti-TB treatment is associated with a 34–68% reduction in mortality rates, with especially good results in patients with CD4+ T-cell counts of <50/μL. Therefore, the main aim in the management of HIV-associated TB is to initiate anti-TB treatment and to immediately consider initiating or continuing ART. All HIV-infected TB patients, regardless of CD4+ T-cell count, are candidates for ART, which optimally is initiated as soon as possible after the diagnosis of TB and with the strong recommendation to start within the first 8 weeks of anti-TB therapy; ART should be started within the first 2 weeks of TB treatment for profoundly immunosuppressed patients with CD4+ T-cell counts of <50/μL. In general, the standard 6-month daily regimen is equally efficacious in HIV-negative and HIV-positive patients with drug-susceptible TB. However, in the uncommon situation

TABLE 178-5 Criteria for Offering a Shorter All-Oral Regimen (9–11 Months) to Patients with Confirmed Multidrug- or Rifampin-Resistant (MDR/RR) Tuberculosis (TB)

- Confirmed absence of resistance to or lack of suspicion of the ineffectiveness of a drug in the shorter MDR-TB regimen (except for isoniazid resistance)
- No history of exposure to one or more second-line drugs used in the shorter MDR-TB regimen (including bedaquiline) for >1 month
- Confirmed fluoroquinolone-susceptible disease
- No intolerance to drugs used in the shorter MDR-TB regimen or risk of toxicity (e.g., drug–drug interactions)
- No pregnancy
- No extensive pulmonary disease
- No disseminated, meningeal, or central nervous system TB
- Availability of all drugs in the shorter MDR-TB regimen

Source: Adapted from the World Health Organization, 2019.

where an HIV-infected patient cannot receive ART, prolongation of the continuation phase of TB treatment by 3 months can be considered. As in any other TB patient, intermittent regimens should not be used in HIV-infected people. As for any other adult living with HIV (**Chap. 202**), first-line ART for TB patients consists of two nucleoside reverse transcriptase inhibitors plus a nonnucleoside reverse transcriptase inhibitor or an integrase or protease inhibitor. Recent guidelines have also considered a two-drug treatment consisting of one nucleoside reverse transcriptase inhibitor plus an integrase inhibitor. Although TB treatment modalities are similar to those in HIV-negative patients, adverse drug reactions may be more pronounced in HIV-infected patients. In this regard, three important considerations are relevant: an increased frequency of paradoxical reactions, interactions between ART components and rifamycins, and development of rifampin mono-resistance with intermittent treatment. IRIS—i.e., the exacerbation of symptoms and signs of TB—has been described above. Rifampin, a potent inducer of enzymes of the cytochrome P450 system, lowers serum levels of many HIV protease inhibitors and some nonnucleoside reverse transcriptase inhibitors—essential drugs used in ART. In such cases, rifabutin, which has much less enzyme-inducing activity, has been used in place of rifampin. However, dosage adjustments for rifabutin and protease or integrase inhibitors are still being assessed. Several clinical trials have found that patients with HIV/TB co-infection whose degree of immunosuppression is advanced (e.g., CD4+ T-cell counts of <100/ μ L) are prone to treatment failure and relapse with rifampin-resistant organisms when treated with “highly intermittent” (i.e., once- or twice-weekly) rifamycin-containing regimens. Consequently, it is now recommended that all TB patients who are infected with HIV, like all other TB patients with rifampin-susceptible disease, receive a rifampin-containing regimen on a daily basis. Because recommendations are frequently updated, consultation of the following websites is advised: www.who.int/hiv, www.who.int/tb, www.cdc.gov/hiv, and www.cdc.gov/tb.

SPECIAL CLINICAL SITUATIONS

Although comparative clinical trials of treatment for extrapulmonary TB are limited, the available evidence indicates that most forms of disease should be treated with a 6-month regimen recommended for patients with pulmonary disease. For TB meningitis, the ATS, the CDC, and the IDSA recommend extension of the continuation phase for 7–10 months. The WHO and the American Academy of Pediatrics recommend that children with bone and joint TB, tuberculous meningitis, or miliary TB receive up to 12 months of treatment (2-month induction treatment followed by 10-month consolidation treatment). Treatment for TB may be complicated by underlying medical problems that require special consideration. As a rule, patients with chronic renal failure should not receive aminoglycosides and should receive ethambutol only if serum drug levels can be monitored. Isoniazid, rifampin, and pyrazinamide may be given in the usual doses in cases of mild to moderate renal failure, but the dosages of isoniazid and pyrazinamide should be reduced for all patients with severe renal failure except those undergoing hemodialysis. Patients with hepatic disease pose a special problem because of the hepatotoxicity of isoniazid, rifampin, and pyrazinamide. Patients with severe hepatic disease may be treated with ethambutol, streptomycin, and possibly another drug (e.g., a fluoroquinolone); if required, isoniazid and rifampin may be administered under close supervision. The use of pyrazinamide by patients with liver failure should be avoided. Silicotuberculosis necessitates the extension of therapy by at least 2 months.

The regimen of choice for pregnant women (Table 178-3) is 9 months of treatment with isoniazid and rifampin supplemented by ethambutol for the first 2 months. Although the WHO has recommended routine use of pyrazinamide for pregnant women in combination with isoniazid and rifampin, this drug has not been recommended for pregnant women in the United States because of insufficient data documenting its safety in pregnancy. Streptomycin is contraindicated because it is known to cause eighth-cranial-nerve

damage in the fetus. The thioamides, bedaquiline, and delamanid should also be avoided in the treatment of pregnant women with MDR-TB. Treatment for TB is not a contraindication to breastfeeding; most of the drugs administered will be present in small quantities in breast milk, albeit at concentrations far too low to provide any therapeutic or prophylactic benefit to the child.

Medical consultation on difficult-to-manage cases is provided by the US CDC Regional Training and Medical Consultation Centers (www.cdc.gov/tb/education/rtrmc/).

PREVENTION

The primary way to prevent TB is to diagnose and isolate infectious cases rapidly and to administer appropriate treatment until patients are rendered noninfectious (usually 2–4 weeks after the start of proper treatment) and the disease is cured. Additional strategies include BCG vaccination and preventive treatment of persons with TB infection who are at high risk of developing active disease.

■ BCG VACCINATION

Historically one of the most used vaccines in the history of medicine, BCG was derived from an attenuated strain of *M. bovis* and was first administered to humans in 1921. Many BCG vaccines are available worldwide; all are derived from the original strain, but the vaccines vary in efficacy, ranging from 80% to nil in randomized, placebo-controlled trials. A similar range of efficacy was found in observational studies (case-control, historic cohort, and cross-sectional) in areas where infants are vaccinated at birth. These studies and a meta-analysis also found higher rates of efficacy in the protection of infants and young children from serious disseminated forms of childhood TB, such as tuberculous meningitis and miliary TB. BCG vaccine is safe and rarely causes serious complications. The local tissue response begins 2–3 weeks after vaccination, with scar formation and healing within 3 months. Side effects—most commonly, ulceration at the vaccination site and regional lymphadenitis—occur in 1–10% of vaccinated persons. Some vaccine strains have caused osteomyelitis in ~1 case per million doses administered. Disseminated BCG infection (“BCGitis”) and death have occurred in 1–10 cases per 10 million doses administered, although this problem is restricted almost exclusively to persons with impaired immunity, such as children with severe combined immunodeficiency syndrome or adults with HIV infection. BCG vaccination induces TST reactivity, which tends to wane with time. The presence or size of TST reactions after vaccination does not predict the degree of protection afforded.

BCG vaccine is recommended for routine use at birth in countries or among populations with high TB prevalence (**Fig. 178-13**). However, because of the low risk of transmission of TB in the United States and other high-income countries, the variability in protection afforded by BCG, and its impact on the TST, the vaccine is not recommended for general use. HIV-infected adults and children should not receive BCG vaccine. Moreover, infants whose HIV status is unknown but who have signs and symptoms consistent with HIV infection or who are born to HIV-infected mothers should not receive BCG.

Over the past decade, renewed research and development efforts have been made toward a new TB vaccine, and several candidates have been developed and tested. The MVA-85A was the first new TB vaccine to be tested in a phase 2B proof-of-concept trial in infants in South Africa. Results published in 2013 showed that MVA-85A was well tolerated and modestly immunogenic but did not confer significant protection against clinical TB or *M. tuberculosis* infection. A second, more promising candidate vaccine, M72/AS01_E, a subunit vaccine pairing two *M. tuberculosis* antigens (32A and 39A) with the adjuvant M72/AS01_E, was recently tested in a randomized trial among 3575 patients with *M. tuberculosis* infection to prevent development of active disease. TB developed in 13 participants among those receiving the vaccine and in 26 among those receiving placebo with an estimated efficacy of 49.7% at 36 months. Adverse events were not different in the two groups. This vaccine is now being considered for further development.

As of the end of 2020, 14 candidate vaccines were in various stages of clinical trials. They included whole-cell or mycobacterial whole-cell

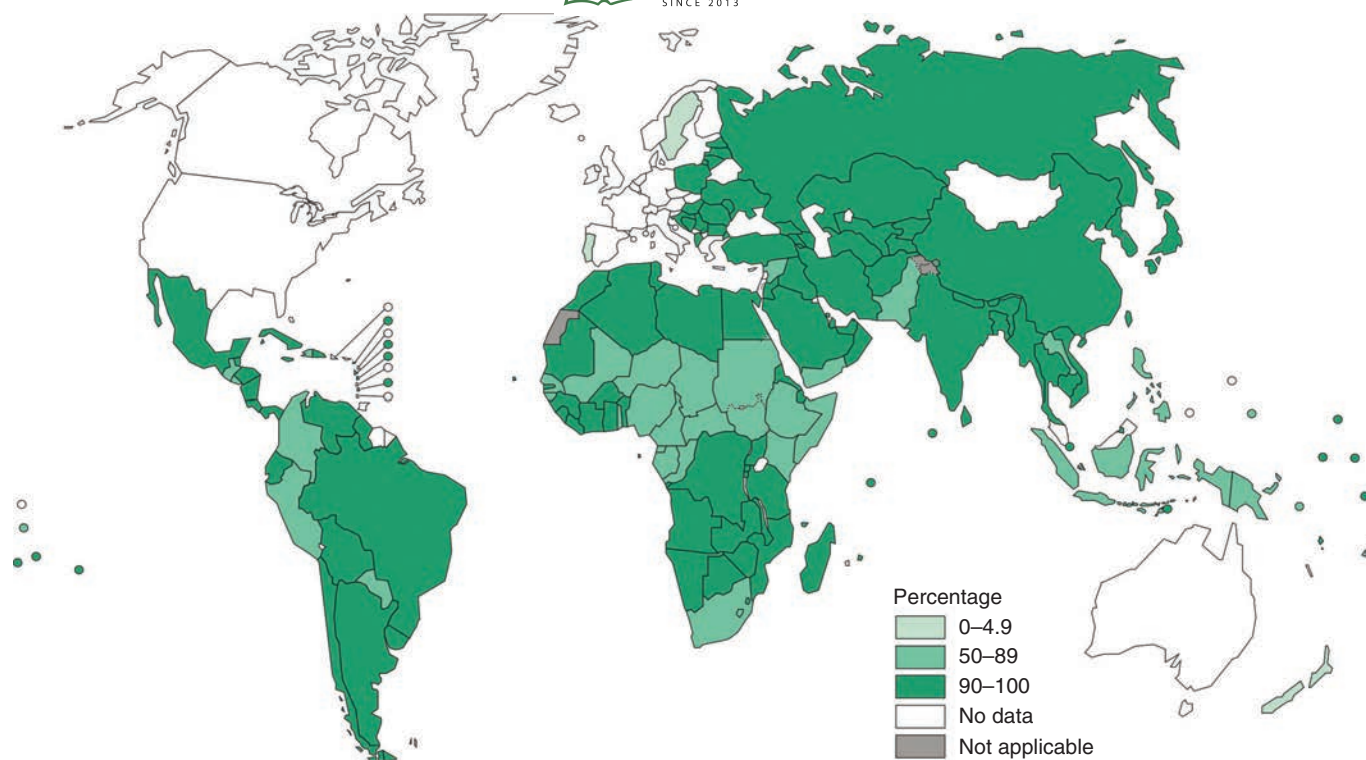


FIGURE 178-13 Coverage of BCG vaccination in 2018. The target population of BCG coverage varies depending on national policies but is typically for the number of live births in the year of reporting. (See disclaimer in Fig. 178-2. Reproduced with permission from *Global Tuberculosis Report 2019*. Geneva, World Health Organization; 2019.)

or lysates, viral vector vaccines, and adjuvant recombinant protein vaccines. Several challenges must be faced in the development of a TB vaccine. For instance, the lack of predictive animal models and protection correlates renders trials long and expensive. Furthermore, the decision about whether a candidate vaccine should be developed for prevention of infection (preexposure) or prevention of reactivation (postexposure) without an exact understanding of its precise mechanism of action is complex. Therefore, introduction of a new vaccine on a large scale is not likely in the near future. This step will require an intensified and much larger investment in research and development.

■ TB PREVENTIVE TREATMENT (TPT)

It is estimated that 1.7 billion people—more than one-quarter of the human population—have been infected with *M. tuberculosis*. Although only a small fraction of these infections will progress toward active disease in a lifetime, new active cases will continue to emerge from this pool of infected individuals. Therefore, TPT (also called *chemoprophylaxis* or *preventive chemotherapy*, and previously referred to as *treatment of latent TB infection*) is a fundamental intervention in TB control and elimination strategies.

Infection can be tested using TST or IGRA, although these tests just measure host immune response to TB antigens. Unfortunately, at present, there is no gold-standard diagnostic test that can confirm true infection (as opposed to immunologic memory of previous exposure) or predict which infected individuals will develop active TB. As a result, decisions to treat infection should include consideration of the risk of progression in an individual. For skin testing, five tuberculin units of polysorbate-stabilized PPD should be injected intradermally into the volar surface of the forearm (i.e., the Mantoux method). Multipuncture tests are not recommended. Reactions are read at 48–72 h as the transverse diameter (in millimeters) of induration; the diameter of erythema is not considered. In some persons, TST reactivity wanes with time but can be recalled by a second skin test administered ≥ 1 week after the first (i.e., two-step testing). For persons periodically undergoing the TST, such as health care workers and individuals admitted to long-term-care institutions, initial two-step testing may preclude subsequent misclassification of those who have boosted reactions as TST converters. The cutoff for a positive TST (and thus for TPT) is related both to the probability that the reaction represents

true infection and to the likelihood that the individual, if truly infected, will develop TB. Table 178-6 suggests possible conventional cutoff by risk group. Thus, positive reactions for persons with HIV infection, recent close contacts of infectious cases, organ transplant recipients,

TABLE 178-6 Tuberculin Reaction Size and Cutoff for Tuberculosis (TB) Preventive Treatment

RISK GROUP	TUBERCULIN REACTION SIZE, mm
HIV-infected persons	≥ 5
Recent contacts of a patient with TB	$\geq 5^a$
Organ transplant recipients	≥ 5
Persons with fibrotic lesions consistent with old TB on chest radiography	≥ 5
Persons who are immunosuppressed—e.g., due to the use of glucocorticoids or tumor necrosis factor α inhibitors	≥ 5
Persons with high-risk medical conditions ^b	≥ 5
Recent immigrants (≤ 5 years) from high-prevalence countries	≥ 10
Injection drug users	≥ 10
Mycobacteriology laboratory personnel; residents and employees of high-risk congregate settings ^c	≥ 10
Children < 5 years of age; children and adolescents exposed to adults in high-risk categories	≥ 10
Low-risk persons ^d	≥ 15

^aTuberculin-negative contacts, especially children, should receive prophylaxis for 2–3 months after contact ends and should then undergo repeat tuberculin skin testing (TST). Those whose results remain negative should discontinue prophylaxis. HIV-infected contacts should receive a full course of treatment regardless of TST results. ^bThese conditions include silicosis and end-stage renal disease managed by hemodialysis. ^cThese settings include correctional facilities, nursing homes, homeless shelters, and hospitals and other health care facilities. ^dExcept for employment purposes where longitudinal TST screening is anticipated, TST is not indicated for these low-risk persons. A decision to treat should be based on individual risk/benefit considerations.

Source: Adapted from Centers for Disease Control and Prevention: TB elimination—treatment options for latent tuberculosis infection (2011). Available at <http://www.cdc.gov/tb/publications/factsheets/testing/skintestresults.pdf>.

previously untreated persons whose chest radiograph shows fibrotic lesions consistent with old TB, and persons receiving drugs that suppress the immune system are defined as an area of induration ≥ 5 mm in diameter. A 10-mm cutoff is used to define positive reactions in most other at-risk persons. For persons with a very low risk of developing TB if infected, a cutoff of 15 mm is used. (Except for employment purposes where longitudinal screening is anticipated, the TST is not indicated for these low-risk persons.) A positive IGRA is based on the manufacturer's recommendations. Good clinical practice requires that, in addition to test results, epidemiologic and clinical factors also guide the decision to implement TPT and that active TB be definitively excluded before the initiation of a prophylactic regimen. The WHO recommends systematic testing for infection and TPT for the following groups at high risk of progression from infection to disease or of exposure and infection: adults, adolescents and children older than 12 months living with HIV; infants with HIV aged <12 months who are contacts of persons with TB; all household contacts of patients with infectious pulmonary TB including children <5 years of age; patients with silicosis, patients starting anti-TNF treatment, patients on dialysis, and patients preparing for organ or hematologic transplantation. In addition, testing and TPT may be considered for persons living or working in at-risk institutional or crowded settings, such as prisoners, health care workers, recent immigrants from high-TB-burden countries, and homeless people who use drugs.

Some TST- and IGRA-negative individuals are also candidates for TPT. Once an appropriate clinical evaluation has excluded active TB, infants and children <5 years of age who were in contact with infectious cases should be offered TPT even in the absence of a positive test for TB infection. HIV-infected persons >1 year of age who have been exposed to an infectious TB patient should receive TPT regardless of the TST result. Any HIV-infected candidate for TPT must be screened carefully to exclude active TB, which would necessitate full disease treatment. The use of a clinical algorithm based on four signs/symptoms (current cough, fever, weight loss, and night sweats) helps to decide which HIV-infected person can start TPT. The absence of all four symptoms tends to exclude active TB in people living with HIV. The presence of one of these four manifestations, on the other hand, warrants further investigation for active TB before TPT is started. Although a test of TB infection is prudent before starting TPT, this test is not an absolute requirement—given the logistical challenges—among contacts aged <5 years and people living with HIV in high-TB-incidence and low-resource settings.

Among people living with HIV and receiving ART, conversion of the TST from negative to positive can occur during the first few months of TPT. Conversions (from negative to positive) and reversions (from positive to negative) are more common with IGRAs than with TSTs among serially tested health care workers in the United States.

TPT in selected persons at risk aims at preventing active disease and, in the absence of an immunizing vaccine, is a critical component of TB elimination strategies. Potential candidates for TPT are listed in Table 178-6. This intervention is based on the results of a large number of randomized, placebo-controlled clinical trials demonstrating that a 6- to 9-month course of isoniazid reduces the risk of active TB in infected people by up to 90%. Analysis of available data indicated that the optimal duration of treatment with this drug was ~ 9 months. In the absence of reinfection, the protective effect is believed to be lifelong. Clinical trials have shown that isoniazid reduces rates of TB among TST-positive persons with HIV infection. Studies in HIV-infected patients have also demonstrated the effectiveness of shorter TPT regimens containing a rifamycin. Several TPT regimens (Table 178-7) can be used. The most widely used has been that based on isoniazid alone at a daily dose of 5 mg/kg (up to 300 mg/d) for 9 months. On the basis of cost-benefit analyses and concerns about feasibility, a 6-month period of treatment at the same dose is considered adequate by the WHO. An alternative regimen for adults is 4 months of daily rifampin, which should also be effective against isoniazid-resistant strains. A 3-month regimen of daily isoniazid and rifampin is used in some countries (e.g., the United Kingdom) for both adults and children who are known not to have HIV infection. A previously recommended 2-month regimen of rifampin and pyrazinamide has been associated with serious or even

TABLE 178-7 Recommended Regimens and Drug Dosages for Tuberculosis Preventive Treatment^a

REGIMEN	DOSE	ADVERSE EVENTS
Isoniazid alone for 6 or 9 months	Adults: 5 mg/kg (max, 300 mg) per day Children <10 years of age: 10 mg/kg per day (range, 7–15 mg)	Drug-induced liver injury, nausea, vomiting, abdominal pain, skin rash, peripheral neuropathy, dizziness, drowsiness, seizure
Rifampin alone for 4 months	Adults: 10 mg/kg per day Children <10 years of age: 15 mg/kg (range, 10–20 mg) per day	Flu-like syndrome, skin rash, drug-induced liver injury, anorexia, nausea, abdominal pain, neutropenia, thrombocytopenia, renal reactions (e.g., acute tubular necrosis and interstitial nephritis)
Isoniazid plus rifampin for 3 months	As above	As above
Rifapentine plus isoniazid for 3 months	Adults and children: Isoniazid: 15 mg/kg (900 mg) weekly Rifapentine: 15–30 mg/kg (900 mg) weekly	Hypersensitivity reactions, petechial skin rash, drug-induced liver injury Anorexia, nausea, abdominal pain Hypotensive reactions
Rifapentine plus isoniazid for 1 month	Age >13 years only: Isoniazid 300 mg and rifapentine 600 mg daily (28 doses)	Essentially like those of isoniazid alone with neutropenia more common and elevation in liver enzyme levels and neuropathy less common

^aSee text for full description of evidence on and limitations of these regimens.

Source: Reproduced with permission from World Health Organization.

fatal hepatotoxicity and is not recommended. The rifampin-containing regimens should be considered for persons who are likely to have been infected with an isoniazid-resistant strain. A clinical trial showed that a regimen of isoniazid (900 mg) and rifapentine (900 mg), given once weekly for 12 weeks, is as effective as the standard 9-month isoniazid regimen. This regimen was associated with higher rates of treatment completion (82% vs 69%) and less hepatotoxicity (0.4% vs 2.7%) than isoniazid alone, although the rate of permanent discontinuation due to an adverse event was higher (4.9% vs 3.7%).

Currently, the isoniazid-rifapentine regimen is not recommended for children <2 years of age or pregnant women. Recently, an open-label, randomized, phase 3 noninferiority trial has shown that among HIV-infected persons, a 1-month regimen of daily rifapentine plus isoniazid was noninferior to the 9-month daily isoniazid regimen and ensured a higher treatment completion. As a result, the WHO has included a 1-month regimen composed of daily isoniazid (300 mg) and rifapentine (600 mg) among the available options for people aged 13 years or more. Rifampin and rifapentine are contraindicated in HIV-infected individuals receiving protease inhibitors, most nonnucleoside reverse transcriptase inhibitors (e.g., nevirapine), and, for those with chronic hepatitis B, tenofovir alafenamide. However, efavirenz and tenofovir disoproxil can be used for simultaneous administration with a rifamycin without dose adjustment. However, the dose of the integrase inhibitor dolutegravir needs to be increased to 50 mg twice daily when given together with rifampin, a dose that is usually well tolerated and gives equivalent efficacy in viral suppression and recovery of CD4⁺ cell count compared with efavirenz. Administration of rifapentine with raltegravir was found to be safe and well tolerated. A recent phase 1/2 trial of the 3-month regimen isoniazid plus rifapentine and dolutegravir in adults with HIV reported good tolerance and viral load suppression, no adverse events of grade >3 , and did not indicate that rifapentine reduced dolutegravir levels sufficiently to require dose adjustment. Clinical trials to assess the efficacy of long-term isoniazid administration (i.e., for at least 3 years) among people living with HIV in high-TB-transmission settings have shown that this regimen can be more effective than 9 months of isoniazid and is therefore

recommended under those circumstances. Studies looking at whether briefer treatment with rifapentine-based regimens could achieve similar efficacies have been undertaken. Isoniazid should not be given to persons with active liver disease. All isoniazid recipients at increased risk of hepatotoxicity (e.g., those abusing alcohol daily and those with a history of liver disease) should undergo baseline and then monthly assessment of liver function; they should be carefully educated about hepatitis and instructed to discontinue use of the drug immediately should any symptoms develop. Moreover, these patients should be seen and questioned monthly during therapy about adverse reactions and should be given no more than a 1-month supply of drug at each visit. Persons receiving high-dose isoniazid and who are at risk of vitamin B6 (pyridoxine) deficiency, as listed above, should receive pyridoxine to prevent peripheral neuropathy.

TPT among persons likely to have been infected by a multidrug-resistant strain is a challenge because no clinical trial results are available to guide treatment. Close observation for early signs of disease is one option. However, in selected high-risk household contacts of patients with MDR-TB (e.g., children, recipients of immunosuppressive therapy), TPT may be considered on the basis of individualized risk assessment and clinical criteria. In the absence of evidence of efficacy of any regimen, important factors in the decision to treat include intensity of exposure, certainty about a source case, information on the drug resistance pattern of the index case, and potential adverse events. Confirmation of infection with available testing is generally required. Drug selection should be based on the drug susceptibility profile of the index case. One regimen recommended by the WHO is daily levofloxacin (750–1000 mg daily among adults) for 6 months.

It may be more difficult to ensure adherence to TPT than when treating those with active TB. If family members of patients with active TB are being treated, adherence and monitoring may be easier. When feasible, supervised therapy may increase the likelihood of completion. As in active cases, the provision of incentives may also be helpful. Currently, no evidence shows that large-scale use of TPT leads to significant development of drug resistance. However, before TPT begins, it is mandatory to carefully exclude active TB in order to prevent undertreatment and promote development of drug resistance.

■ PRINCIPLES OF TB CONTROL

The highest priority in any TB control program is the prompt detection of cases and the provision of treatment to all TB patients under proper case-management conditions, including DOT and social support. In addition, screening of high-risk groups, including immigrants from high-prevalence countries, migrant workers, prisoners, homeless individuals, substance abusers, and HIV-seropositive persons, is recommended. TST- or IGRA-positive high-risk persons should receive TPT as described above. Contact investigation is an important component of efficient TB control. In the United States and other countries worldwide, a great deal of attention has been given to the transmission of TB (particularly in association with HIV co-infection) in institutional settings such as hospitals, homeless shelters, and prisons. Measures to limit such transmission include respiratory isolation of persons with suspected TB until they are proven to be noninfectious (at least by sputum AFB smear negativity), proper ventilation in rooms of patients with infectious TB, use of ultraviolet irradiation in areas of increased risk of TB transmission, correct use of personal protective equipment, and periodic screening of personnel who may come into contact with known or unsuspected cases of TB. In the past, radiographic surveys, especially those conducted with portable equipment and miniature films, were advocated for case finding. Today, however, the prevalence of TB in industrialized countries is sufficiently low that “mass miniature radiography” is not cost effective.

In high-prevalence countries, most TB control programs have made remarkable progress in reducing morbidity and mortality since the mid-1990s by adopting and implementing the standards and strategies internationally promoted by the WHO. Between 2000 and 2018, more than 60 million lives are estimated to have been saved. The essential elements of good TB care and control were established in the mid-1990s and consist of well-defined interventions that were the basis

of the “DOTS strategy”: early detection of cases and bacteriologic confirmation of the diagnosis; administration of standardized short-course chemotherapy, with direct supervision to ensure adherence to treatment and the provision of social support to patients; availability of drugs of proven quality, with an effective supply and management system; and a monitoring and evaluation system, including assessment of treatment outcomes—e.g., cure, completion of treatment without bacteriologic proof of cure, death, treatment failure, and default—in all cases registered and notified as well as measurement of the impact of control methods on classical TB indicators such as mortality, incidence, prevalence, and drug resistance. In 2006, the WHO indicated that, besides pursuing these essential elements that remain the fundamental components of any control strategy, additional steps had to be undertaken in order to reach international TB control targets. These steps included addressing HIV-associated TB and MDR-TB with additional measures; operating in harmony with general health services; engaging all care providers beyond the public providers; empowering people with TB and their communities; and enabling and promoting research. Evidence-based International Standards for Tuberculosis Care—focused on diagnosis, treatment, and public health responsibilities—were introduced for wide adoption by medical and professional societies, academic institutions, and all practitioners worldwide.

Care and control of HIV-associated TB are particularly challenging in poor countries because existing interventions require collaboration between HIV/AIDS and TB programs as well as standard services. TB programs must test every patient for HIV in order to provide access to trimethoprim-sulfamethoxazole prophylaxis against common infections and ART. HIV/AIDS programs must regularly screen persons living with HIV/AIDS for active TB, provide TPT, and ensure infection control in settings where people living with HIV congregate.

Early and active case detection is considered an important intervention not only among persons living with HIV/AIDS but also among other vulnerable populations, as it reduces transmission in a community and provides early effective care. Additional measures are indicated for the management of MDR-TB, RR-TB, and other forms of drug-resistant TB; they include upgrades of laboratory capacity to perform rapid DST and ensure surveillance of drug resistance; availability of drug regimens that are recommended for RR/MDR-TB, with assured quality of drugs; and infection control measures in all settings where patients with drug-resistant forms of TB may congregate. In the new era of the United Nations Sustainable Development Goals (2016–2030), the approach to TB control and care needs to evolve further and become multisectoral and more holistic. Engagement beyond dedicated programs and even the health sector is now essential. Therefore, the new “End TB” strategy promoted by the WHO since 2016 builds on three pillars and relies on increased investments and efforts by all governments, their national programs, and a multitude of partners within and beyond the health sector: (1) integrated, patient-centered care and prevention; (2) bold policies and supportive systems; and (3) intensified research and innovation. The first pillar incorporates all technological innovations, such as early diagnostic approaches (including universal DST and systematic screening of identified, setting-specific, high-risk groups); well-designed treatment regimens for all forms of TB; proper management of HIV-associated TB and other comorbidities; and preventive treatment of persons at high risk. The second pillar is fundamental and is normally beyond the scope of dedicated programs, relying on policies forged by the highest-level health and governmental authorities: availability of adequate human and financial resources; engagement of civil organizations and all relevant public and private providers to pursue proper care for all patients and prevention for all people at risk; a policy of universal health coverage (which, together with social protection, implies avoidance of catastrophic expenditures caused by TB among the poorest); regulatory frameworks for case notifications, vital registration, quality and rational use of medicines, and infection control; social protection mechanisms as part of poverty alleviation strategies; and promotion of interventions against the broader determinants of TB. Finally, the third pillar of the new strategy emphasizes intensification of research and development on new tools and interventions as well as optimal

1382 implementation and rapid adoption of new tools in endemic countries. Besides specific clinical care and control interventions as described in this chapter, elimination of TB in a society ultimately will require control and mitigation of the multitude of direct risk factors (e.g., HIV infection, smoking, alcohol abuse, diabetes) and socioeconomic determinants (e.g., extreme poverty, inadequate living conditions and poor housing, malnutrition, indoor air pollution) with clearly implemented policies within the health sector and other sectors linked to human development and welfare.

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that accompanies leprosy has left marks on society that have endured long after the disease's disappearance in many countries. In everyday language, leprosy has become a metaphor for a horrible condition that warrants social exclusion. Leprosy is a neglected disease and is often thought no longer to exist. However, 202,185 new cases from 150 countries were reported in 2019. A general lack of awareness among both the public and medical practitioners often delays diagnosis and treatment and thus results in irreversible impairments. Early diagnosis and treatment of leprosy and leprosy reactions can cure the disease and prevent most chronic complications.

ETIOLOGY

M. leprae is an obligate, intracellular, acid-fast staining, rod-shaped bacterium, measuring 1–8 μm in length and 0.3 μm in diameter. *M. leprae* mostly appears irregularly stained and fragmented or granular, in which case the organism is usually considered to be dead. The few bacteria that are brightly and uniformly stained are thought to be solid, viable bacilli. The morphologic index is a measure of uniformly stained solid bacilli on slit-skin smear examination and is calculated as the percentage of viable bacilli among the total number of bacilli counted under oil-immersion microscopy. On slit-skin smear examination at the lepromatous end of the disease spectrum, *M. leprae* is predominantly found in clumps or globi within macrophages (*lepra cells*). Inside these cells, *M. leprae* multiplies in unrestricted fashion, and hundreds of bacilli may be present; the organisms are arranged in parallel arrays placed side by side as a result of the presence of surface lipids (*glial substances*). The bacteriologic index is a logarithmic-scaled measure of the density of bacilli of all forms found in the dermis upon slit-skin smear examination, varying from 0 to 6+ (with or without globi) from the tuberculoid to the lepromatous end of the disease spectrum. The bacteriological index falls an average of 1 log unit per year with multidrug therapy. *M. leprae* infects mainly macrophages and Schwann cells. It has never been grown in artificial media. Reproduction occurs by binary fission, and the organism grows slowly (over 12–14 days) in the footpads of mice. The temperature required for survival and proliferation—between 27°C and 30°C—explains the greater impact of the disease on surface areas such as the skin, peripheral nerves, testicles, and upper airways, with less inner visceral involvement. *M. leprae* remains viable for 9 days in the environment.

Ultrastructural Characteristics of *M. leprae* Electron microscopy reveals that *M. leprae* has a cytoplasm, plasma membrane, cell wall, and capsule. The cytoplasm contains structures common in gram-positive microorganisms. The plasma membrane has a permeable lipid bilayer containing interacting proteins—the protein surface antigens. Similar to that of other mycobacteria, *M. leprae*'s cell wall, which is attached to the plasma membrane, is composed of peptidoglycans bound to branched-chain polysaccharides; these peptidoglycans are arabinogalactans, which support mycolic acids, and lipoarabinomannan (LAM). The capsule—the outermost structure—contains lipids, particularly phthiocerol dimycocerosate and phenolic glycolipid (PGL-1), which has a trisaccharide bound to lipid by a molecule of phenol. Because this trisaccharide is antigenically specific for *M. leprae*, its detection is helpful in serologic diagnosis of leprosy.

Genome of *M. leprae* Comparative analysis of the genomics of single-nucleotide polymorphisms indicates that four distinct strains of *M. leprae* originated in East Africa or Central Asia. A mutation spread to Europe and subsequently underwent two separate mutations that were then followed by spread to West Africa and the Americas. The genome of *M. leprae* is circular. Its estimated molecular mass is 2.2×10^9 Da, with 3,268,203 base pairs and a guanine-plus-cytosine content of 57.8%.

Culture Difficulties Compared to the genome of *Mycobacterium tuberculosis*, that of *M. leprae* underwent reductive evolution, resulting in a smaller genome rich in inactive or entirely deleted genes. This reductive evolution, gene decay, and genome downsizing all may explain the unusually long generation time and may account for the inability to culture the leprosy bacillus in artificial media. As a result,

179 Leprosy

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Leprosy, also referred to as Hansen's disease, is a chronic infectious disease caused by *Mycobacterium leprae*. The clinical manifestations are largely confined to the skin, peripheral nervous system, eyes, and upper respiratory tract. The differing immune responses to *M. leprae* result in a spectrum of disease ranging from tuberculoid to lepromatous leprosy. *M. leprae* has a predilection for peripheral nerves, and immunologically mediated reactional states can cause nerve damage to the face, arms, and legs; this damage often results in disability, which in turn can lead to stigma and social exclusion. The physical disfigurement

181 Antimycobacterial Agents

Divya Reddy, Sebastian G. Kurz,
Max R. O'Donnell

Agents used for the treatment of mycobacterial infections, including tuberculosis (TB), leprosy, and infections due to nontuberculous mycobacteria (NTM), are administered in multiple-drug regimens for prolonged courses. Currently, >160 species of mycobacteria have been identified, the majority of which do not cause disease in humans. While the incidence of disease caused by *Mycobacterium tuberculosis* has been declining in the United States, TB remains a leading cause of morbidity and mortality in low- and middle-income countries—for example, in sub-Saharan Africa and Asia, where the HIV epidemic rages. Well-organized infrastructure for early diagnosis, treatment of TB infection and disease, and development of effective drug regimens and vaccines remain vital to the global strategies for TB control (Chaps. 472 and 474). Infections with NTM have gained in clinical prominence in the United States and other developed countries. These largely environmental organisms often establish infection in immunocompromised patients or in persons with structural lung disease.

TUBERCULOSIS

■ GENERAL PRINCIPLES

The earliest recorded human case of TB dates back 9000 years. Early treatment modalities, such as bloodletting, were replaced by the sanatorium movement in the late nineteenth century, which focused on fresh air, nutrition, and bed rest to treat consumptive patients and came with the benefit of isolating infected individuals. The discovery of streptomycin in 1943 launched the era of antibiotic treatment for TB. Over subsequent decades, the discovery of additional agents and the use of multiple-drug regimens allowed progressive shortening of the treatment course from years to as little as 6 months for drug-susceptible TB. Latent TB infection (LTBI) and active TB disease are diagnosed by history, physical examination, radiographic imaging, tuberculin skin test, interferon- γ release assays, acid-fast staining, mycobacterial cultures, and/or new molecular diagnostics. LTBI is treated with isoniazid plus rifapentine (weekly for 3 months), rifampin (daily for 4 months), isoniazid plus rifampin (daily for 3 months), or isoniazid (optimally daily or twice weekly for 6–9 months) (Table 181-1). The 3-month, weekly regimen of isoniazid with rifapentine is currently the regimen of choice in children >2 years of age and in all adults including HIV-positive individuals. The regimen is not recommended for pregnant women and for persons with hypersensitivity reactions to isoniazid or rifampin. Shorter duration rifamycin-based regimens (rifampin alone for 4 months or for 3 months in combination with isoniazid) are currently preferred for the treatment of LTBI over isoniazid for 6–9 months in adults and children due to their effectiveness, safety, and tolerability. Caution is advised in

HIV-positive individuals due to potential for drug interactions, lack of definitive data on effectiveness, and the possibility of subclinical TB disease that could facilitate the development of rifampin resistance.

Completion rates of a self-administered, once-weekly regimen of isoniazid plus rifapentine for 3 months with monthly monitoring were found to be noninferior to those seen with directly observed therapy (DOT) in the United States, and thus, this regimen is considered an acceptable strategy for treating LTBI in countries with a focus on secondary prevention of TB disease. Recently, a 1-month daily regimen of rifapentine and isoniazid in HIV-positive individuals was found to be noninferior to 9 months of isoniazid; this regimen will be included in the new World Health Organization (WHO) LTBI treatment guidelines.

For active or suspected TB disease, clinical factors, including HIV co-infection, symptom duration, radiographic appearance, and public health concerns about TB transmission, drive diagnostic testing and treatment initiation. Confirmation of active TB relies on detection of *M. tuberculosis* via culture or molecular testing. A combination of drugs is used for the treatment of TB disease (Table 181-2). For drug-susceptible disease, a standardized regimen is used with an intensive phase consisting of four drugs—isoniazid, rifampin, pyrazinamide, and ethambutol—given for 2 months, which is followed by a continuation phase of isoniazid and rifampin for 4 months, for a total treatment duration of 6 months. U.S. guidelines recommend extension of the continuation phase to 7 months (for a total treatment duration of 9 months) for patients with cavitory disease; if the 2-month course of pyrazinamide is not completed; or if sputum cultures remain positive beyond 2 months of treatment (delayed culture conversion), which also warrants evaluation for development of drug resistance.

Treatment of TB in patients co-infected with HIV poses significant challenges, but some progress is being made. To improve survival, current recommendations include initiation of antiretroviral therapy (ART) in HIV patients co-infected with *M. tuberculosis* within 2 weeks of the initiation of treatment for TB (except TB meningitis) if the CD4+ T-cell count is $\leq 50/\mu\text{L}$ and by 8–12 weeks of TB treatment initiation if the CD4+ T-cell count is $\geq 50/\mu\text{L}$. Interactions of rifampin with protease inhibitors or nonnucleoside reverse transcriptase inhibitors can be significant and require close monitoring and dose adjustments. Reassuringly, a recent study comparing the safety and efficacy of rifampin for 4 months in patients with LTBI showed that it was as effective as isoniazid for 9 months and was also well tolerated and safe for treatment in persons living with HIV. Rifabutin is an alternative drug of choice in HIV patients co-infected with *M. tuberculosis*, as it is a less potent cytochrome P3A inhibitor than rifampin. The TB immune reconstitution inflammatory syndrome (IRIS) may appear as early as 1 week after initiation of ART and manifests as paradoxical worsening or unmasking of existing TB infection. Conservative management consists of continued administration of ART and TB medications. However, severe or debilitating IRIS has been treated in reported case series with varying doses of glucocorticoids. A randomized, double-blind, placebo-controlled trial showed that a 4-week course of prednisone significantly reduced need for hospitalization and hastened symptom improvement and quality of life in TB IRIS. Intermittent antimycobacterial therapy

TABLE 181-1 Regimens for the Treatment of Latent Tuberculosis Infection in Adults

REGIMEN	SCHEDULE	DURATION	COMMENTS
Isoniazid plus rifapentine	900 mg (15 mg/kg) weekly plus 900 mg (for weight >50 kg) weekly	3 months	Directly observed therapy is recommended for once-weekly treatment in HIV-positive and -negative individuals. This regimen may be supplemented with pyridoxine (25–50 mg/d).
Rifampin	600 mg/d (10 mg/kg)	4 months	Recommended in HIV-negative individuals and in children. Data on effectiveness in HIV-positive patients are unavailable.
Isoniazid plus rifampin	300 mg/d (5 mg/kg) plus 600 mg/d (10 mg/kg)	3 months	Risk of hepatotoxicity may be higher with the combination regimen compared to that of the individual drugs.
Isoniazid	300 mg/d (5 mg/kg) Alternative: 900 mg twice weekly (15 mg/kg)	6–9 months (6 months acceptable)	Supplement with pyridoxine (25–50 mg daily) 6 months' duration strongly recommended for HIV-negative patients and conditional for HIV-positive patients. 9 months may be more effective but with higher risk of hepatic toxicity. Twice-weekly regimens require directly observed therapy.

Source: Sterling TR et al. Guidelines for the treatment of latent tuberculosis infection: Recommendations from the National Tuberculosis Controllers Association and CDC, 2020. MMWR Recomm Rep 69(No. RR-1):1, 2020.

TABLE 181-2 Simplified Approach to Treatment of Active Tuberculosis (TB) in Adults

CULTURE RESULTS	INTENSIVE PHASE	CONTINUATION PHASE	EXTENSION OF TOTAL TREATMENT
Culture-positive, drug-susceptible	HRZE for 2 months, daily ^a or 3 times per week (with dose adjustment)	HR for 4 months, daily or 5 days per week or HR for 4 months, 3 times per week ^b (with dose adjustment)	Continuation phase extended to 7 months if 2 months of Z is not completed, if the patient is infected with HIV and is not receiving antiretroviral therapy, or if culture conversion is prolonged and/or cavitation is evident on chest radiography (U.S. guidelines) ^c
Culture-negative	HRZE for 2 months	HR for 2 months, daily or 2 or 3 times per week ^d	Continuation phase extended to 4 months if the patient is infected with HIV
Extrapulmonary, drug-susceptible	HRZE for 2 months	HR for 4–7 months, daily or 5 days per week ^e	Continuation phase extended to 10 months in TB meningitis; 7 months recommended by some authorities for bone/joint TB

^aDaily treatment is preferred; however, thrice-weekly therapy in the intensive phase (with or without an initial 2 weeks of daily therapy) may be considered in patients who are not infected with HIV and are at low risk of relapse (i.e., in pulmonary tuberculosis caused by drug-susceptible organisms that, at the start of treatment, is noncavitary and/or smear negative). ^bUse regimen with caution in HIV patients and/or those with cavitory disease, as missed doses can lead to treatment failure, relapse, and acquired drug resistance. ^cCulture conversion is prolonged if it occurs beyond 2 months. ^dTwice-weekly treatment regimens are not recommended in patients infected with HIV and those with cavitory pulmonary disease suspected to be TB. ^eStandard daily 6-month TB treatment regimen is considered to be adequate for most forms of extrapulmonary TB. For TB meningitis, the addition of glucocorticoids is recommended.

Abbreviations: E, ethambutol; H, isoniazid; R, rifampin; Z, pyrazinamide.

Sources: Official American Thoracic Society/Centers for Disease Control and Prevention/Infectious Diseases Society of America: Clinical practice guidelines: Treatment of drug-susceptible tuberculosis. Clin Infect Dis 63:e147, 2016.

in patients infected with HIV and *M. tuberculosis* has been associated with low plasma levels of several key TB drugs and with higher rates of treatment failure or relapse; therefore, intermittent twice-weekly therapy for TB in HIV-co-infected individuals is not recommended.

Adherence to medications is critical in achieving a cure with antimycobacterial therapy. In addition to DOT by trained staff, either in the clinic or at home, case management interventions such as patient education/counseling, field/home visits, and patient reminders are also recommended to improve treatment adherence. Use of mobile health

technologies, including video DOT, text messaging, and next-generation electronic pillboxes, shows promise in promoting TB adherence. In drug-susceptible TB, monthly dispensing of TB medications is also advised for all patients to allow essential clinical monitoring for hepatotoxicity due to these medications. Clinical monitoring includes at least monthly assessment for symptoms (nausea, vomiting, abdominal discomfort, and unexplained fatigue) and signs (jaundice, dark urine, light stools, diffuse pruritus) of hepatotoxicity, although the latter represent comparatively late manifestations (Table 181-3). The presence

TABLE 181-3 Monitoring and Clinical Management of Tuberculosis (TB) Treatment in Adults^a

DRUG	ASSESSMENT	MANAGEMENT
LTBI Treatment		
With hepatic risk factors ^b , check ALT and bilirubin at baseline. If ALT is $\geq 3 \times$ ULN or total bilirubin is $>2 \times$ ULN, defer treatment and reevaluate.		
Isoniazid	Determine whether hepatic risk factors are present. If so, obtain baseline and periodic ALT and bilirubin values	If ALT is $5 \times$ ULN (or $3 \times$ ULN with symptoms) ^c or if bilirubin reaches jaundice levels (usually $>2 \times$ ULN), interrupt treatment. With normalization, consider an alternative agent.
Rifampin	Same as above	Same as above
TB Treatment		
Check ALT, bilirubin, platelets, creatinine, and hepatitis panel for all patients at baseline. If hepatic risk factors are present, check ALT and bilirubin monthly.		
Isoniazid	If ALT is $>5 \times$ ULN (or $>3 \times$ ULN with hepatitis symptoms) ^c	Obtain history of alcohol consumption and concomitant drug use. In most instances, discontinue H, Z, R, and other hepatotoxic drugs. Consider alternative agents. Obtain viral hepatitis serologies. Rechallenge: With normalization of liver enzymes, R and H may be sequentially reintroduced. With no recurrence of hepatotoxicity, Z is not resumed in many cases. Alternative rechallenge protocols have been used.
Rifampin	If primary elevation is in bilirubin and alkaline phosphatase, most likely due to rifampin	Discontinue R if total bilirubin reaches jaundice levels (usually $>2 \times$ ULN). May try to reintroduce; if not tolerated, may substitute Q.
Ethambutol	Decrease in visual acuity or color vision on monthly screening	Discontinue ethambutol and repeat ocular examination. Peripheral neuropathy may be a precursor of ocular toxicity; if it occurs, consider repeat ocular examination.
Pyrazinamide	If ALT is $>5 \times$ ULN (or $>3 \times$ ULN with symptoms) ^c	Same as for H.
Fluoroquinolone, bedaquiline, delamanid	QTc prolongation is a concern and should be monitored, especially if drugs are used in combination	Asymptomatic QTc prolongation should prompt consideration of stopping known QT-prolonging drugs and/or close monitoring, depending on the clinical situation and degree of prolongation. Symptomatic QTc prolongation (e.g., palpitations or arrhythmias) should prompt discontinuation of drugs.
Linezolid	Visual impairment; monitor for peripheral neuropathy and bone marrow suppression including anemia, thrombocytopenia, and leukopenia	Discontinue linezolid if visual toxicity develops. Rechallenge after complete resolution, especially with a lower dose, is an option. Stop if peripheral neuropathy or bone marrow suppression develops.

^aAll regimens require monthly clinical monitoring. ^bHepatic risk factors: chronic alcohol use, viral hepatitis, preexisting liver disease, pregnancy or ≤ 3 months postpartum, hepatotoxic medications. ^cRelevant manifestations include nausea, vomiting, abdominal pain, jaundice, or unexplained fatigue.

Abbreviations: ALT, alanine aminotransferase; H, isoniazid; LTBI, latent tuberculosis infection; Q, fluoroquinolone; QTc, corrected QT interval; R, rifampin; ULN, upper limit of normal; Z, pyrazinamide.

Sources: JJ Saukkonen et al: An official ATS statement: Hepatotoxicity of antituberculosis therapy. Am J Respir Crit Care Med 174:935, 2006; American Thoracic Society/Centers for Disease Control and Prevention/Infectious Diseases Society of America: Treatment of tuberculosis. Am J Respir Crit Care Med 167:603, 2003; WHO consolidated guidelines on drug-resistant tuberculosis treatment. Geneva: World Health Organization; 2019. License: CC BY-NC-SA 3.0 IGO.

of such symptoms and signs mandates provisional discontinuation of potentially hepatotoxic agents; discontinuation at the onset of hepatitis symptoms reduces the risk of progression to fatal hepatitis. Although biochemical monitoring is not routinely recommended, baseline assessment of liver function is recommended in adults including testing of at least serum alanine aminotransferase (ALT) and total bilirubin levels (Table 181-3). (See Chap. 178 for further details.) For patients with active TB, monthly mycobacterial cultures of sputum are recommended until it is certain that the organisms have been cleared and the patient has responded to therapy or until no sputum is available for culture.

If significant clinical improvement does not occur or the patient's condition deteriorates over the course of therapy, possibilities include treatment failure due to nonadherence, poor medication absorption, or the development of resistance. For patients co-infected with HIV and *M. tuberculosis*, IRIS, which is a diagnosis of exclusion, should also be a consideration. Drug susceptibility testing should be repeated at this point. If resistance is documented or strongly suspected, at least two efficacious drugs to which the isolate is susceptible or which the patient has not already taken should be added to the therapeutic regimen.

Multidrug-resistant tuberculosis (MDR-TB) is defined as disease caused by a strain of *M. tuberculosis* that is resistant to both isoniazid and rifampin—the most efficacious of the first-line TB drugs. The risk of MDR-TB is elevated in patients presenting from geographic areas in which $\geq 5\%$ of incident cases are MDR-TB and in patients previously treated for TB. Treatment regimens for MDR-TB are rapidly evolving, and in 2019, the WHO issued a new classification of second-line agents to treat drug-resistant disease (See Table 178-4). Current WHO recommendations are emphasizing an all-oral bedaquiline-containing regimen with the goal to limit treatment duration to 9–11 months compared to conventional durations of 18–20 months (Table 181-4).

Results from several recent large clinical trials have formed the basis of these recommendations. The “Bangladesh regimen” was the first short-course MDR-TB regimen systematically studied in the STREAM-1 trial and was able to reduce treatment duration to 9–12 months with favorable outcomes in up to 90% of patients. It consists of a seven-drug intensive phase (kanamycin, prothionamide, isoniazid, fluoroquinolone, ethambutol, pyrazinamide, and clofazimine) and a four-drug continuation phase (fluoroquinolone, ethambutol, pyrazinamide, and clofazimine). In 2018, a large meta-analysis, which pooled individual data from >12,000 patients enrolled in 50 trials, assessed the role of individual drugs to treat MDR-TB. This analysis showed an association of significantly better treatment outcomes with

the use of linezolid, bedaquiline, clofazimine, carbapenems, and later generation fluoroquinolones and worse outcomes with kanamycin and capreomycin in these patients. As a result of this analysis, oral drug combinations are now prioritized, while several traditional second-line drugs, including kanamycin and capreomycin, are no longer recommended. A further step toward a shortened all-oral regimen was the Nix-TB study, which showed that a 6-month regimen of bedaquiline, pretomanid, and linezolid (BPaL regimen) for treatment of highly drug-resistant TB was associated with favorable outcomes (absence of clinical or bacteriologic treatment failure or relapse within 6 months of treatment completion) in 89% of patients. While a major breakthrough, caution has been raised regarding higher rate of side effects mostly due to linezolid and lack of a control arm. The BPaL regimen is currently recommended under operational research conditions only.

The shift toward all-oral regimens of shortened duration has been made possible by the introduction of novel drugs, most prominently bedaquiline and pretomanid, as well as the repurposing of existing agents for MDR-TB treatment (e.g., linezolid, clofazimine). High cost and limited access to these new drugs are barriers that need to be addressed to facilitate global adaptation of these new regimens.

■ FIRST-LINE ANTITUBERCULOSIS DRUGS

The following discussion of individual anti-TB agents focuses on treatment of TB in adults, unless otherwise noted. Several agents are being actively investigated during the current remarkable period of drug development for TB treatment.

Isoniazid Isoniazid is a critical drug for treatment of both TB disease and LTBI. Isoniazid has excellent bactericidal activity against both intracellular and extracellular, actively dividing *M. tuberculosis*. This drug is bacteriostatic against slowly dividing organisms. In treatment of LTBI, isoniazid is generally well tolerated, has well-established efficacy, and is inexpensive. In this setting, the drug is taken daily, which is the preferred dosing schedule, or intermittently (i.e., twice weekly) using DOT for 6 months, which has been found to be equivalent to the traditional 9 months in most settings. A weekly isoniazid and rifapentine regimen, administered over 3 months under DOT, has been shown to be noninferior to daily isoniazid given for 9 months and had a higher treatment completion rate than the single-drug regimen. More recent evidence also suggests that completion rates of a self-administered 3-month regimen of weekly isoniazid and rifapentine are noninferior to those seen with DOT in the United States. It is expected that a 1-month daily regimen in combination with rifapentine will be added to new WHO guidelines.

CULTURE RESULTS	INTENSIVE PHASE	CONTINUATION PHASE	EXTENSION OF TOTAL TREATMENT
Resistant to H	Lfx RZE ^b for 6 months	...	Prolonged culture conversion and/or evidence of cavitation on chest radiography
Resistant to HR (MDR) ^c			
WHO short course regimen ^d	Bdq, Lfx or Mfx, Eto, E, Z, Hh, Cfz for 4–6 months	Lfx or Mfx, Cfz, Z, E for 5 months	Prolonged culture conversion, delayed response, and/or evidence of cavitation on chest radiography
WHO extended regimen ^e	At least five effective second-line agents, including all group A and at least one group B, add group C if intolerant to A or B drugs for 5–7 months	4 drugs for a total of 18–20 months or for 15–17 months after culture conversion	Prolonged culture conversion, delayed response, and/or evidence of cavitation on chest radiography

^aDrug-resistant TB treatment regimens should be constructed and care provided in close consultation with experienced drug-resistant TB clinicians. Surgical management should also be considered in appropriate cases. ^bProlonged pyrazinamide duration may be associated with increased risk of liver toxicity. ^cMono-resistance to R is rare and should be treated as MDR. ^dThe WHO short-course regimen is for patients with no prior exposure to second-line drugs and documented fluoroquinolone susceptibility only. Patients with treatment intolerance to antimycobacterial agents, disseminated TB, or pregnancy should be excluded from short-course regimens. It is currently not endorsed by U.S. societies. ^ePatients who do not qualify for WHO short-course regimens should be treated using extended MDR-TB treatment regimens. The construction of extended regimens is guided by the requirement for selection of effective antimycobacterial agents, the need to combine sufficient medicines to maximize relapse-free survival, and the need to minimize toxicity.

Abbreviations: Bdq, bedaquiline; Cfz, clofazimine; E, ethambutol; Eto, ethionamide; H, isoniazid; Hh, high-dose isoniazid; Lfx, levofloxacin; MDR, multidrug resistant; Mfx, moxifloxacin; Pa, pretomanid; Q, fluoroquinolone; R, rifampin; WHO, World Health Organization; Z, pyrazinamide.

Sources: Official American Thoracic Society/Centers for Disease Control and Prevention/Infectious Diseases Society of America: Clinical practice guidelines: Treatment of drug-resistant tuberculosis. *Am J Respir Crit Care Med* 200:e93, 2019; World Health Organization consolidated guidelines on drug-resistant tuberculosis treatment. WHO 2019; Rapid Communication: Key changes to the treatment of drug-resistant tuberculosis, WHO December 2019.

For treatment of TB disease, isoniazid is used in combination with other agents to ensure killing of both actively dividing *M. tuberculosis* and slowly growing “persister” mycobacteria. Unless the organism is resistant, the standard regimen includes isoniazid, rifampin, ethambutol, and pyrazinamide (Table 181-2). Isoniazid is often given together with 25–50 mg of pyridoxine daily to prevent drug-related peripheral neuropathy.

MECHANISM OF ACTION Isoniazid is a prodrug activated by the mycobacterial KatG catalase-peroxidase; isoniazid is coupled with reduced nicotinamide adenine dinucleotide (NADH). The resulting isonicotinic acyl–NADH complex blocks the mycobacterial ketoenoylreductase known as InhA through binding to its substrate and inhibiting fatty acid synthase and ultimately mycolic acid synthesis. Mycolic acids are essential components of the mycobacterial cell wall. KatG activation of isoniazid also results in the release of free radicals that have antimycobacterial activity, including nitric oxide.

The minimal inhibitory concentrations (MICs) of isoniazid for wild-type (untreated) susceptible strains are <0.1 µg/mL for *M. tuberculosis* and 0.5–2 µg/mL for *M. kansasii*.

PHARMACOLOGY Isoniazid is the hydrazide of isonicotinic acid, a small, water-soluble molecule. The usual adult oral daily dose of 300 mg results in peak serum levels of 3–5 µg/mL within 30 min to 2 h after ingestion—well in excess of the MICs for most susceptible strains of *M. tuberculosis*. Both oral and IM preparations of isoniazid reach effective levels in the body, although antacids and high-carbohydrate meals may interfere with oral absorption. Isoniazid diffuses well throughout the body, reaching therapeutic concentrations in body cavities and fluids, with concentrations in cerebrospinal fluid (CSF) comparable to those in serum.

Isoniazid is metabolized in the liver via acetylation by *N*-acetyltransferase 2 (NAT2) and hydrolysis. Both fast- and slow-acetylation phenotypes occur; patients who are “fast acetylators” may have lower serum levels of isoniazid, whereas “slow acetylators” may have higher levels and experience more toxicity. Satisfactory isoniazid levels are attained in the majority of homozygous fast NAT2 acetylators given a dose of 6 mg/kg and in the majority of homozygous slow acetylators given only 3 mg/kg. Genotyping is increasingly being used to characterize isoniazid-related pharmacogenomic responses.

Isoniazid’s interactions with other drugs are due primarily to its inhibition of the cytochrome P450 system. Among the drugs with significant isoniazid interactions are warfarin, carbamazepine, benzodiazepines, acetaminophen, clopidogrel, maraviroc, dronedarone, salmeterol, tamoxifen, eplerenone, and phenytoin.

DOSING The recommended daily dose of isoniazid for the treatment of TB is 5 mg/kg for adults and 10 mg/kg for children (U.S. guidelines recommend 10–15 mg), with a maximal daily dose of 300 mg for both. For intermittent therapy in adults (usually twice per week), the dose is 15 mg/kg, with a maximal daily dose of 900 mg. Isoniazid does not require dosage adjustment in patients with renal disease. When the 12-dose, 3-month weekly LTBI regimen is used, the dose of isoniazid is 15 mg/kg, with a maximal dose of 900 mg, and the drug is co-administered with rifapentine. The novel 1-month regimen uses isoniazid 300 mg in conjunction with rifapentine for people aged >13 years without weight adjustment.

RESISTANCE Although isoniazid, along with rifampin, is the mainstay of TB treatment regimens, ~7% of clinical *M. tuberculosis* isolates in the United States are resistant. Rates of primary isoniazid resistance among untreated patients are significantly higher in many populations born outside the United States. Five separate pathways for isoniazid resistance have been elucidated. Most strains have amino acid changes in either the catalase-peroxidase gene (*katG*) or the mycobacterial ketoenoylreductase gene (*inhA*). Less frequently, alterations in *kasA*, the gene for an enzyme involved in mycolic acid elongation, and loss of NADH dehydrogenase 2 activity confer isoniazid resistance. In 20–30% of isoniazid-resistant *M. tuberculosis* isolates, increased expression of efflux pump genes, such as *efpA*, *mmpL7*, *mmr*, *p55*, and the Tap-like gene *Rv1258c*, has been implicated as the underlying mechanism of resistance.

ADVERSE EFFECTS Although isoniazid is generally well tolerated, drug-induced liver injury and peripheral neuropathy are significant adverse effects associated with this agent. Isoniazid may cause asymptomatic transient elevation of aminotransferase levels (often termed *hepatic adaptation*) in up to 20% of recipients. Other adverse reactions include rash (2%), fever (1.2%), anemia, acne, arthritic symptoms, a systemic lupus erythematosus-like syndrome, optic atrophy, seizures, and psychiatric symptoms. Symptomatic hepatitis occurs in <0.1% of persons treated with isoniazid alone for LTBI, and fulminant hepatitis with hepatic failure occurs in <0.01%. Isoniazid-associated hepatitis is idiosyncratic, but its incidence increases with age, with daily alcohol consumption, and in women who are within 3 months postpartum.

In patients who have liver disorders or HIV infection, who are pregnant or in the 3-month postpartum period, who have a history of liver disease (e.g., hepatitis B or C, alcoholic hepatitis, or cirrhosis), who use alcohol regularly, who have multiple medical problems, or who have other risk factors for chronic liver disease, the risks and benefits of isoniazid treatment for LTBI should be weighed. If treatment is undertaken, these patients should have serum concentrations of ALT determined at baseline. Routine baseline hepatic ALT testing based solely on an age of >35 years is optional and depends on individual concerns. Monthly biochemical monitoring during isoniazid treatment is indicated for patients whose baseline liver function tests yield abnormal results and for persons at risk for hepatic disease, including the groups just mentioned. Guidelines recommend that isoniazid be discontinued in the presence of hepatitis symptoms or jaundice and an ALT or AST level three times the upper limit of normal or in the absence of symptoms with an ALT or AST level five times the upper limit of normal (Table 181-3).

Peripheral neuropathy associated with isoniazid occurs in up to 2% of patients given 5 mg/kg. Isoniazid appears to interfere with pyridoxine (vitamin B₆) metabolism. The risk of isoniazid-related neurotoxicity is greatest for patients with preexisting disorders that also pose a risk of neuropathy, such as HIV infection; for those with diabetes mellitus, alcohol abuse, or malnutrition; and for those simultaneously receiving other potentially neuropathic medications, such as stavudine. These patients should be given prophylactic pyridoxine (25–50 mg/d).

Rifampin Rifampin is a semisynthetic derivative of *Amycolatopsis rifamycinica* (formerly known as *Streptomyces mediterranei*). The most active antimycobacterial agent available, rifampin is the keystone of first-line treatment for TB. Introduced in 1968, this drug eventually permitted dramatic shortening of the TB treatment course. Rifampin has both sterilizing and bactericidal activity against dividing and nondividing *M. tuberculosis*. The drug is also active against an array of other organisms, including some gram-positive and gram-negative bacteria, *Legionella*, *M. kansasii*, and *Mycobacterium marinum*.

MECHANISM OF ACTION Rifampin exerts both intracellular and extracellular bactericidal activities. Like other rifamycins, rifampin specifically binds to and inhibits mycobacterial DNA-dependent RNA polymerase, blocking RNA synthesis. Susceptible strains of *M. tuberculosis* as well as *M. kansasii* and *M. marinum* are inhibited by rifampin concentrations of 1 µg/mL.

PHARMACOLOGY Rifampin is a fat-soluble, complex macrocyclic molecule readily absorbed after oral administration. Serum levels of 10–20 µg/mL are achieved 2.5 h after the usual adult oral dose of 10 mg/kg (given without food). Rifampin has a half-life of 1.5–5 h. The drug distributes well throughout most body tissues, including CSF. Rifampin turns body fluids such as urine, saliva, sputum, and tears a reddish-orange color—an effect that offers a simple means of assessing patients’ adherence to this medication. Rifampin is excreted primarily through the bile and enters the enterohepatic circulation; <30% of a dose is renally excreted.

As a potent inducer of the hepatic cytochrome P450 system, rifampin can decrease the half-life of some drugs, such as digoxin, warfarin, phenytoin, prednisone, cyclosporine, methadone, oral contraceptives, clarithromycin, azole antifungal agents, quinidine, antiretroviral

protease inhibitors, and nonnucleoside reverse transcriptase inhibitors. The Centers for Disease Control and Prevention (CDC) has issued guidelines for the management of drug interactions during treatment of HIV and *M. tuberculosis* co-infection (www.cdc.gov/tb/).

DOSING The daily dosage of rifampin is 10 mg/kg for adults and 10–20 mg/kg for children, with a maximum of 600 mg/d for both. The drug is given once daily, twice weekly, or three times weekly. No adjustments of dose or frequency are necessary in patients with renal insufficiency.

RESISTANCE Resistance to rifampin in *M. tuberculosis*, *M. leprae*, and other organisms is the consequence of spontaneous, mostly missense point mutations in a core region of the bacterial gene coding for the β subunit of RNA polymerase (*rpoB*). RNA polymerase altered in this manner is no longer subject to inhibition by rifampin. Most rapidly and slowly growing NTM harbor intrinsic resistance to rifampin, for which the mechanism has yet to be determined.

ADVERSE EFFECTS Adverse events associated with rifampin are infrequent and generally mild. Hepatotoxicity due to rifampin alone is uncommon in the absence of preexisting liver disease and often consists of isolated hyperbilirubinemia rather than aminotransferase elevation. Other adverse reactions include rash, pruritus, gastrointestinal symptoms, and pancytopenia. Rarely, a hypersensitivity reaction may occur with intermittent therapy, manifesting as fever, chills, malaise, rash, and—in some instances—renal and hepatic failure.

Pyrazinamide A nicotinamide analog, pyrazinamide is an important bactericidal drug used in the initial phase of TB treatment. Its administration for the first 2 months of therapy with rifampin and isoniazid allows treatment duration to be shortened from 9 to 6 months and decreases rates of relapse.

MECHANISM OF ACTION Pyrazinamide's antimycobacterial activity is essentially limited to *M. tuberculosis*. The drug is more active against slowly replicating organisms than against actively replicating organisms. Pyrazinamide is a prodrug that is converted by the mycobacterial pyrimidase to the active form, pyrazinoic acid (POA). This agent is active only in acidic environments (pH <6.0), as are found within phagocytes or granulomas. The exact mechanism of action of POA is unclear, but fatty acid synthetase I may be the primary target in *M. tuberculosis*. Susceptible strains of *M. tuberculosis* are inhibited by pyrazinamide concentrations of 16–50 μ g/mL at pH 5.5.

PHARMACOLOGY AND DOSING Pyrazinamide is well absorbed after oral administration, with peak serum concentrations of 20–60 μ g/mL at 1–2 h after ingestion of the recommended adult daily dose of 15–30 mg/kg (maximum, 2 g/d). It distributes well to various body compartments, including CSF, and is an important component of treatment for tuberculous meningitis. The serum half-life of the drug is 9–11 h with normal renal and hepatic function. Pyrazinamide is metabolized in the liver to POA, 5-hydroxypyrazinamide, and 5-hydroxy-POA. A high proportion of pyrazinamide and its metabolites (~70%) is excreted in the urine. The dosage must be adjusted according to the level of renal function in patients with reduced creatinine clearance.

ADVERSE EFFECTS At the higher dosages used previously, hepatotoxicity was seen in as many as 15% of patients treated with pyrazinamide. However, at the currently recommended dosages, hepatotoxicity now occurs less commonly when this drug is administered with isoniazid and rifampin during the treatment of TB. Older age, active liver disease, HIV infection, and low albumin levels may increase the risk of hepatotoxicity. The use of pyrazinamide with rifampin for the treatment of LTBI is no longer recommended because of unacceptable rates of hepatotoxicity and death in this setting. Hyperuricemia is a common adverse effect of pyrazinamide therapy that usually can be managed conservatively. Clinical gout is rare.

Although pyrazinamide is recommended by international TB organizations for routine use in pregnancy, it is not recommended in the United States because of inadequate teratogenicity data.

RESISTANCE The basis of pyrazinamide resistance in *M. tuberculosis* is a mutation in the *pncA* gene coding for pyrazinamidase, the enzyme that converts the prodrug to active POA. Resistance to pyrazinamide is associated with loss of pyrazinamidase activity, which prevents conversion of pyrazinamide to POA. Of pyrazinamide-resistant *M. tuberculosis* isolates, 72–98% have mutations in *pncA*. Conventional methods of testing for susceptibility to pyrazinamide may produce both false-negative and false-positive results because the high-acidity environment required for the drug's activation also inhibits the growth of *M. tuberculosis*. There is some controversy as to the clinical significance of in vitro pyrazinamide resistance.

Ethambutol Ethambutol is a bacteriostatic antimycobacterial agent first synthesized in 1961. A component of the standard first-line regimen, ethambutol provides synergy with the other drugs in the regimen and is generally well tolerated. Susceptible species include *M. tuberculosis*, *M. marinum*, *M. kansasii*, and organisms of the *Mycobacterium avium* complex (MAC); however, among first-line drugs, ethambutol is the least potent against *M. tuberculosis*. This agent is also used in combination with other agents in the continuation phase of treatment when patients cannot tolerate isoniazid or rifampin or are infected with organisms resistant to either of the latter drugs.

MECHANISM OF ACTION Ethambutol is bacteriostatic against *M. tuberculosis*. Its primary mechanism of action is the inhibition of the arabinosyltransferases involved in cell wall synthesis, which probably inhibits the formation of arabinogalactan and lipoarabinomannan. The MIC of ethambutol for susceptible strains of *M. tuberculosis* is 0.5–2 μ g/mL.

PHARMACOLOGY AND DOSING From a single dose of ethambutol, 75–80% is absorbed within 2–4 h of administration. Serum levels peak at 2–4 μ g/mL after the standard adult daily dose of 15 mg/kg. Ethambutol is well distributed throughout the body except in the CSF; a dosage of 25 mg/kg is necessary for attainment of a CSF level half of that in serum. For intermittent therapy, the dosage is 25–35 mg/kg thrice weekly. To prevent toxicity, the dosage must be lowered and the frequency of administration reduced for patients with renal insufficiency.

ADVERSE EFFECTS Ethambutol is usually well tolerated and has no significant interactions with other drugs. Optic neuritis, the most serious adverse effect reported, typically presents as reduced visual acuity, central scotoma, and loss of the ability to see green (or, less commonly, red). The cause of this neuritis is unknown, but it may be due to an effect of ethambutol on the amacrine and bipolar cells of the retina. Symptoms typically develop several months after initiation of therapy, but ocular toxicity soon after initiation of ethambutol has been described. The risk of ocular toxicity is dose dependent, with occurrence in 1–5% of patients, and can be increased by renal insufficiency. The routine use of ethambutol in younger children is not recommended because monitoring for visual complications can be difficult. If drug-resistant TB is suspected, ethambutol can be used for treatment of children.

All patients starting therapy with ethambutol should have a baseline test for visual acuity, visual fields, and color vision and should undergo an examination of the optic fundus. Visual acuity and color vision should be monitored monthly or less often as needed. Cessation of ethambutol in response to early symptoms of ocular toxicity usually results in reversal of the deficit within several months. Recovery of all visual function may take up to 1 year. In the elderly and in patients whose symptoms are not recognized early, deficits may be permanent. Some experts think that supplementation with hydroxycobalamin (vitamin B₁₂) is beneficial for patients with ethambutol-related ocular toxicity. Other adverse effects of ethambutol are rare. Peripheral sensory neuropathy occurs in rare instances.

RESISTANCE Ethambutol resistance in *M. tuberculosis* and NTM is associated primarily with missense mutations in the *embB* gene that encodes for arabinosyltransferase. Mutations have been found in resistant strains at codon 306 in 50–70% of cases. Mutations at *embB*306 can cause significantly increased MICs of ethambutol, resulting in clinical resistance.

OTHER RIFAMYCIN DRUGS

Rifabutin Rifabutin, a semisynthetic derivative of rifamycin S, inhibits mycobacterial DNA-dependent RNA polymerase. Rifabutin is recommended in place of rifampin for the treatment of TB in HIV-co-infected individuals who are taking protease inhibitors or nonnucleoside reverse transcriptase inhibitors, particularly nevirapine. A study in India showed better TB treatment outcomes in HIV-co-infected patients given daily rifabutin plus atazanavir/ritonavir than in those given thrice-weekly rifabutin plus atazanavir/ritonavir. Rifabutin's effect on hepatic enzyme induction is less pronounced than that of rifampin. Protease inhibitors may cause significant increases in rifabutin levels through inhibition of hepatic metabolism. Rifabutin is more active in vitro than rifampin against MAC organisms and other NTM, but its clinical superiority has not been established.

PHARMACOLOGY Like rifampin, rifabutin is lipophilic and is absorbed rapidly after oral administration, reaching peak serum levels 2–4 h after ingestion. Rifabutin distributes best to tissues, reaching levels 5–10 times higher than those in plasma. Unlike rifampin, rifabutin and its metabolites are partially cleared by the hepatic microsomal system. Rifabutin's slow clearance results in a mean serum half-life of 45 h—much longer than the 3- to 5-h half-life of rifampin. Clarithromycin (but not azithromycin) and fluconazole appear to increase rifabutin levels by inhibiting hepatic metabolism.

ADVERSE EFFECTS The most common adverse effects of rifabutin treatment are gastrointestinal; other reactions include rash, headache, asthenia, chest pain, myalgia, and insomnia. Less common adverse reactions include fever, chills, a flulike syndrome, anterior uveitis, hepatitis, *Clostridium difficile*-associated diarrhea, a diffuse polymyalgia syndrome, and yellow skin discoloration (“pseudo-jaundice”). Laboratory abnormalities include neutropenia, leukopenia, thrombocytopenia, and increased levels of liver enzymes. Rifabutin appears to be better tolerated by the majority (72%) of adult TB patients who have developed rifampin-related adverse effects. Female patients, those co-infected with hepatitis B or hepatitis C, and those with rifampin-related arthralgias, dermatologic reactions, and cholestasis are more likely to develop mild to severe rifabutin-related adverse effects.

RESISTANCE Similar to rifampin resistance, rifabutin resistance is mediated by mutations in *rpoB*.

Rifapentine Rifapentine is a semisynthetic cyclopentyl rifamycin, sharing a mechanism of action with rifampin. Rifapentine is lipophilic and has a prolonged half-life that permits weekly or twice-weekly dosing. Therefore, rifapentine is the subject of intensive clinical investigation aimed at determining optimal dosing and frequency of administration. Currently, it is an alternative to rifampin in the continuation phase of treatment for noncavitary drug-susceptible pulmonary TB in HIV-seronegative patients who have negative sputum smears at completion of the initial phase of treatment. When administered in these specific circumstances, rifapentine (10 mg/kg, up to 600 mg) is given once weekly with isoniazid. Because of higher rates of relapse, this regimen is not recommended for patients with TB disease and HIV co-infection; moreover, it has not been approved for children <12 years of age. In a phase 2 study, substituting daily rifapentine for rifampin yielded higher rates of sputum sterilization after 2 months of intensive treatment. Higher doses of rifapentine (20 mg/kg vs 10 mg/kg) had better results and were safe and well tolerated. Regimens containing high doses of rifapentine are being evaluated to see whether they can shorten the TB treatment course to <6 months.

PHARMACOLOGY Rifapentine's absorption is improved when the drug is taken with food. After oral administration, rifapentine reaches peak serum concentrations in 5–6 h and achieves a steady state in 10 days. The half-life of rifapentine and its active metabolite, 25-desacetyl rifapentine, is ~13 h. The administered dose is excreted via the liver (70%).

ADVERSE EFFECTS The adverse effects profile of rifapentine is similar to that of other rifamycins. Rifapentine is teratogenic in animal models and is relatively contraindicated in pregnancy.

RESISTANCE Rifapentine resistance is mediated by mutations in *rpoB*. Mutations that cause resistance to rifampin also cause resistance to rifapentine.

SECOND-LINE ANTITUBERCULOSIS DRUGS

Second-line anti-TB agents are indicated for treatment of drug-resistant TB, for patients who are intolerant or allergic to first-line agents, and when first-line supplemental agents are unavailable. According to their usability, they are divided into three WHO groups.

Group A • FLUOROQUINOLONES Fluoroquinolones inhibit mycobacterial DNA gyrase and topoisomerase IV, preventing cell replication and protein synthesis, and are bactericidal. Given their excellent activity, they have been investigated for their potential to shorten the course of treatment for drug-susceptible TB from 6 to 4 months. In contrast to prior trials, a recent large, open-label randomized controlled trial (TBTC Study 31) yielded promising results for shortening of TB treatment. Patients with drug susceptible TB disease were randomized to receive either standard six-month TB regimen or 4-month regimen containing rifapentine (8 weeks of once-daily rifapentine, isoniazid, pyrazinamide, and ethambutol followed by 9 weeks of once-daily rifapentine and isoniazid) or 4-month regimen containing rifapentine and moxifloxacin (8 weeks of once-daily rifapentine, isoniazid, pyrazinamide, and moxifloxacin followed by 9 weeks of once-daily rifapentine, isoniazid, and moxifloxacin). The trial demonstrated that a four-month regimen using daily rifapentine and moxifloxacin (but not the rifapentine only regimen) was non-inferior to the standard six-month TB treatment regimen using an end point of TB-free survival 12 months after randomization. Combining once daily rifapentine with moxifloxacin allows for synergistic action on sputum conversion in a compliance-friendly once-daily option. Current recommendations continue to be for a standard six-month regimen though it is anticipated that these results will inform future guidelines. Gatifloxacin has fallen out of favor because of significant dysglycemia. Ciprofloxacin and ofloxacin are no longer recommended for the treatment of TB because of poor efficacy. Despite documented resistance to early-generation fluoroquinolones (e.g., ofloxacin and ciprofloxacin), use of a later-generation fluoroquinolone in patients with drug-resistant TB has been associated with favorable outcomes. Fluoroquinolones are also considered safe alternatives for patients who develop treatment-limiting adverse effects from first-line agents. Levofloxacin and moxifloxacin have both been used effectively in the treatment of MDR-TB. The optimal dose of levofloxacin for this indication is being actively studied, but doses of at least 750 mg are commonly used. High-dose moxifloxacin (800 mg) is recommended for standardized shorter MDR-TB regimens.

The fluoroquinolones are well absorbed orally, reach high serum levels, and distribute well into body tissues and fluids. Their absorption is decreased by co-ingestion with products containing multivalent cations, such as antacids. Adverse effects are relatively infrequent (0.5–10% of patients) and include gastrointestinal intolerance, rashes, dizziness, and headache. Most studies of fluoroquinolone side effects have been based on relatively short-term administration for bacterial infections, but trials have now shown the relative safety and tolerability of fluoroquinolones administered for months during TB treatment in adults. Although the potential to prolong the QTc interval, leading to cardiac arrhythmias, has been a source of concern with fluoroquinolones, cessation of treatment due to this adverse effect is rare. Because the benefits may outweigh the risks in treatment of drug-resistant TB, there is increasing interest in the use of fluoroquinolones in children, which has traditionally been avoided because of the risks of tendon rupture and cartilage damage.

Multiple courses of empirical fluoroquinolone therapy for presumed community-acquired pneumonia are associated with delayed diagnosis of active pulmonary TB and increased fluoroquinolone resistance in *M. tuberculosis*. Mutations in the genes encoding for DNA gyrase (*gyrA* and *gyrB*) are implicated in the majority of cases—but not all cases—of clinical resistance to fluoroquinolones.

DIARYLQUINOLINES Bedaquiline (TMC207 or R207910) is a diarylquinoline with a novel mechanism of action: inhibition of the

mycobacterial ATP synthetase proton pump. Bedaquiline is bactericidal for *M. tuberculosis*. Resistance has been reported due to point mutations in the *atpE* gene encoding for subunit c of ATP synthetase. Clinical bedaquiline resistance has also been reported due to nontarget mutations in Rv0678 (a negative repressor of the MmpL5 efflux pump) and PepQ (a cytoplasmic peptidase), both of which may cause cross-resistance to clofazimine. Bedaquiline is metabolized by the hepatic cytochrome CYP3A4. Rifampin lowers bedaquiline levels by 50%, and protease inhibitors also interact significantly with this drug. Because efavirenz induces CYP3A4, there is concern about lower bedaquiline levels with co-administration. In a study of co-treatment with bedaquiline and efavirenz in healthy volunteers, bedaquiline levels were reduced by only 20%; however, in a study modeling chronic co-administration of these two drugs, the reduction in bedaquiline levels was estimated to be 50%, leading many national TB programs to avoid efavirenz co-administration with bedaquiline.

The oral bioavailability of bedaquiline appears to be excellent. The dosage is 400 mg/d for the first 2 weeks and then 200 mg thrice weekly typically for 6 months total. The elimination half-life is long (>14 days). A single dose of this drug can inhibit the growth of *M. tuberculosis* for up to 1 week through a combination of long plasma half-life, high-level tissue penetration, and long tissue half-life. Bedaquiline added to a background regimen improved the 2-month sputum culture-conversion rate in multicenter, randomized placebo-controlled trials, and these results led to approval by the U.S. Food and Drug Administration (FDA). However, the death rate in one trial was higher in the bedaquiline arm than in the control arm (11.4% vs 2.5%); the result was a “black box” warning from the FDA, which also included QT prolongation. Subsequent studies have not found an association with significant mortality. The CDC has made a provisional recommendation for the use of bedaquiline for 24 weeks in adults with laboratory-confirmed pulmonary MDR-TB when no other effective treatment regimen can be provided. Bedaquiline is an integral part of all shorter course, oral MDR treatment regimens endorsed by the WHO.

OXAZOLIDINONES Linezolid is an oxazolidinone used primarily for the treatment of drug-resistant gram-positive bacterial infections. However, this drug is active in vitro against *M. tuberculosis* and NTM. Several case series have suggested that linezolid may help clear mycobacteria relatively rapidly when included in a regimen for the treatment of complex cases of drug-resistant TB. Linezolid’s mechanism of action is disruption of protein synthesis by binding to the 50S bacterial ribosome. Linezolid has nearly 100% oral bioavailability, with good penetration into tissues and fluids, including CSF. Clinical resistance to linezolid has been reported and is typically associated with mutations in the 23S rRNA and in two ribosomal proteins, L3 (*rplC*) and L4 (*rplD*). Adverse effects may include optic and peripheral neuropathy, pancytopenia, and lactic acidosis and are usually associated with higher doses. Linezolid is a weak monoamine oxidase inhibitor and can be associated with the serotonin syndrome when given concomitantly with serotonergic drugs (primarily antidepressants such as selective serotonin reuptake inhibitors). It has been shown that ~80% of patients with MDR-TB can be successfully treated with linezolid-containing, individualized anti-TB regimens based on drug sensitivity testing. Replacement of ethambutol with linezolid for 2–4 weeks during the intensive phase of treatment of drug-susceptible TB is currently being evaluated for possible faster sputum conversion and a shorter treatment regimen. For MDR-TB treatment, linezolid is usually administered at a dose of 600 mg (or less in some cases) once daily, which appears to be effective. A single daily dose is associated with fewer adverse events than twice-a-day dosing.

Sutezolid, a modified version of oxazolidinones and protein synthesis inhibitor, is found to have higher early bactericidal activity compared to linezolid and is currently undergoing phase 2A trials. It is currently FDA approved for complex skin infections and appears to have less frequent side effects compared to linezolid; the adverse effects profile of long-term exposure compared with that of linezolid needs further investigation.

Group B • CLOFAZIMINE Clofazimine is a fat-soluble riminophenazine dye used primarily in the treatment of leprosy worldwide.

It is currently gaining popularity in the management of drug-resistant TB because of its low cost and its intracellular and extracellular activity. By increasing reactive oxidant species and causing membrane destabilization, clofazimine may promote killing of antibiotic-tolerant *M. tuberculosis* “persister” organisms. In addition to antimicrobial activity, the drug has other pharmacologic activities, such as anti-inflammatory, pro-oxidative, and immunopharmacologic properties. Clofazimine has a half-life of ~70 days in humans, and average steady-state concentrations are achieved at ~1 month. Intake with fatty meals can improve its low and variable rates of absorption (45–62%). Common side effects include gastrointestinal intolerance, and reversible orange to brownish discoloration of skin, bodily fluids, and secretions. Dose adjustment may be necessary in patients with severe hepatic impairment. Clofazimine was studied as part of a regimen developed in Bangladesh for potential shortening of the MDR-TB treatment course. A meta-analysis suggested that inclusion of clofazimine in a multidrug regimen for treatment of MDR-TB was associated with a favorable outcome. Newer analogues with improved pharmacokinetics and alternative formulations of clofazimine (liposomal, nanosuspension, inhalational) are being studied.

CYCLOSERINE Cycloserine is an analog of the amino acid D-alanine and prevents bacterial cell-wall synthesis. It inhibits the action of enzymes, including alanine racemase, that are involved in the production of peptidoglycans. Cycloserine is active against a range of bacteria, including *M. tuberculosis*. Mechanisms of mycobacterial resistance are not well understood, but overexpression of alanine racemase can confer resistance in *Mycobacterium smegmatis*. Cycloserine is well absorbed after oral administration and is widely distributed throughout body fluids, including CSF. The usual adult dosage is 250 mg two or three times per day. Serious potential side effects include seizures and psychosis (with suicide in some cases), peripheral neuropathy, headache, somnolence, and allergic reactions. Drug levels are monitored to achieve optimal dosing and to reduce the risk of adverse effects, especially in patients with renal failure. Cycloserine should be administered as DOT only with caution and with support from experienced TB physicians to patients with epilepsy, active alcohol abuse, severe renal insufficiency, or a history of depression or psychosis.

Group C • NITROIMIDAZOLES The prodrugs delamanid (OPC-67683) and pretomanid (PA 824) are novel nitro-dihydroimidazo[4,5-b]pyridine derivatives that are activated by *M. tuberculosis*-specific flavin-dependent nitroreductases and whose antimycobacterial activity is attributable to inhibition of mycolic acid biosynthesis. Delamanid was shown in a randomized, placebo-controlled, multinational clinical trial to significantly improve the culture conversion rate at 2 months. QT prolongation occurred significantly more often in delamanid-treated patients, but no clinically relevant events were reported. In a subsequent randomized phase 3 trial, there was no significant difference in 6 months sputum conversion between delamanid and placebo among patients with an optimized background regimen. Currently, it is part of several ongoing clinical trials including combination with bedaquiline. It is recommended for the use in children younger than 6 years with rifampicin-resistant TB. Usual adult dose is 100 mg twice daily.

Pretomanid, the second novel agent from this class, has shown promising results in the treatment of drug-resistant TB in combination with bedaquiline. A combination of pretomanid with moxifloxacin and pyrazinamide for treatment of drug-susceptible TB was found to have higher culture conversion rates at 8 weeks compared to HRZE; however, a subsequent phase 3 study raised concern for higher frequency of potentially fatal hepatotoxicity. It is currently being evaluated in several phase 3 clinical trials in various combinations, including with fluoroquinolones and pyrazinamide. Based on the previously mentioned results with the BPaL regimen (Nix-TB study), the FDA has granted approval for specific highly resistant TB cases. Adult treatment dose is 200 mg administered daily.

AMOXICILLIN-CLAVULANATE AND CARBAPENEMS β -Lactam agents are largely ineffective for the treatment of *M. tuberculosis* because of

resistance conferred by a hydrolyzing class A β -lactamase, BlaC. Carbapenems are poor substrates of BlaC, and clavulanic acid leads to irreversible inhibition. While the use of either amoxicillin-clavulanic acid or carbapenems alone for highly resistant forms of TB has been anecdotally reported with unclear results, the combination of meropenem and clavulanic acid turned out to be highly active in vitro. Recently, the combination was found to have effective early bactericidal activity, and in a large individual patient data meta-analysis, the combination was associated with positive outcomes. Nevertheless, the need to administer these carbapenems intravenously and the lack of information on the drugs' long-term side effects have restricted their use to certain severe cases only. Recommended daily doses are either imipenem-cilastatin 1 g (each component) IV twice daily or meropenem 1 g IV three times daily, each in combination with clavulanic acid 125 mg oral twice daily, which is only available in combination with amoxicillin.

AMINOGLYCOSIDES Aminoglycosides have played a time-honed role in the treatment of mycobacterial infections. Amikacin and streptomycin are aminoglycosides that exert mycobactericidal activity by binding to the 16S ribosomal subunit. The spectrum of antibiotic activity for amikacin and streptomycin includes *M. tuberculosis*, several NTM species, and aerobic gram-negative and gram-positive bacteria. Due to the need of intravenous or painful intramuscular injections and their serious side effect profile, the WHO recommends limiting their use with the increased availability of novel oral agents. Kanamycin and capreomycin, a cyclic polypeptide similar to aminoglycosides, are no longer recommended due to worse treatment outcomes and increased mortality. This recommendation is based on a large individual patient-level meta-analysis of observational cohort studies and is likely due to increased toxicity seen with these agents. Streptomycin was the first antimycobacterial agent used for the treatment of TB. Derived from *Streptomyces griseus*, streptomycin is bactericidal against dividing *M. tuberculosis* organisms but has only low-level early bactericidal activity. In developing countries, it continues to be widely used due to its low cost. The usual daily dose of streptomycin (given IM either daily or 5 days per week) is 15 mg/kg for adults and 20–40 mg/kg for children, with a maximum of 1 g/d for both with dose reduction recommended for patients ≥ 60 years of age or with renal impairment. Central nervous system penetration is poor.

Amikacin resistance is less widespread, and streptomycin-resistant strains may still be susceptible. The usual daily adult dosage is 15–30 mg/kg given IM or IV (maximal daily dose, 1 g). It is frequently used to treat severe nontuberculous mycobacterial infections.

Mycobacterial resistance to aminoglycosides is due to mutations in the genes encoding the 16S ribosomal RNA gene (*rrs*). Adverse effects of both amikacin and streptomycin include ototoxicity (in up to 10% of recipients, with auditory dysfunction occurring more commonly than vestibulotoxicity), nephrotoxicity, and neurotoxicity.

ETHIONAMIDE Ethionamide is a derivative of isonicotinic acid. Its mechanism of action is through inhibition of the *inhA* gene product enoyl-acyl carrier protein (acp) reductase, which is involved in mycolic acid synthesis. Ethionamide is bacteriostatic against metabolically active *M. tuberculosis* and some NTM. It is used in the treatment of drug-resistant TB, but its use is limited by severe gastrointestinal reactions (including abdominal pain, nausea, and vomiting) as well as significant central and peripheral neurologic side effects, reversible hepatitis (in ~5% of recipients), hypersensitivity reactions, and hypothyroidism. Ethionamide should be taken with food to reduce gastrointestinal effects and with pyridoxine (50–100 mg/d) to limit neuropathic side effects.

PARA-AMINOSALICYLIC ACID Para-aminosalicylic acid (PAS; 4-aminosalicylic acid) is an oral agent used in the treatment of drug-resistant TB. Its bacteriostatic activity is due to inhibition of folate synthesis and of iron uptake. PAS has relatively little activity as an anti-TB agent. Adverse effects may include high-level nausea, vomiting, and diarrhea. PAS may cause hemolysis in patients with glucose-6-phosphate dehydrogenase deficiency. The drug should be taken with acidic foods to improve absorption. Enteric-coated PAS granules (4 g orally every 8 h) appear to be better tolerated than other formulations and produce

higher therapeutic blood levels. PAS has a short half-life (1 h), and 80% of the dose is excreted in the urine.

■ DRUGS IN DEVELOPMENT

The pipeline of novel TB drugs is rapidly changing. We direct the reader to the Working Group on New TB Drugs for the most up-to-date information (<https://www.newtbdrugs.org/pipeline/clinical>).

NONTUBERCULOUS MYCOBACTERIA

More than 150 species of NTM have been identified. Only a minority of these environmental organisms, which are extensively found in soil and water, are important human pathogens. NTM cause extensive disease primarily in persons with preexisting pulmonary disease or immunocompromise but can also cause nodular/bronchiectatic disease in otherwise seemingly healthy hosts. Disseminated infections with NTM are common in immunocompromised individuals. NTM are also important causes of skin and soft tissue infections in surgical settings. The two major classes of NTM are the slow-growing and rapidly growing species; subcultures of the latter grow within 1 week. The growth characteristics of NTM have diagnostic, therapeutic, and prognostic implications. The rate of growth can provide useful preliminary information within a specific clinical context, in that growth within 2–3 weeks is much more likely to indicate an NTM than *M. tuberculosis*. When NTM do grow from cultures, colonization should be distinguished from active disease in order to optimize the risk and benefit of prolonged treatment with multiple medications. According to the recommendations of the American Thoracic Society and the Infectious Diseases Society of America, significant clinical manifestations and/or radiographic evidence of progressive disease consistent with NTM infection as well as either reproducible sputum culture results or a single positive culture from bronchoscopy are required for the diagnosis of NTM pulmonary disease. Isolation of NTM from blood or from an infected extrapulmonary site, such as soft tissue or bone, is usually indicative of disseminated or local NTM infection (**Chap. 180**). Treatment of NTM disease is prolonged and requires multiple medications. Side effects of the regimens employed are common, and intermittent therapy is often used to mitigate these adverse events. Treatment regimens depend on the NTM species, the extent or type of disease, and—to some degree—drug susceptibility test results.

■ THERAPEUTIC CONSIDERATIONS FOR SPECIFIC NTM

Slowly Growing Mycobacteria Slowly growing mycobacteria can be divided into three categories based on their pigment-producing capabilities and—if they do produce pigment—their requirement for light to do so. *Photochromogens*, including *M. marinum* and *M. kansasii*, can produce yellowish-orange pigment only when exposed to light. *Scotochromogens*, including *Mycobacterium goodii* and *Mycobacterium scrofulaceum*, can make pigment regardless of light exposure. MAC organisms and *Mycobacterium ulcerans* are *nonchromogens*—i.e., are incapable of making pigment irrespective of light exposure.

MYCOBACTERIUM AVIUM COMPLEX Among the NTM, MAC organisms most commonly cause human disease. In immunocompetent hosts, MAC species are most often found in conjunction with underlying significant lung disease, such as chronic obstructive pulmonary disease or bronchiectasis. For patients with nodular or bronchiectatic MAC lung disease, an initial regimen consisting of clarithromycin or azithromycin, rifampin or rifabutin (the latter is preferred for HIV patients receiving ART), and ethambutol is given three times per week for at least 12 months after culture conversion. A daily regimen of these three drugs, with consideration of amikacin or streptomycin in the initial treatment phase, is recommended for patients with fibrocavitary MAC lung disease or severe nodular/bronchiectatic disease. Routine initial testing for macrolide resistance is recommended, as is testing at 6 months with a failing regimen (i.e., with cultures persistently positive for NTM). Interpretation of susceptibility tests to drugs other than macrolides and aminoglycosides is hampered by poor correlation with clinical outcomes. Amikacin has been reformulated as a liposomal suspension with increased penetration into airway biofilms. The

CONVERT trial showed that addition of inhaled liposomal amikacin to standard three-drug regimen of azithromycin or clarithromycin, rifampin, and ethambutol in treatment-refractory (persistent sputum positivity after at least 6 months) MAC lung disease significantly increases culture conversion rates from 9 to 26% at 6 months. Respiratory adverse events (primarily dysphonia, cough, and dyspnea) were reported in 87.4% of patients receiving inhaled liposomal amikacin compared to 50% in the standard therapy group; however, rates of serious adverse events were not different between the regimens. Inhaled liposomal amikacin is now approved for use in refractory pulmonary MAC infections (persistent positive cultures after at least 6 months of treatment). It is currently being evaluated as a first-line agent and as a replacement for rifampin in the treatment of MAC lung disease.

Surgical resection should be considered for individuals whose infection is localized to one lung, who have adequate lung function to tolerate lung resection, who have had a poor response to medical therapy, and/or who have developed macrolide-resistant MAC disease.

Treatment of MAC in persons living with HIV should be initiated in consultation with an infectious diseases specialist. For HIV-infected patients with well-controlled HIV disease and CD4 T-cell counts in the normal range, MAC treatment is identical to patients without HIV disease except that drug-drug interactions between antimycobacterial agents and ART should be carefully considered. HIV-infected patients with low CD4 count (CD4+ T-cell count >100/ μ L) are at risk for disseminated MAC infection. MAC disease in these patients is generally treated with clarithromycin, ethambutol, and rifabutin. Azithromycin may be preferred to clarithromycin depending on adverse effects and patient tolerance. Amikacin and fluoroquinolones are often used in salvage regimens. Treatment for disseminated MAC infection in AIDS patients may be lifelong in the absence of immune reconstitution. Therapy is recommended for at least 12 months after culture conversion and at least 6 months of effective immune reconstitution with ART (CD4+ T-cell count >100/ μ L).

MYCOBACTERIUM KANSASII *M. kansasii* is the second most common NTM causing human disease in the United States. It is also the second most common cause of NTM pulmonary disease in the United States, where it is most commonly reported in the southeastern region. *M. kansasii* infection can be treated with rifampin, ethambutol, and either isoniazid or macrolide; therapy continues for at least 18 months or for 12 months after culture conversion. The American Thoracic Society and the Infectious Diseases Society of America recommend routine susceptibility testing to rifampin only. Resistance to isoniazid and ethambutol can be acquired during therapy but is usually associated with rifampin resistance as well. Rifampin-resistant *M. kansasii* is treated with a three-drug regimen including agents such as ciprofloxacin, azithromycin, ethambutol, rifabutin, amikacin, trimethoprim-sulfamethoxazole, and streptomycin after drug susceptibility testing.

MYCOBACTERIUM MARINUM *M. marinum* is an NTM found in salt water and freshwater, including swimming pools and fish tanks. It is a cause of localized soft tissue infections, which may require surgical management. Combination regimens include clarithromycin and either ethambutol or rifampin. Other agents with activity against *M. marinum* include doxycycline, minocycline, and trimethoprim-sulfamethoxazole. Drug susceptibility testing is recommended only if the swab remains culture positive after 3 months of appropriate therapy.

Rapidly Growing Mycobacteria Rapidly growing mycobacteria causing human disease include *Mycobacterium abscessus*, *Mycobacterium fortuitum*, and *Mycobacterium chelonae*. Treatment of these mycobacteria is complex and should be undertaken with input from experienced clinicians. It is important to note that testing rapidly growing mycobacteria for macrolide resistance is tricky, as an inducible *erm* gene may confer in vivo macrolide resistance to isolates that are susceptible in vitro.

M. abscessus is the third most common NTM pathogen in the United States. It is endemic in the southeastern states between Texas and Florida. Skin, soft tissue, and bone infections occur, usually after

accidental trauma or surgery. This organism appears to have a predilection to cause lung infections in white nonsmoking women aged >60 who have no preexisting lung disease. *M. abscessus* isolates are usually resistant to standard anti-TB regimens. Skin and soft tissue infections are usually treated for a minimum of 4 months with a macrolide (clarithromycin or azithromycin) and a parenteral agent such as amikacin, cefoxitin, or imipenem. Bone infections are treated for at least 6 months. This regimen can be used for the treatment of lung infections but is often unsuccessful because of drug adverse effects and toxicities. A regimen comprising a combination of at least three active drugs (amikacin, linezolid, tigecycline, imipenem, azithromycin, provided the organism is macrolide susceptible) is recommended on the basis of in vitro drug susceptibility testing. A recent meta-analysis has shown that overall therapeutic efficiency rates in *M. abscessus* lung infection are low at ~35%; however, incorporation of amikacin, imipenem, linezolid, and/or tigecycline was associated with improved outcomes. Conversely, macrolide resistance has been associated with worse outcomes. Surgical resection should be considered in all patients with good lung reserve and a localized infection.

■ DRUGS FOR THE TREATMENT OF NTM

Clarithromycin Clarithromycin is a macrolide antibiotic with broad activity against many gram-positive and gram-negative bacteria as well as NTM. This drug is active against MAC organisms and many other NTM species, inhibiting protein synthesis by binding to the 50S mycobacterial ribosomal subunit. NTM resistance to macrolides is probably caused by overexpression of the gene *ermB*, with consequent methylation of the binding site. Strains of *M. abscessus* subsp. *abscessus* harbor an inducible macrolide resistance mechanism coded by *erm41*, which leads to ribosomal methylation and becomes apparent after macrolide incubation of 3–5 days, significantly hampering treatment success. Twenty percent of strains have a nonfunctional *erm41* gene. Clarithromycin is well absorbed orally and distributes well to tissues. It is cleared both hepatically and renally; the dosage should be reduced in renal insufficiency. Clarithromycin is a substrate for and inhibits cytochrome 3A4 and should not be administered with cisapride, pimozide, or terfenadine because cardiac arrhythmias may occur. Numerous drugs interact with clarithromycin through the CYP3A4 metabolic pathway. Rifampin lowers clarithromycin levels; conversely, rifampin levels are increased by clarithromycin. However, the clinical relevance of this interaction does not appear to be great.

For patients with nodular/bronchiectatic MAC infection, the dosage of clarithromycin is 500 mg, given morning and evening three times a week. For the treatment of fibrocavitary or severe nodular/bronchiectatic MAC infection, a dose of 500–1000 mg is given daily. Disseminated MAC infection is treated with 1000 mg daily. Clarithromycin is used in combination regimens that typically include ethambutol and a rifamycin in order to avoid the development of macrolide resistance. Adverse effects include frequent gastrointestinal intolerance, hepatotoxicity, headache, rash, and rare instances of hypoglycemia. Clarithromycin is contraindicated during pregnancy because of its teratogenicity in animal models.

Azithromycin Azithromycin is a derivative of erythromycin. Although technically an azalide and not a macrolide, it works similarly to macrolides, inhibiting protein synthesis through binding to the 50S ribosomal subunit. Azithromycin is preferred over clarithromycin due to once-daily dosing, better tolerability, fewer drug interactions, and equal efficacy. Resistance to azithromycin is almost always associated with complete cross-resistance to clarithromycin. Azithromycin is well absorbed orally, with good tissue penetration and a prolonged half-life (~48 h). The usual dosage for treatment of MAC infection is 250 mg daily or 500 mg three times per week. Azithromycin is used in combination with other agents to avoid the development of resistance. For prophylaxis against disseminated MAC infection in immunocompromised individuals, a dose of 1200 mg once per week is given. Because azithromycin is not metabolized by cytochrome P450, it interacts with few drugs. Adjustment of the dosage on the basis of renal function is not necessary.

1406 Amikacin Liposome Inhalation Suspension (ALIS) ALIS is a new formulation of the aminoglycoside amikacin, which allows for improved penetration in the lung with reduced toxicity. It is now approved for treatment of refractory MAC lung infection with persistent sputum positivity at 6 months while on appropriate background regimen. Typical dose is 590 mg (one vial once a day) for 6 months along with the standard three-drug regimen of macrolide, rifampin, and ethambutol. Dosage adjustments in patients with hepatic and renal dysfunction are not required. Half-life elimination typically occurs in ~5.9–9.5 h. Respiratory side effects such as bronchospasm, cough, dysphonia, and dyspnea are common. Monitoring for systemic aminoglycoside toxicity should be considered.

Imipenem Imipenem primarily inhibits cell-wall biosynthesis by binding to the penicillin-binding proteins. It is rapidly gaining importance for the treatment of *M. abscessus* with a meta-analysis showing improved outcomes with its inclusion in a multidrug regimen. It is dosed at 500 mg to 1 g twice to three times a day as part of a combination regimen for the treatment of *M. abscessus*. Half-life of imipenem is ~1 h, and because it is metabolized in the kidneys, dosing adjustment is needed with renal dysfunction. Adverse effects include anemia, thrombocytopenia, and liver dysfunction.

Cefoxitin Cefoxitin is a second-generation parenteral cephalosporin with activity against rapidly growing NTM, particularly *M. abscessus* and *M. chelonae*. Its mechanism of action against NTM is unknown but may involve inactivation of cell-wall synthesis enzymes. High doses are used for treatment of NTM: 200 mg/kg IV three or four times per day, with a maximal daily dose of 12 g. The half-life of cefoxitin is ~1 h, with primary renal clearance that requires adjustment in renal insufficiency. Adverse effects are uncommon but include gastrointestinal manifestations, rash, eosinophilia, fever, and neutropenia.

Newer Drugs Three newer classes of drugs—the oxazolidinones, the glycyclines, and the ketolides—are currently being evaluated for possible use in the treatment of NTM infections, especially those caused by *M. abscessus*. Approximately 50% of *M. abscessus* isolates have shown some degree of susceptibility in vitro to linezolid, an oxazolidinone. Tigecycline, which is a glycycline and a tetracycline derivative, and telithromycin, a ketolide, also appear to have in vitro activity against *M. abscessus*. These drugs, however, have not yet been clinically tested in patients.

In addition, some anti-TB drugs, including clofazimine and bedaquiline, are being evaluated as alternative agents for the treatment of refractory NTM infections. In particular, clofazimine appears to act synergistically in combination with amikacin, bedaquiline, or tigecycline. The exact role of these agents in the treatment of refractory NTM infections remains unclear. Suppressive therapy with periodic parenteral/oral drugs to limit disease progression and control symptoms may be an appropriate alternative to curative treatment.

CONCLUSION

Treatment of mycobacterial infections requires multiple-drug regimens that often exert significant side effects with the potential to limit tolerability. The prolonged duration of treatment has vastly improved results over those obtained in past decades, but drugs and regimens that will shorten treatment duration and limit adverse drug effects and interactions are needed.

FURTHER READING

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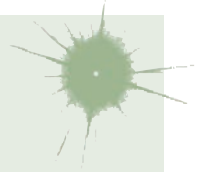
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Section 9 Spirochetal Diseases

182 Syphilis

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DEFINITION

Syphilis, a chronic systemic infection caused by *Treponema pallidum* subspecies *pallidum*, is usually sexually transmitted and is characterized by episodes of active disease interrupted by asymptomatic periods (latency). After an incubation period averaging 2–6 weeks, a primary lesion appears—often associated with regional lymphadenopathy—and then resolves without treatment. The secondary stage, with generalized mucosal and cutaneous lesions and generalized lymphadenopathy, also resolves spontaneously and is followed by a latent period of subclinical infection lasting years or decades. Central nervous system (CNS) invasion may occur early in infection, and CNS involvement may be symptomatic or asymptomatic. In the preantibiotic era, one-third of untreated patients developed tertiary syphilis, characterized by destructive mucocutaneous, skeletal, or parenchymal lesions; aortitis; or late CNS manifestations.

ETIOLOGY

The Spirochaetales include five genera that are pathogenic for humans and for a variety of other animals: *Leptospira* species (leptospirosis, **Chap. 184**); *Borrelia* and *Borrelia* species (relapsing fever and Lyme disease, respectively; **Chaps. 185 and 186**); *Brachyspira* species (gastrointestinal infections); and *Treponema* species (syphilis and the endemic treponematoses; **see also Chap. 183**). The *Treponema* subspecies include *T. pallidum* subsp. *pallidum* (venereal syphilis); *T. pallidum* subsp. *pertenue* (yaws); *T. pallidum* subsp. *endemicum* (endemic syphilis or bejel); and *T. carateum* (pinta). Historically, the subspecies were distinguished by the clinical syndromes they produce, but phylogenetic analyses of whole genome sequences from a relatively small number of strains (excluding *T. carateum*) yield the three named subspecies groupings. Whether these groupings represent geographical variation or true biological differences is unclear. The crossing of subspecies boundaries by some “molecular signatures” and the recent recognition of treponemes of the *endemicum* genotype in sexually acquired genital ulcers (chancres) and secondary rashes (**Chap. 183**) support the concept of a genetic and clinical “continuum” among strains and subspecies of the pathogenic treponemes.

T. pallidum subspecies are thin spiral organisms, with a cell body surrounded by a trilaminar cytoplasmic membrane, a delicate peptidoglycan layer, and a lipid-rich outer membrane. Endoflagella wind around the cell body in the periplasmic space and are responsible for motility.



Historically, *T. pallidum* could not be cultured in vitro, but long-term propagation of the Nichols strain of *T. pallidum* in complex medium with eukaryotic cells was recently reported. To date, the *pertenue* and *endemicum* subspecies have not been cultured. All *T. pallidum* subspecies have severely limited metabolic capabilities and are highly dependent on host-derived amino acids, carbohydrates, and lipids. Genetic analyses have revealed the existence of a 12-member